

IMPERIAL

ADVANCED PARTICLE PHYSICS

IMPERIAL COLLEGE LONDON

DEPARTMENT OF PHYSICS

Comprehensive Lecture Summaries on Advanced Particle Physics

*Based on Lectures
by Dr. Michael McCann*

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These summaries draw upon the lecturer's outstanding presentations and are further enhanced by my own insights and carefully curated materials from respected sources, including:

- Nuclear and Particle Physics by Martin & Shaw
- An Introduction to Quantum Field Theory by Peskin & Schroeder
- Quantum Field Theory in a Nutshell by Anthony Zee
- Quantum Field Theory by Mark Srednicki
- David Skinner's lecture notes on Quantum Field Theory
- Quantum Field Theory Lectures of Sidney Coleman by Sidney Coleman

I hope these compiled summaries prove beneficial and insightful for students who aspire to learn advanced particle physics.

– Yoogeun Song.

APP LI

Dr. Michael McCann ①
Experimentalist 7/01/2025

He'll show us the 'experimental' POV.

Quiz

#1.

Given the op.s $E = i \frac{\partial}{\partial t}$ & $p = -i \nabla$,

how is the usual form of the Schrödinger Eq. formed?

- By plugging in the op.s into the classical KE formula, $E = \frac{p^2}{2m}$.

#2. How is the Schrödinger Eq. extended to describe a particle's interaction w/ a mag. fld. B ?

- By introducing a term $\underline{\sigma} \cdot \underline{B}$.

(2)

"Reminder"

SM — Matter

Generations			EM	Weak	Strong
1	2	3			
u	c	t	$2/3$	✓	✓
d	s	b	$-1/3$	✓	✓
e	μ	τ	-1	✓	X
ν_e	ν_μ	ν_τ	0	✓	X

Forces → EM, Weak, Strong, Gravity

Higgs → "What's its mass?"

Annotate on the LNs. → uploaded each wk.

① L2-5 → Fermions

② Next → EM

③ SM → Strong & Weak

④ Higgs Mechanism

⑤ BSM → "Destroy the SM"
↳ That's his goal

Prb.s → algebra he doesn't do in class.

[try to do 'em on time]

9AM Tuesdays

ACP-style RF session

CERN Summer Sch	Jan 26
DESY //	Jan 31
ICL UROP	March 1

[Particle Physics]

How do particles work? How do they interact w/ e.o.?

To understand how nature works, we build up to the SM.

We use 'bottom-up' approach, Experiments, MFs!
instead of some theory courses using "top-down" approach.

SM — Matter

Generations			EM	Weak	Strong
1	2	3			
u	c	t	$2/3$	✓	✓
d	s	b	$-1/3$	✓	✓
e	μ	τ	-1	✓	X
ν_e	ν_μ	ν_τ	0	✓	X

Quarks interact w/ the strong force.

Leptons do NOT interact w/ the strong force.

To a good approximation, these 3 generations are nearly identical.

Matter particles $\rightarrow$ organized into 3 generations.

neutral lepton $\rightarrow$ charge zero

Fermions $\rightarrow$ Spin $-\frac{1}{2}$ particles

④

We can count the # of neutrinos that are light.
The matter particles themselves do not interact directly.

4 fundam. forces of nature.

The SM is a qtm theory.

W & Z bosons $\xrightarrow{\text{quantize}}$ Weak force
(hideously massive)

gluons $\xrightarrow{\text{quantize}}$ Strong force

particles interact w/ fld.s $\rightarrow$ fld.s interact w/ particles again

done smth similar
in the NPP module

Higgs $\rightarrow$ responsible for mass.

EM $\rightarrow$ has an infinite range.

S.F. > W.F. > E.M. > Gravity

The SM cannot explain Gravity.

$\hookrightarrow$ Beyond SM (BSM) $\rightarrow$ may explain gravity.

We use natural units: $\hbar = c = 1 = E_0 = M_0$

$$[c] = \frac{[D]}{[t]} = \frac{[\text{dist.}]}{[\text{time}]}$$

$$= 1$$

$$[\hbar] = [E][\text{time}]$$

$$= 1$$

simplifies
relativistic
formulas

$$E^2 = p^2 c^2 + m^2 c^4$$

$$= p^2 + m^2$$

$$\hat{E} = i\hbar \frac{\partial}{\partial t} \rightarrow i \frac{\partial}{\partial t}$$

(also 1.)
 $\rightarrow$ simplify
Maxwell's Eqns

We accelerated
things

like e^- s or protons

(w/ $q_e = -e$)

(w/ $q_p = +e$)

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$[E] = 1 \text{ eV} \simeq 1.6 \times 10^{-19} \text{ J}$$

A good rule of thumb is to think of protons.

The proton's En. is about 1 GeV

$$E_p = 939 \text{ MeV}$$

$$m_p = 939 \text{ MeV}/c^2$$

(5)

3-vec. (bf. or $\underline{x}$)

$$\underline{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x^i$$

4-vec.

$$x^\mu = \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x^0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} : \text{contravariant vec.}$$

$$x_\mu = \begin{pmatrix} t \\ -x \\ -y \\ -z \end{pmatrix} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} : \text{Covariant vec.}$$

$$x^\mu = \sum_\nu g^{\mu\nu} x_\nu = g^{\mu\nu} x_\nu$$

 $g_{\mu\nu}$ = metric

$$\eta^{\mu\nu} = \begin{pmatrix} 1 & & & 0 \\ & -1 & & \\ & & -1 & \\ 0 & & & -1 \end{pmatrix} : \text{Minkowski metric}$$

* He uses $g^{\mu\nu}$ as the Minkowski metric.In Dr. McCann's notation, $g^{\mu\nu} = \eta^{\mu\nu}$.

$$x^\mu x_\mu = x^0 x_0 + x^1 x_1 + \dots \\ = t^2 - x^2 - y^2 - z^2$$

(+ - - -)

Like QFT!

$$p^\mu p_\mu = E^2 - p_x^2 - p_y^2 - p_z^2 = m^2$$

We just say mass when we mean invariant mass.Standard Wave Fn. for a free particle:
(WF)

$$\Psi = N e^{-i(Et - \underline{p} \cdot \underline{x})} = N e^{-i p^\mu x_\mu}$$

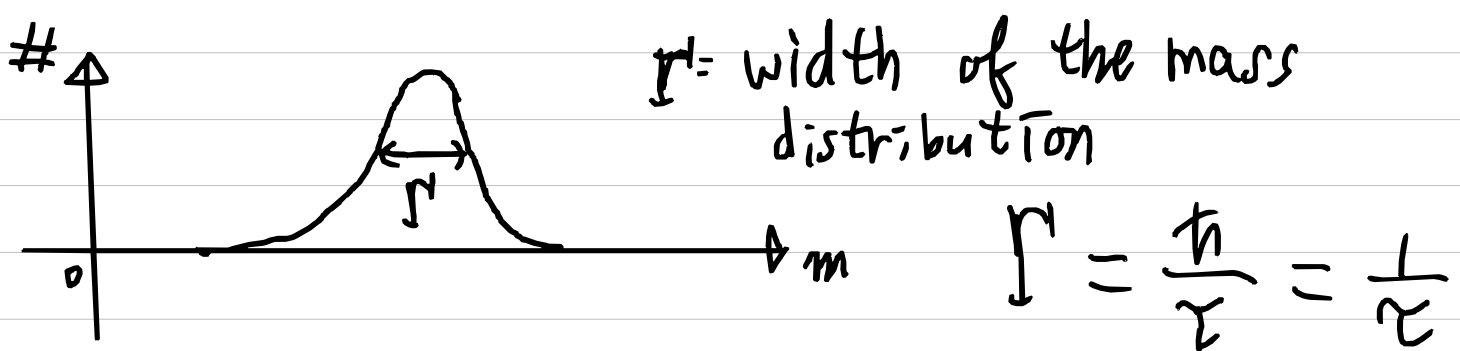
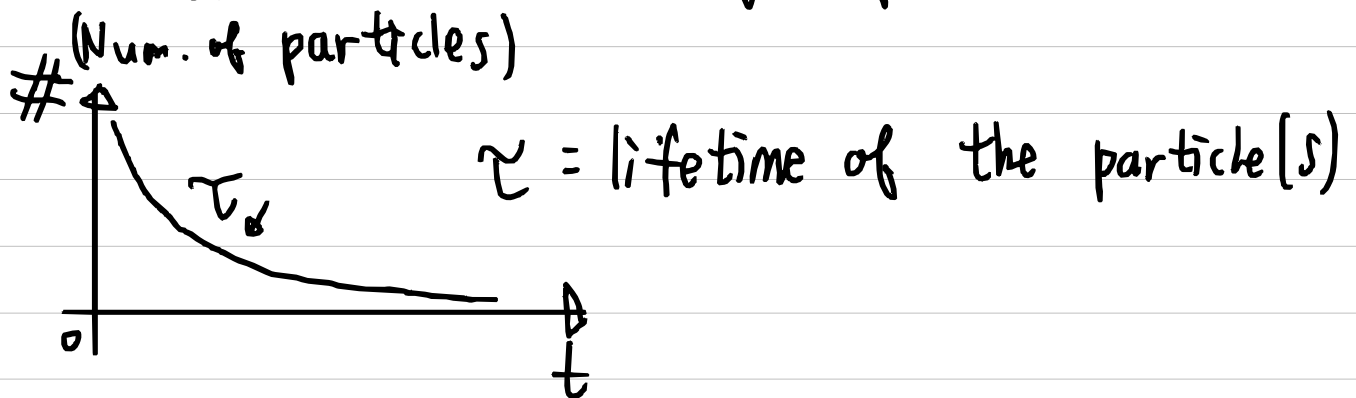
$$\frac{\partial}{\partial x^\mu} (x^\nu x_\nu) \rightarrow \text{do the deriv. w.r.t. } y.$$

$$\frac{\partial}{\partial x^2} = \frac{\partial}{\partial y} \quad \frac{\partial}{\partial y} (t^2 - x^2 - y^2 - z^2) \\ = -2y = 2x_2$$

index $y = x^2$ upstairs downstairs

$$\frac{\partial}{\partial x^\mu} = \partial_\mu \quad \& \quad \partial_\mu = \begin{pmatrix} \frac{\partial}{\partial t} \\ -\nabla \end{pmatrix} \quad \& \quad \partial^\mu = \begin{pmatrix} \frac{\partial}{\partial t} \\ \nabla \end{pmatrix} \quad (6)$$

So, diff. En. levels in diff. particles.



We can measure lifetime τ or width Γ .

Experimentally, easier to measure lifetimes or widths.

Very short-lived particles have very wide width.
(small τ) (big Γ)

One more deg. of fr. m (DOF) $\rightarrow$ particle spin

Reminder:

$$\partial_\mu = \begin{pmatrix} \frac{\partial}{\partial t} \\ \underline{\nabla} \end{pmatrix}, \quad \partial^\mu = \begin{pmatrix} \frac{\partial}{\partial t} \\ -\underline{\nabla} \end{pmatrix}$$

Schrödinger Eqn.

$$i \frac{\partial}{\partial t} \psi = -\frac{1}{2m} \underline{\nabla}^2 \psi + V \psi$$

time comes in w/ one power.

space comes in w/ a different power.

Free soln. ($V=0$): $\psi = N e^{-i(Et - \underline{p} \cdot \underline{x})}$

$$\hat{E} = i \frac{\partial}{\partial t}, \quad \hat{\underline{p}} = -i \underline{\nabla}$$

$$\hat{E} \psi = \frac{\hat{\underline{p}}^2}{2m} \psi, \quad E^2 = p^2 + m^2$$

$$-\frac{\partial^2}{\partial t^2} \psi = -\frac{\partial^2}{\partial x^2} \psi + m^2 \psi \quad (1D)$$

$$\Leftrightarrow (K - G E)$$

$$\partial^\mu \partial_\mu \psi + m^2 \psi = 0$$

Physics is descriptive, not prescriptive.

We do them bc they work, not bc of some grand design.

The Pauli Matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\sigma_i \sigma_j + \sigma_j \sigma_i = \{ \sigma_i, \sigma_j \} = 2 \delta_{ij}$$

(2)

$$E^2 = p^2 + m^2$$

$$= p_1^2 + p_2^2 + p_3^2 + m^2 \rightarrow E = \pm \sqrt{p^2 + m^2}$$

$$\hat{H}\Psi = \hat{E}\Psi$$

If it's linear in E or m as a necessary, insufficient condition, what powers does m have come in with?

Remembering E is a power of time.

$$\hat{H}\Psi = \hat{E}\Psi = \alpha_1 p_1 + \alpha_2 p_2 + \alpha_3 p_3 + \beta m$$

$$= \underline{\alpha} \cdot \underline{p} + \beta m$$

$$\underline{\alpha} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}$$

$$i \frac{\partial}{\partial t} \Psi = -i \underline{\alpha} \cdot \underline{\nabla} \Psi + \beta m \Psi$$

$$\hat{E}^2 \Psi = (\underline{\alpha} \cdot \underline{p} + \beta m)^2 \Psi$$

$$= \alpha_1^2 \hat{p}_1^2 + \alpha_2^2 \hat{p}_2^2 + \alpha_3^2 \hat{p}_3^2 + \beta^2 m^2$$

$$+ (\alpha_1 \alpha_2 + \alpha_2 \alpha_1) \hat{p}_1 \hat{p}_2 + \dots$$

$$+ (\alpha_1 \beta + \beta \alpha_1) m \hat{p}_1 + \dots$$

$$\alpha_1^2 = 1 = \alpha_2^2 = \alpha_3^2 = \beta^2$$

$$\begin{cases} \alpha_1 \alpha_2 + \alpha_2 \alpha_1 = 0 \\ \alpha_1 \beta + \beta \alpha_1 = 0 \end{cases}$$

$$\alpha_1 \alpha_2 + \alpha_2 \alpha_1 = 0 \rightarrow \alpha_1 \alpha_2 = -\alpha_2 \alpha_1$$

$$\alpha_1 \beta + \beta \alpha_1 = 0$$

$$i \beta \frac{\partial}{\partial t} = -i \beta \underline{\alpha} \cdot \underline{\nabla} + \beta^2 m$$

$$\beta^2 = 1$$

$$\gamma^\mu = \begin{pmatrix} \beta \\ \beta \underline{\alpha} \end{pmatrix}$$

$$i \gamma^\mu \partial_\mu \Psi = m \Psi$$

covariant form of Dirac Eq

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

$$i \gamma^0 \frac{\partial}{\partial t} + i \underline{\gamma} \cdot \underline{\nabla} = m \Psi$$

$$\hat{H} = i \frac{\partial}{\partial t} = -i \gamma^0 \underline{\gamma} \cdot \underline{\nabla} + m \gamma^0 \Psi$$

4D

* The Gamma Matrices in the Standard (Dirac) Representation

(3)

$$\gamma^0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\gamma^1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_1 \\ -\sigma_1 & 0 \end{pmatrix}$$

$$\gamma^2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_2 \\ -\sigma_2 & 0 \end{pmatrix}$$

$$\gamma^3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_3 \\ -\sigma_3 & 0 \end{pmatrix}$$

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

$$i[\hat{H}, \hat{O}] = \frac{\partial \hat{O}}{\partial t}$$

$$\hat{\mathcal{J}} = \hat{\mathcal{L}} + \hat{\mathcal{S}}$$

$$[\hat{H}, \hat{\mathcal{J}}] = 0 \stackrel{?}{=} [\hat{H}, \hat{\mathcal{L}}]$$

$$\hat{\mathcal{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = -i\hat{\mathbf{r}} \times \nabla$$

$$[\hat{H}, \hat{\mathcal{L}}] = -\gamma^0 \underline{\gamma} \cdot \nabla$$

$$[\hat{H}, \hat{\mathcal{J}}] = 0 = [\hat{H}, \hat{\mathcal{L}} + \hat{\mathcal{S}}] \\ = [\hat{H}, \hat{\mathcal{L}}] + [\hat{H}, \hat{\mathcal{S}}]$$

$$[\hat{H}, \hat{\mathcal{S}}] = \gamma^0 \underline{\gamma} \times \nabla$$

$$\gamma^5 = i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$(\gamma^5)^2 = 1, \quad \{\gamma^5, \gamma^\mu\} = 0 \\ \mu=0,1,2,3$$

$$i\gamma^0 \frac{\partial}{\partial t} + i\underline{\gamma} \cdot \nabla = m\psi$$

$$\hat{H} = i\frac{\partial}{\partial t} = -i\gamma^0 \underline{\gamma} \cdot \nabla + m\gamma^0 \psi$$

$$\hat{\mathcal{S}} = \frac{1}{2} \gamma^5 \gamma^0 \underline{\gamma} = \frac{1}{2} \begin{pmatrix} \underline{\sigma} & 0 \\ 0 & \underline{\sigma} \end{pmatrix}$$

$$\hat{\mathcal{S}} = \frac{\hbar}{2} \underline{\sigma} = \begin{pmatrix} \frac{1}{2} \sigma_1 \\ \frac{1}{2} \sigma_2 \\ \frac{1}{2} \sigma_3 \end{pmatrix}$$

APP L3

Dirac Eqn.

$$i \gamma^\mu \partial_\mu \psi = m \psi \quad \text{w/ } \psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$

$$\hat{H} = -i \gamma^0 \underline{\sigma} \cdot \underline{\nabla} + m \gamma^0$$

$$Spin: \quad \hat{J} = \frac{1}{2} \gamma^5 \gamma^0 \underline{\sigma}$$

$$J^0 = \bar{\psi} \gamma^0 \psi$$

$$J^\mu = \bar{\psi} \gamma^\mu \psi$$

$$N = \int \rho dV \quad \text{dV}$$

$$\rho \sim E \quad \& \quad \psi \sim \rho^{\frac{1}{2}} \sim \sqrt{E}$$

$$\psi = u e^{-i(Et - \underline{p} \cdot \underline{x})} = u e^{-p_\mu x^\mu}$$

ψ, u : spinors

$$i \gamma^\mu (-i p_\mu \psi) = m \psi$$

$$i \gamma^\mu (-i p_\mu) u = m u$$

$$\gamma^\mu p_\mu u = m u \\ = (\gamma^0 E - \underline{\sigma} \cdot \underline{p}) u$$

$$u = \sqrt{E + m} \begin{pmatrix} \phi \\ \chi \end{pmatrix}$$

$$\left[\begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} - \begin{pmatrix} 0 & \underline{\sigma} \cdot \underline{p} \\ -\underline{\sigma} \cdot \underline{p} & 0 \end{pmatrix} \right] \begin{pmatrix} \phi \\ \chi \end{pmatrix}$$

$$= \begin{pmatrix} M & 0 \\ 0 & M \end{pmatrix} \begin{pmatrix} \phi \\ \chi \end{pmatrix}$$

$$\begin{pmatrix} E - M & 0 \\ 0 & -(E + M) \end{pmatrix} \begin{pmatrix} \phi \\ \chi \end{pmatrix} = \begin{pmatrix} 0 & \underline{\sigma} \cdot \underline{p} \\ -\underline{\sigma} \cdot \underline{p} & 0 \end{pmatrix} \begin{pmatrix} \phi \\ \chi \end{pmatrix}$$

$$(E - M) \phi = \underline{\sigma} \cdot \underline{p} \chi$$

$$-(E + M) \chi = -\underline{\sigma} \cdot \underline{p} \phi$$

$$(E + M) \chi = \underline{\sigma} \cdot \underline{p} \phi$$

(2)

$$\varphi = \frac{\underline{\sigma} \cdot \underline{p}}{E-M} \frac{\underline{\sigma} \cdot \underline{p}}{E+M} \varphi = \frac{(\underline{\sigma} \cdot \underline{p})^2}{E^2 - M^2} \varphi$$

$$(\underline{\sigma} \cdot \underline{p})^2 = \begin{pmatrix} p_x & p_x + i p_y \\ p_x - i p_y & -p_z \end{pmatrix}^2$$

$$= p_x^2 + p_y^2 + p_z^2 = p^2$$

$$\frac{p^2}{E^2 - m^2} = 1 \quad \rightarrow \quad p^2 = E^2 - m^2$$

$$E^2 = p^2 + m^2$$

$$\text{then, } (p \rightarrow 0, E \rightarrow m)$$

$$\varphi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \varphi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\varphi_1^2 = \varphi_2^2 = 1$$

$$\psi = \psi^+ \psi = \sqrt{E+m} \left(\varphi \frac{\underline{\sigma}^+ \cdot \underline{p}}{E+m} \right)$$

Rewatch APP L1, L2, RF

Recap:

Dirac Eq.: $i \sigma^\mu \partial_\mu \psi = m \psi$

Soln. $\psi = u e^{-i p_\mu x^\mu}$

$$u = \sqrt{E+m} \begin{pmatrix} \phi \\ \chi \end{pmatrix}$$

$$\phi = \frac{\underline{\sigma} \cdot \underline{p}}{E-m} \chi, \quad \chi = \frac{\underline{\sigma} \cdot \underline{p}}{E+m} \phi \quad \rightarrow \quad E^2 = p^2 + m^2$$

$$\rightarrow u = \sqrt{E+m} \begin{pmatrix} \phi \\ \frac{\underline{\sigma} \cdot \underline{p}}{E+m} \phi \end{pmatrix}, \quad \phi = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\hat{S} = \frac{1}{2} \gamma^5 \gamma^0 \underline{\gamma} = \frac{1}{2} \begin{pmatrix} \underline{\sigma} & 0 \\ 0 & \underline{\sigma} \end{pmatrix}$$

$$\phi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \phi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \text{Can this be associated w/ spin up \& down particles?}$$

$$\hat{S}_2 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad \begin{matrix} p \rightarrow 0 \\ E \rightarrow m \end{matrix}$$

$$u_1 = \sqrt{E+m} \begin{pmatrix} 1 \\ 0 \\ \frac{p_z}{E+m} \\ \frac{p_x - i p_y}{E+m} \end{pmatrix}, \quad u_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\hat{S}_2 u_1 = \frac{1}{2} u_1$$

$$\hat{S}_2 u = \begin{pmatrix} 1 \\ 0 \\ \frac{p_z}{E+m} \\ -\frac{p_x + i p_y}{E+m} \end{pmatrix} \neq \frac{1}{2} u_1$$

$$p_x = p_y = 0$$

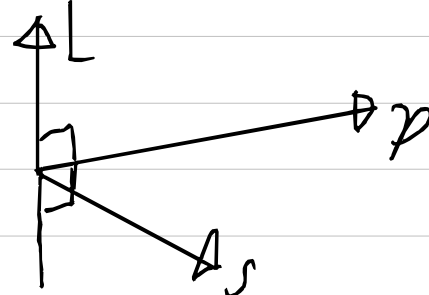
$$[\hat{H}, \hat{S}] \neq 0 = \gamma^0 \gamma^5 \underline{\gamma}$$

$$\hat{S}_2 u_1 = \frac{1}{2} u_1$$

$$\hat{J} = \hat{L} + \hat{S}$$

$$\hat{L} = \hat{r} \times \hat{p} : \text{ang. mom.}$$

$$\hat{L}_2 = i(x \partial_y - y \partial_x)$$

 $\hat{L}$ is always $\perp \hat{p}$ 

§4.2.

$$J_{||} = S_{||}$$

parallel

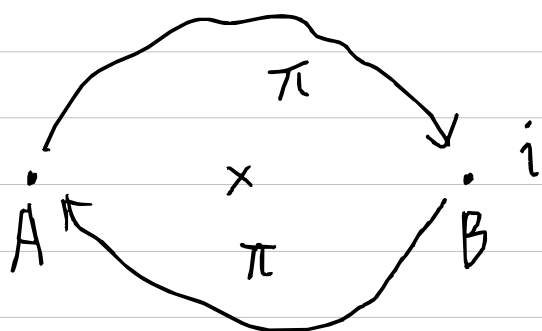
$$J_{\perp} = L + S_{\perp}$$

$$\text{Helicity} : \hat{h} = \frac{\hat{J} \cdot \underline{p}}{|\underline{p}|} \rightarrow \text{Exercise}$$

$$\phi_1 = \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{+i\varphi/2} \end{pmatrix}, \quad \phi_2 = \begin{pmatrix} -\sin \frac{\theta}{2} e^{-i\varphi/2} \\ \cos \frac{\theta}{2} e^{+i\varphi/2} \end{pmatrix}$$

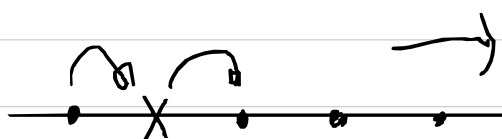
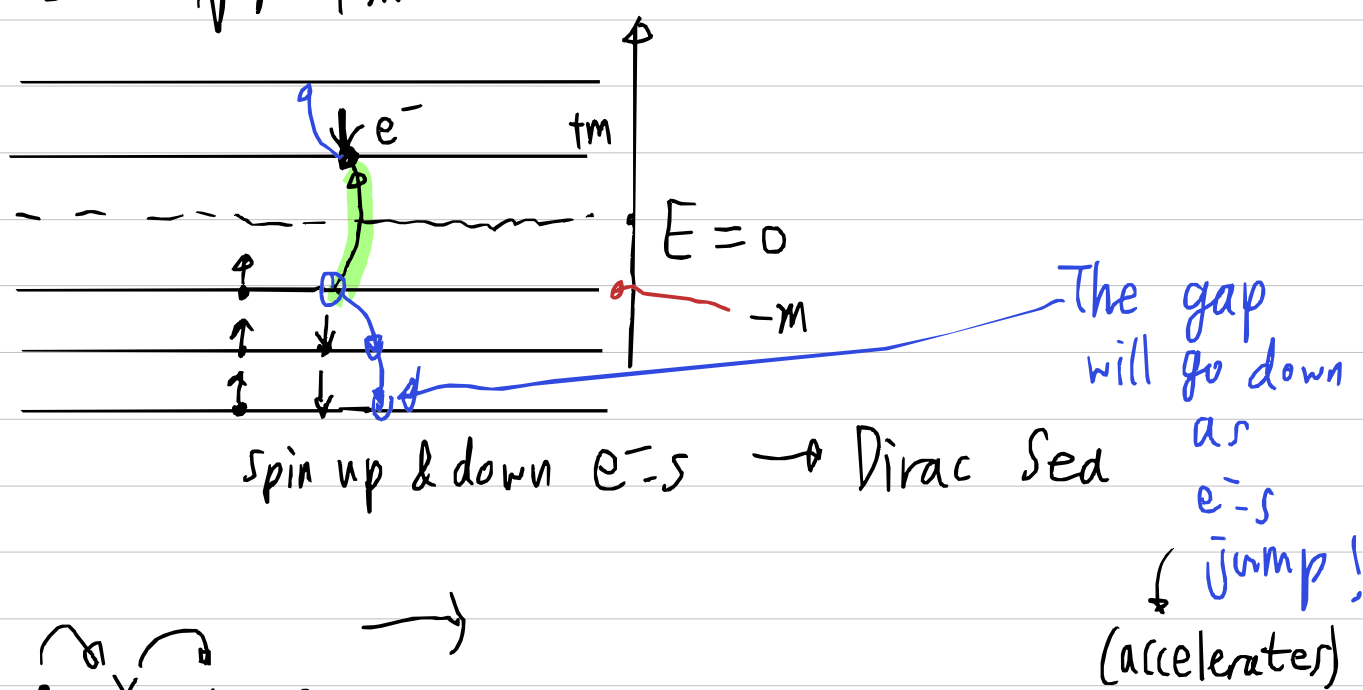
$$\theta \rightarrow \theta + 2\pi \rightarrow \cos \frac{\theta}{2} \rightarrow \cos \left(\frac{\theta}{2} + \pi \right)$$

Remember Riemann sheets
& Branch cut?



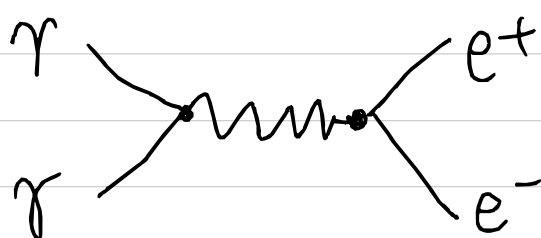
§4.4. Negative En. Solns

$$E = -\sqrt{p^2 + m^2}$$



Antimatter

$$\gamma\gamma \rightarrow e^+e^-$$



1/2 spin $\rightarrow$ Fermions $\rightarrow$ Pauli Exclusion Principle
 $e^\pm, \mu^\pm, \nu^\pm$

* Feynman's Interpretation $\rightarrow$ bosons (mesons) $\left\{ \begin{matrix} W^\pm \\ \pi^\pm, K^\pm \end{matrix} \right\}$

$$-i \boxed{E} t \rightarrow -i \boxed{-|E|} t \rightarrow -i \boxed{|E|} \boxed{-t}$$

$\uparrow$
neg. En.
(antimatter soln.)

$\rightarrow$ antimatter like e^+ 's

$\downarrow$
positive En. states travelling backwards in time mathematically!

negative En. states travelling forward in time

(3)

$$\psi = v e^{+i p_\mu x^\mu}$$

$$\left(\text{As } p \rightarrow 0, \quad E \rightarrow -m. \right)$$

$$v = \sqrt{m + |E|} \begin{pmatrix} \frac{\underline{\sigma} \cdot \underline{p}}{-|E| - m} \chi \\ \chi \end{pmatrix}$$

$$= \sqrt{m + |E|} \begin{pmatrix} \frac{\underline{\sigma} \cdot \underline{p}}{|E| + m} \chi \\ \chi \end{pmatrix}$$

Soln.

$$v_1 = \sqrt{E + m} \begin{pmatrix} \frac{\underline{\sigma} \cdot \underline{p}}{E + m} \varphi_2 \\ \varphi_2 \end{pmatrix}$$

for fermions
w/ diff. m 's

$$\text{Soln.}$$

$$v_2 = \sqrt{E + m} \begin{pmatrix} \frac{\underline{\sigma} \cdot \underline{p}}{E + m} \varphi_1 \\ \varphi_1 \end{pmatrix}$$

$e^{\pm}$'s,
 $\mu^{\pm}$'s, ...

§4.6.
★ Parity

$$\underline{r} \rightarrow -\underline{r}$$

$$\underline{p} \rightarrow -\underline{p}$$

$$\underline{\hat{L}} = \underline{\hat{r}} \times \underline{\hat{p}} \rightarrow (-\underline{\hat{r}}) \times (-\underline{\hat{p}}) = \underline{\hat{L}}$$

$\underline{r}$ & $\underline{p}$ → "polar" vectors

$\underline{L}$ → "axial" vector
↳ do not flip sign

$\underline{B}$ → "axial" vec.

$$\psi^* \hat{O} \psi \quad \bar{\psi} \hat{O} \psi \quad \underline{\gamma}^\mu, \underline{\gamma}^5 \underline{\gamma}^\mu \quad (4\text{-vec.s})$$

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^1 = \begin{pmatrix} 0 & \sigma_1 \\ -\sigma_1 & 0 \end{pmatrix}$$

$$\gamma^2 = \begin{pmatrix} 0 & \sigma_2 \\ -\sigma_2 & 0 \end{pmatrix}, \quad \gamma^3 = \begin{pmatrix} 0 & \sigma_3 \\ -\sigma_3 & 0 \end{pmatrix}$$

↑
Pauli Matrices

$$\hat{P} V = P(V^0, \underline{V}) = (V^0, -\underline{V})$$

$$\hat{P} A = \hat{P}(A^0, \underline{A}) = (-A^0, \underline{A})$$

* Maxwell's Eqns

$$\underline{\nabla} \cdot \underline{B} = 0, \quad \frac{\partial \underline{B}}{\partial t} - \underline{\nabla} \times \underline{E} = 0$$

$$\underline{\nabla} \cdot \underline{E} = \rho, \quad \frac{\partial \underline{E}}{\partial t} + \underline{\nabla} \times \underline{B} = \underline{J}$$

* Vector Calculation:

$$\left\{ \begin{array}{l} \underline{\nabla} \cdot (\underline{\nabla} \times \underline{a}) = 0 \\ \underline{\nabla} \times \underline{\nabla} b = 0 \\ \underline{\nabla} \times \underline{\nabla} \times \underline{a} = \underline{\nabla} (\underline{\nabla} \cdot \underline{a}) - \underline{\nabla}^2 \underline{a} \end{array} \right.$$

$$\underline{B} = \underline{\nabla} \times \underline{A} \quad \& \quad \frac{\partial}{\partial t} (\underline{\nabla} \times \underline{A}) - \underline{\nabla} \times \underline{E} = 0$$

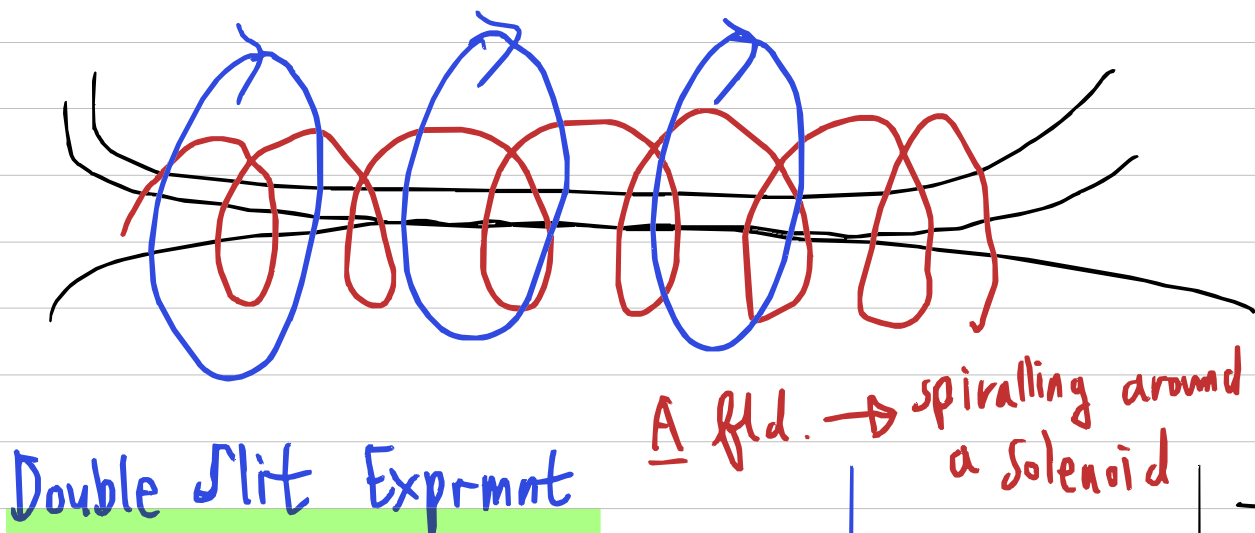
$$-\underline{\nabla} \times \left(\underline{E} + \frac{\partial \underline{A}}{\partial t} \right) = 0 \iff -\underline{\nabla} \times (-\underline{\nabla} \phi) = 0$$

$$-\underline{\nabla} \phi = \underline{E} + \frac{\partial \underline{A}}{\partial t} \iff \underline{E} = -\underline{\nabla} \phi - \frac{\partial \underline{A}}{\partial t}$$

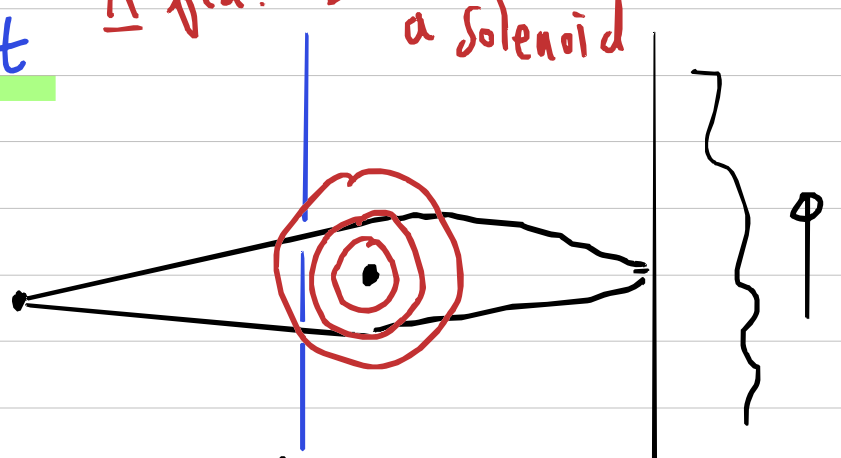
$$\frac{\partial^2 \phi}{\partial t^2} - \underline{\nabla}^2 \phi - \frac{\partial}{\partial t} \left(\frac{\partial \phi}{\partial t} + \underline{\nabla} \cdot \underline{A} \right) = \rho$$

$$\frac{\partial^2 \underline{A}}{\partial t^2} - \underline{\nabla}^2 \underline{A} + \underline{\nabla} \left(\frac{\partial \phi}{\partial t} + \underline{\nabla} \cdot \underline{A} \right) = \underline{J}$$

$$\phi \rightarrow \phi + \frac{\partial \Lambda}{\partial t} \quad \& \quad \underline{A} \rightarrow \underline{A} - \underline{\nabla} \Lambda$$



Double Slit Experiment



$$\frac{\partial \phi}{\partial t} + \underline{\nabla} \cdot \underline{A} = 0$$

$$e^{i\phi} \sim \int \underline{A} \cdot d\mathbf{l}$$

$$\frac{\partial^2 \phi}{\partial t^2} - \underline{\nabla}^2 \phi = \rho$$

$$\frac{\partial^2 \underline{A}}{\partial t^2} - \underline{\nabla}^2 \underline{A} = \underline{J}$$

$$A^\mu = (\phi, \underline{A})$$

$$J^\mu = (\rho, \underline{J})$$

$$\partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = J^\nu$$

$$= \partial_\mu F^{\mu\nu}$$

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

$$\text{Lorenz gauge} \rightarrow \partial_\mu A^\mu = 0$$

$$\text{gauge transform: } A^\mu \rightarrow A^\mu + \partial^\mu \Lambda$$

In this gauge, Maxwell's eqns will reduce to:

$$\partial_\mu \partial^\mu A^\nu = J^\nu$$

$$\partial_\mu \partial^\mu A^\nu = 0 \quad : \text{K-G E}$$

$$\partial_\mu \partial^\mu A^\nu + m^2 A^\nu = 0$$

$$\phi = \frac{q}{4\pi r} \quad \longleftrightarrow \quad -\nabla^2 \phi = 0$$

$$\nabla^2 \phi + m^2 \phi = 0 \longrightarrow \phi = \frac{q}{4\pi r} e^{-mr}$$

Yukawa potential

②

$$\partial_\nu \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = \partial_\nu J^\mu$$

$$\partial_\nu \partial_\mu \partial^\mu A^\nu - \partial_\nu \partial_\mu \partial^\nu A^\mu = 0 = \partial_\nu J^\mu$$

$$\partial_\mu (q \bar{\psi} \gamma^\mu \psi) = 0 \quad \& \quad J^\mu = q \bar{\psi} \gamma^\mu \psi$$

$$E = \frac{p^2}{2m} + V = \frac{p^2}{2m} + q \phi$$

$$= \frac{(p - qA)^2}{2m} + q \phi$$

$$E - q \phi = \frac{(p - qA)^2}{2m}$$

$$p^\mu \longrightarrow p^\mu - q A^\mu$$

$$\gamma^\mu p_\mu \psi = m \psi \longrightarrow \gamma^\mu (p_\mu - q A_\mu) \psi = m \psi$$

$$\hookrightarrow i \gamma^\mu (\partial_\mu + i q A_\mu) \psi = m \psi$$

$$D_\mu = \partial_\mu + i q A_\mu$$

* Reminder

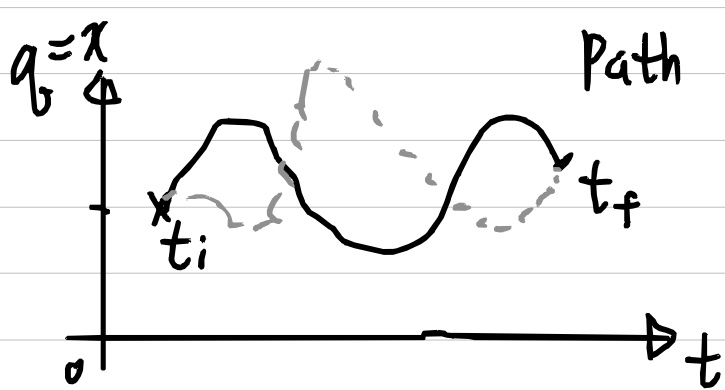
$$\text{Dirac Eqn.: } \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi - q A_\mu \bar{\psi} \gamma^\mu \psi$$

$$\text{EM Eqn.: } \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = J^\nu$$

$$\partial_\mu F^{\mu\nu} = q \bar{\psi} \gamma^\nu \psi$$

§6.1. Lagrangians

$$L(q, \dot{q}) = KE - PE$$



"princ. of least action"

$$\text{Action } A = \int_{t_i}^{t_f} L(q, \dot{q}) dt$$

$$\delta q : \delta q(t_i) = \delta q(t_f) = 0 \quad : \text{B.C.s}$$

$$\delta A = \int \delta L dt = \int_{t_i}^{t_f} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) dt$$

$\quad \quad \quad u \quad dv$

$$= \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_i}^{t_f} - \int_{t_i}^{t_f} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q dt$$

B.C.s

Then,

$$\delta A = \int_{t_i}^{t_f} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q dt$$

We need $\delta A = 0$ at the minimum.For $\forall \delta q(t)$, holds true iff

$$\frac{\partial L}{\partial q} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = E - L E,$$

which allows us to
calcul. the path
 $q(t)$ EX. When $q=x$.

$$q=x, \quad L = \frac{1}{2} m \dot{x}^2 - V(x) = T - V$$

$$\frac{\partial L}{\partial x} = -\frac{dV}{dx}, \quad \frac{\partial L}{\partial \dot{x}} = m \dot{x}$$

$$-\frac{dV}{dx} = m \ddot{x} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right)$$

 $(F = ma)$ NII Law

$$\text{For multi-dim.: } \frac{\partial L}{\partial q_i} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right)$$

$$q(t, \underline{r})$$

$$L = \int \mathcal{L} \left(q, \frac{\partial q}{\partial t}, \frac{\partial q}{\partial x}, \frac{\partial q}{\partial y}, \frac{\partial q}{\partial z} \right) d^3 r$$

↑
Integrating over space

$$\begin{aligned} A &= \int L dt = \iint \mathcal{L} d^3 r dt = \int \mathcal{L} d^4 x \\ &= \int \mathcal{L}(q, \partial_\mu q) d^4 x \end{aligned}$$

Fld. ver. of E-L E:

$$\frac{\partial L}{\partial q} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \rightarrow \frac{\partial \mathcal{L}}{\partial q} = \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu q)} \right]$$

Consider the following w/ $q = \varphi$:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi - \frac{1}{2} m^2 \varphi^2$$

$$q = \varphi : \quad \frac{\partial \mathcal{L}}{\partial \varphi} = -m^2 \varphi$$

$$\left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \right) = \partial^\mu \varphi$$

$$\downarrow$$

$$-m^2 \varphi = \partial_\mu \partial^\mu \varphi$$

$$\text{or } \partial_\mu \partial^\mu \varphi + m^2 \varphi = (\square + m^2) \varphi$$

$$= 0 : \text{K-G E}$$

§ 6.3.1. Dirac Lagrangian Density

(3)

$$\mathcal{L} = \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi$$

$$\left(\frac{\partial \mathcal{L}}{\partial q_i} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu q_i)} \right) \right)$$

Re $\psi \rightarrow \psi$ & Im $\psi \rightarrow \psi^*$
pretend pretend

$$\left(\frac{\partial \mathcal{L}}{\partial \psi} \right)_{\psi^*} \quad \& \quad (\psi, \psi^*) \quad \& \quad (\psi, \bar{\psi})$$

If $q = \bar{\psi}$: $\frac{\partial \mathcal{L}}{\partial \bar{\psi}} = i \gamma^\mu \partial_\mu \psi - m \psi$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu \bar{\psi})} = 0$$

$\therefore$ The E-L E : $i \gamma^\mu \partial_\mu \psi - m \psi = 0$
"Dirac Eq."

If we do it for $q = \psi$,
we get $i \gamma^\mu \partial_\mu \bar{\psi} - m \bar{\psi} = 0$.

§ 6.3.2. EM Lagrangian density

$$\mathcal{L} = -\frac{1}{4} (\partial^\mu A^\nu - \partial^\nu A^\mu) (\partial_\mu A_\nu - \partial_\nu A_\mu)$$

$$= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{2} (E^2 - B^2)$$

Here, $q = A_\nu$:

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = 0, \quad \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} = -(\partial^\mu A^\nu - \partial^\nu A^\mu)$$

$\downarrow$

E-L E : $-\partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = 0$

Do the P.D. (similar probs are there)

$$\partial_\mu F^{\mu\nu} = 0$$

$A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$: gauge invariant

(4)

$$\mathcal{L}_{\text{int}} = -q A_\mu \bar{\psi} \gamma^\mu \psi$$

↑

Interacting Lagr.

§ 6.4. The QED Lagr.

$$\mathcal{L} = i \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi - \frac{1}{2} F^{\mu\nu} F_{\mu\nu} - q A_\mu \bar{\psi} \gamma^\mu \psi$$

* Conserved quantities (§ 6.5. & § 6.6.)

$$\begin{aligned} \frac{dL}{d\alpha} &= \frac{\partial L}{\partial q} \frac{dq}{d\alpha} + \frac{\partial L}{\partial \dot{q}} \frac{d\dot{q}}{d\alpha} \\ &= \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \frac{dq}{d\alpha} + \frac{\partial L}{\partial \dot{q}} \frac{d}{dt} \left(\frac{dq}{d\alpha} \right) \\ &= \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \frac{dq}{d\alpha} \right) \end{aligned}$$

If L is invariant to changes in α , i.e., $\frac{dL}{d\alpha} = 0$,

$$\frac{\partial L}{\partial \dot{q}} \frac{dq}{d\alpha} = \text{const.} \rightarrow \text{conserved.}$$

§ 6.6.

$$\psi \rightarrow \psi e^{-i\alpha}, \quad \bar{\psi} \rightarrow \bar{\psi} e^{i\alpha}$$

↳ ∴ the Lagr. density is unchanged under this transformation.

$$J^\mu = \bar{\psi} \gamma^\mu \psi : \text{Noether current}$$

$$\alpha(x^\mu)$$

$$\mathcal{L}_{\text{Dir}} \rightarrow \mathcal{L} + (\partial_\mu \alpha) \bar{\psi} \gamma^\mu \psi$$

$$\text{Under } A^\mu \rightarrow A^\mu + \partial^\mu \Lambda$$

$$\mathcal{L}_{\text{EM}} \rightarrow \mathcal{L} - q (\partial_\mu \Lambda) \bar{\psi} \gamma^\mu \psi$$

gauge thy → breaks down → for very massive fld.s.

• Measurable rates & Feynman Rules

§7.1. QED Lagr.

$$\mathcal{L} = i\bar{\psi} \gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi - q A_\mu \bar{\psi} \gamma^\mu \psi - \frac{1}{4} (\partial^\mu A^\nu - \partial^\nu A^\mu) (\partial_\mu A_\nu - \partial_\nu A_\mu)$$

$$\mathcal{L} = i\bar{\psi} \gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi$$

$$\psi \rightarrow \psi e^{i\alpha(x^\mu)}, \quad \bar{\psi} \rightarrow \bar{\psi} e^{-i\alpha(x^\mu)}$$

U(1) : $\mathcal{L}' = \mathcal{L} + i\partial_\mu \alpha \bar{\psi} \gamma^\mu \psi$

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iq A_\mu$$

$$A_\mu \rightarrow A_\mu + \partial_\mu \Lambda, \quad \Lambda = \frac{i\alpha}{q}$$

§7.3. Fermi's Golden Rule

$$\text{Rate} = \int |M|^2 d\rho \prod_{\text{in}} \frac{1}{2E_{\text{in}}}$$

• M : "matrix elem." or "amplitude"
b/t an init. & final state,
& depends on the Lagr. density.

$$d(\text{Rate}) = |M|^2 d\rho \prod_i \frac{1}{2E_i}$$

$$\text{Rate} = |M|^2 \rho \prod_i \frac{1}{2E_i} \quad (\text{for 2-body})$$

§7.4. Particle Decays $X \rightarrow c + d$.

For a particle X decaying, Fermi's Golden Rule tells us

$$\text{Rate} = |M|^2 \rho \frac{1}{2E_X} \equiv \Gamma$$

↳ not Lorentz invar. bc of $\frac{1}{E_X}$ factor.

↳ High En. particles decay slower.

Then,

Decay Rate :

$$\text{Rate} = \Gamma = -\frac{1}{N} \frac{dN}{dt}$$

The # of X particles left as a fn. of time :

$$N = N_0 e^{-t/\tau}, \quad \frac{1}{\tau} = \Gamma$$

$$\Gamma = \sum_j \Gamma_j = |M_j|^2 e_j \frac{1}{2m_X}$$

(2)

$$\Gamma_i = |M_i|^2 \rho \frac{1}{2m_x}$$

§7.5. Particle Reactions



$$\text{Rate} = \text{Flux} \times (\text{cross section})$$

$$\text{Cross sec.: } \sigma [\text{m}^2] = \frac{\text{Rate} [\text{s}^{-1}]}{\text{Flux} [\text{m}^{-2} \text{s}^{-1}]}$$

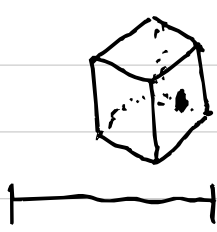
Consider the reaction of a & b , w/ 1 particle per unit vol. :

$$\sigma = \frac{1}{\text{Flux}} \left(|M|^2 \rho \frac{1}{4E_a E_b} \right)$$

The flux per particle for just 2 incoming particles
 $\hookrightarrow$ their relative vel. :

$$\text{Flux} = |\underline{\beta}_a - \underline{\beta}_b|$$

β_a & β_b : velocity 3-vec.s of the incoming particles a & b .



Flux for 1-particle = β

$$\text{Flux for 2-particles} = |\underline{\beta}_a - \underline{\beta}_b|$$

In the COM sys., $\underline{p}_b = -\underline{p}_a$

$$\beta = \frac{p}{E}$$

$$\text{Flux} = \left| \frac{\underline{p}_a}{E_a} - \frac{\underline{p}_b}{E_b} \right| = |\underline{p}_a| \left(\frac{1}{E_a} + \frac{1}{E_b} \right)$$

$$= |\underline{p}_a| \left| \frac{E_a + E_b}{E_a E_b} \right| = \frac{p_a E_{\text{com}}}{E_a E_b}$$

$$\therefore \sigma = \frac{\cancel{E_a E_b}}{p_a (E_a + E_b)} |M|^2 \rho \frac{1}{4\cancel{E_a E_b}}$$

$$E_{\text{com}} = E_a + E_b = \sqrt{s}$$

$$s = (p_a + p_b)^2$$

$$\therefore \sigma = |M|^2 \rho \frac{1}{4p_a E_{\text{com}}} = |M|^2 \rho \frac{1}{2s}$$

$$E \sim p \gg m$$

$$E_{a,b} = p_{a,b} = \frac{1}{2} \sqrt{s}$$

$$\text{Flux} = 2 \quad \& \quad \sigma = \frac{|M|^2 \rho}{2s}$$

(3)

$$dN = \frac{d^3p}{h^3} = \frac{d^3p}{(2\pi h)^3} \rightarrow \frac{d^3p}{(2\pi)^3}$$

§7.7. Lorentz Invariant Phase Space (LIPS)

$$d\rho = (2\pi)^4 \int \delta^{(4)}(\underbrace{p_{in}^\mu}_{\text{init. mtm.}} - \underbrace{p_{out}^\mu}_{\text{fin. mtm.}}) \prod_{out} \frac{d^3p_{out}}{(2\pi)^3} \frac{1}{2E_{out}}$$

2-body LIPS (c & d)

$$d\rho = \frac{(2\pi)^4}{(2\pi)^6} \int \delta(E_{in} - E_{out}) \delta^{(3)}(\underline{p}_{in} - \underline{p}_{out}) \frac{d^3p_c d^3p_d}{4E_c E_d}$$

$$= \frac{1}{4(2\pi)^2} \int \delta(E_{in} - E_{out}) \delta^{(3)}(\underline{p}_{in} - \underline{p}_{out}) \frac{d^3p_c d^3p_d}{E_c E_d}$$

$$\underline{p}_d = -\underline{p}_c \quad \& \quad E_d = E_c$$

$$\underline{p}_{in} = 0$$

means

$$\underline{p}_{out} = \underline{p}_c + \underline{p}_d = 0$$

$$= \frac{1}{4(2\pi)^2} \int \delta(E_{com} - 2E_c) \frac{d^3p_c}{E_c^2}$$

$$d^3p_c = p_c^2 dp_c d\Omega = E_c^2 dE_c d\Omega$$

$$d\Omega = d(\cos\theta) d\varphi : \text{Solid Angle}$$

$$d\rho_2 = \frac{1}{16\pi^2} \int \delta(E_{com} - 2E_c) dE_c d\Omega = \frac{1}{32\pi^2} d\Omega$$

$$\frac{d\rho_2}{d\Omega} = \frac{1}{32\pi^2}$$

2 means "2-particle"

If masses are included:

$$\frac{d\rho_2}{d\Omega} = \frac{1}{32\pi^2} \sqrt{1 - \frac{2(m_c^2 + m_d^2)}{s} + \frac{(m_c^2 - m_d^2)^2}{s^2}}$$

In the HE limit, for a 2-body decay $X \rightarrow c + d$, then

$$\frac{d\Gamma}{d\Omega} = \frac{|M|^2}{64\pi^2 m_X}$$

For a cross sec. $a+b \rightarrow c+d$, then

$$\frac{d\sigma}{d\Omega} = \frac{|M|^2}{64\pi^2 s}$$

QED Lagr.

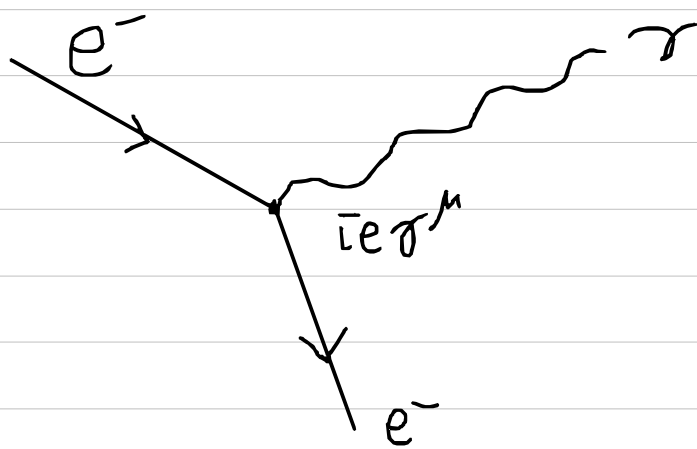
$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi - \frac{1}{4}F^{\mu\nu}F_{\mu\nu} - qA_\mu\bar{\psi}\gamma^\mu\psi$$

$$\mathcal{L} \rightarrow |M|^2 ?$$

$$\frac{d\Gamma}{d\Omega} = \frac{|M|^2}{64\pi^2 m} \quad \& \quad \frac{d\sigma}{d\Omega} = \frac{|M|^2}{64\pi^2 s}$$

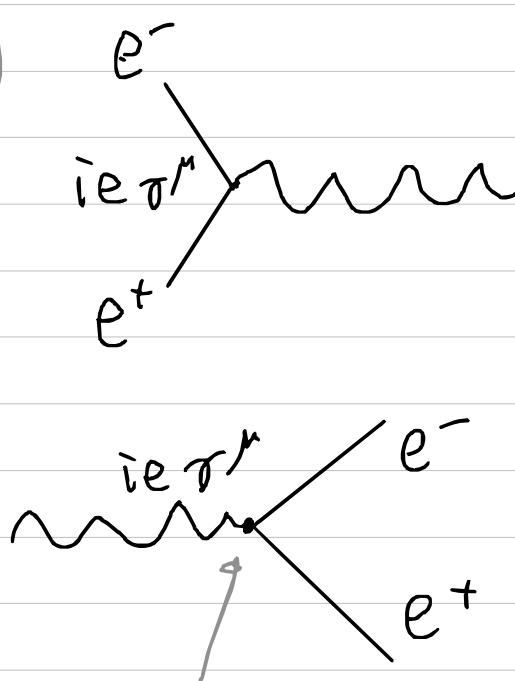
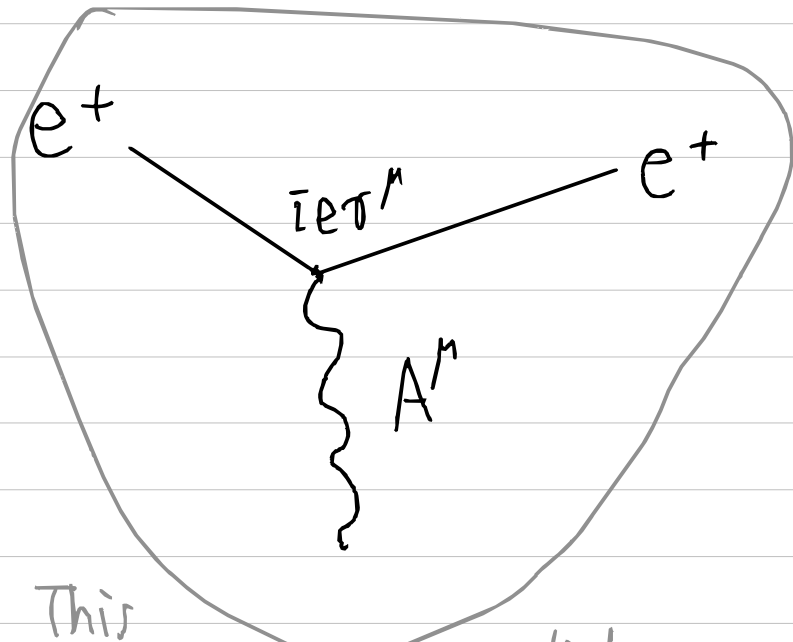
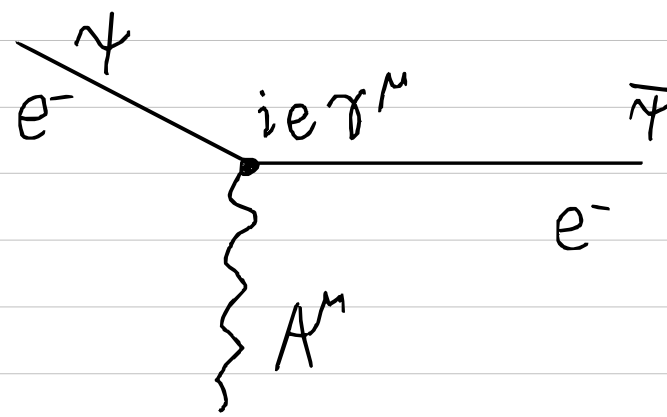
* Feynmann Diagrams

- External Lines
- Vertices
- Internal Lines



$$-eA_\mu\bar{\psi}\gamma^\mu\psi$$

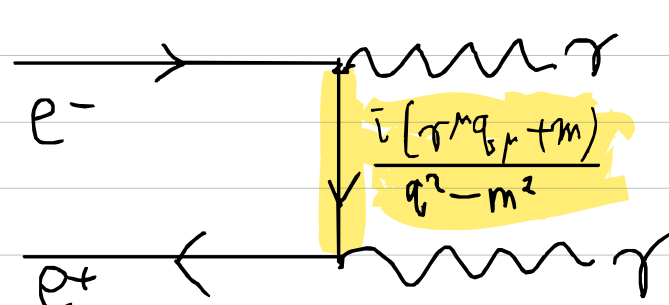
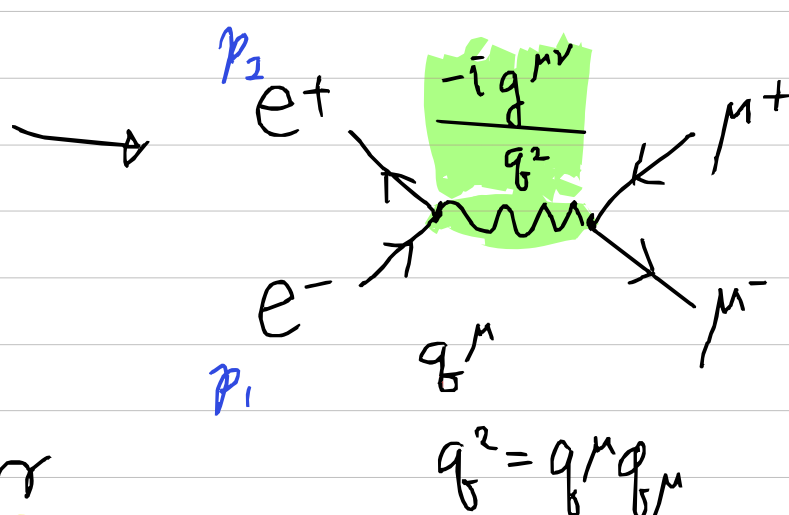
photon fld



This would not be a valid diagram on its own, bc it cannot conserve E & p.

Energy & mtn are always conserved at the vertex.

We can put in 2 or more vertices



$$\partial_\alpha \partial^\alpha x + \omega_0^2 x = F$$

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0, \quad J$$

$$F = F_0 e^{i\omega t}$$

$$x_0 = \frac{F_0}{\sqrt{(\omega^2 - \omega_0^2)^2}}$$

$$\partial_\alpha \partial^\alpha x + \Gamma \partial_\alpha x + \omega_0^2 x = F$$

$$x_0 = \frac{F_0}{\sqrt{(\omega^2 - \omega_0^2)^2 + \Gamma^2 \omega^2}}$$

$$A^\mu \sim \frac{1}{q^2} \quad \& \quad Z \sim \frac{1}{(q^2 - m^2)^2 + \Gamma^2 m^2}$$

↑ ↑

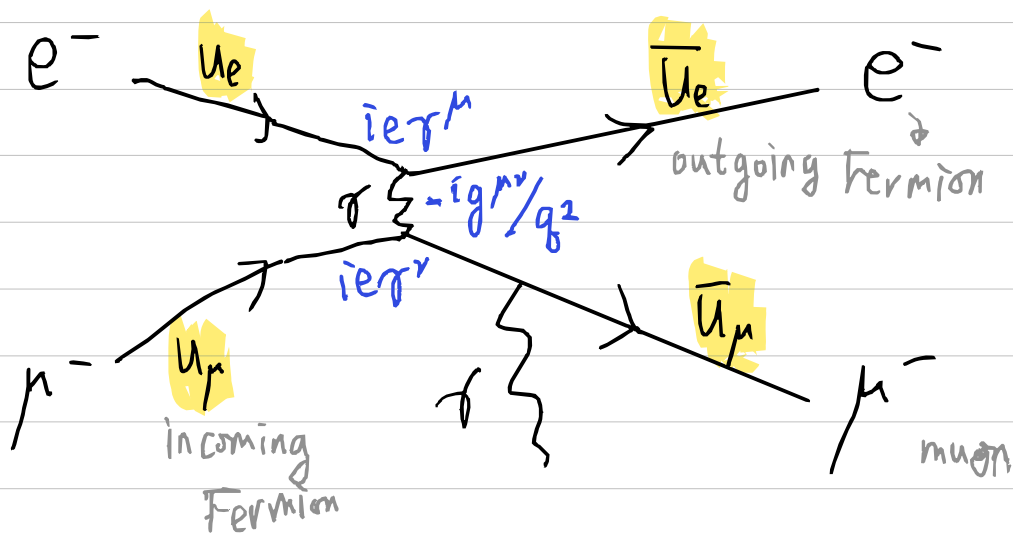
For the cases where something decays

$$Z \rightarrow \text{mass } 91 \text{ GeV}$$

We need to get Green's fns.

②

$$e^- \mu^- \rightarrow e^- \mu^-$$



Incoming

Outgoing

Fermions u

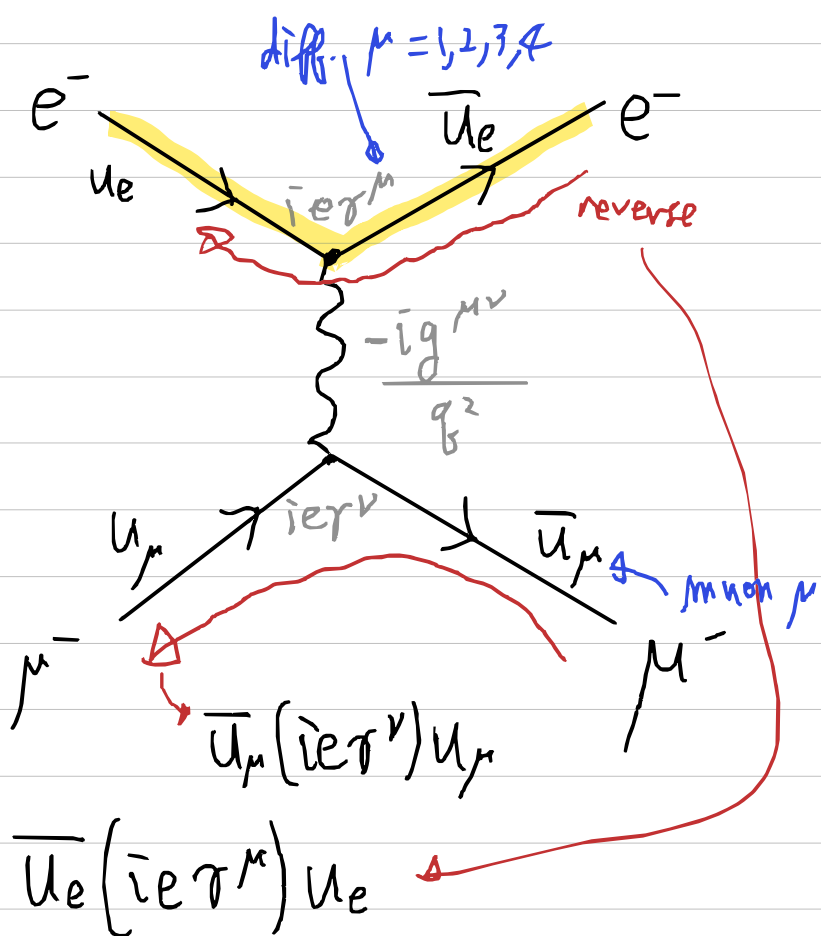
$$\bar{u} = u^\dagger \gamma^0$$

Anti-Fermions $\bar{v}$

v

Photons ϵ^μ

ϵ^μ^*

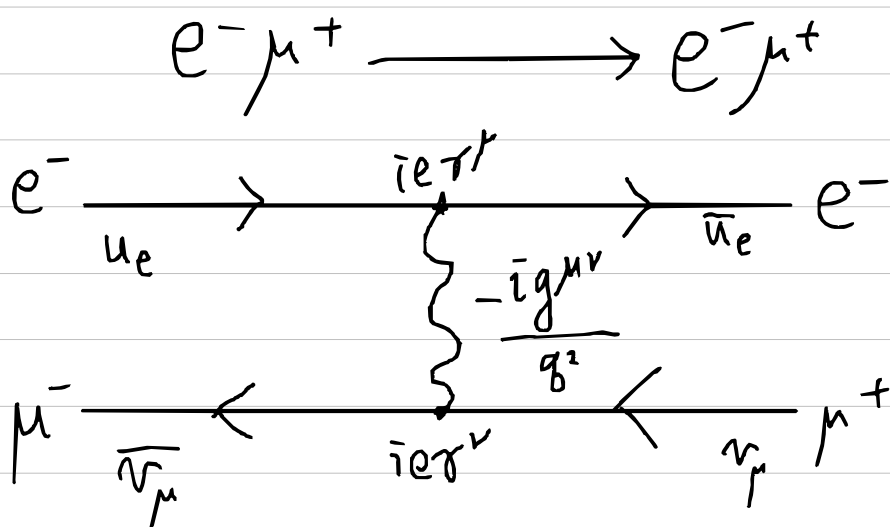


We need to sandwich γ^μ b/t vectors

& transpose vectors,

→ We need to get multiplication order right, by going the reverse dir. of the arrows.

$$\bar{u}_e (i\gamma^\mu) u_e \quad i g^{\mu\nu} \frac{1}{q^2} \quad \bar{u}_\mu (i\gamma^\nu) u_\mu$$



$$-iM = \bar{u}_e \bar{u}_e \gamma^\mu u_e \left(-\frac{i g^{\mu\nu}}{q^2} \right) \bar{v}_\mu (i\gamma^\nu) v_\mu$$

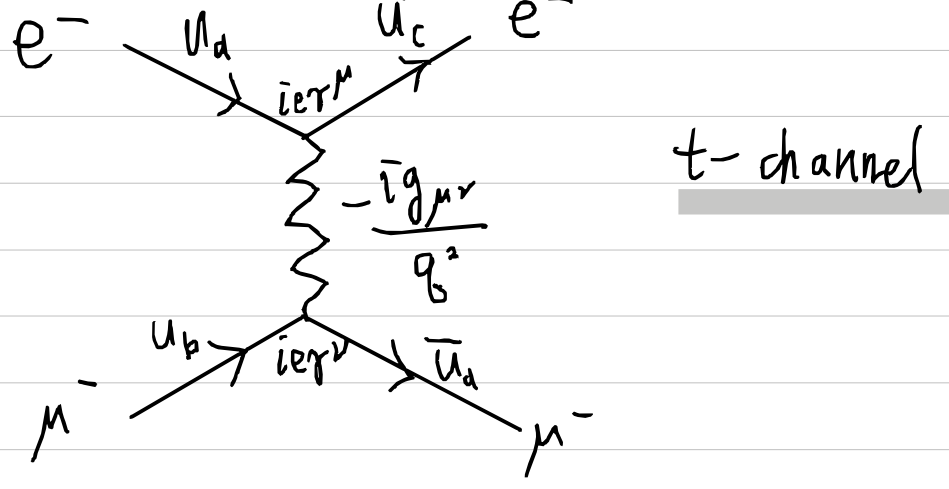
APP L9

L9: QED Cross Sections

Recap.

$$\text{Cross Section} = \frac{d\sigma}{d\Omega} = \frac{|M|^2}{64\pi^2 s}$$

$$e^- \mu^- \rightarrow e^- \mu^-$$



$$-iM = \bar{u}_c (i\epsilon\gamma^\alpha) u_a \left(-\frac{i g_{\alpha\beta}}{q^2} \right) \bar{u}_d (i\epsilon\gamma^\beta) u_b$$

current J^α_i current J^β_i

$$\bar{u} \gamma u = \begin{pmatrix} a & b & c & d \end{pmatrix} \begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix} \begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix}$$

= const.

$$e \bar{u} \gamma^\mu u = J^\mu$$

$$M = -\frac{J_e^\alpha J_{\mu\alpha}}{q^2}$$

$$M_T = \frac{1}{4} \sum_{i=1}^{16} |M_i|^2$$

$$J_{++}^\mu = J_+^\mu \quad J_{+-}^\mu = J_-^\mu = 0$$

chirality is conserved.

$$J_+^\alpha J_{+\alpha} = J_-^\alpha J_{-\alpha} = -2ie^2 s$$

$$J_+^\alpha J_-^\alpha = J_-^\alpha J_+^\alpha = -2ie^2 s \cos^2\left(\frac{\theta}{2}\right)$$

$$u_1 = \sqrt{E+m} \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) e^{-i\varphi/2} \\ \sin\left(\frac{\theta}{2}\right) e^{i\varphi/2} \\ \frac{p}{E+m} \cos\left(\frac{\theta}{2}\right) e^{-i\varphi/2} \\ \frac{p}{E+m} \sin\left(\frac{\theta}{2}\right) e^{i\varphi/2} \end{pmatrix}$$

$$u_2 = \sqrt{E+m} \begin{pmatrix} -\sin\left(\frac{\theta}{2}\right) e^{-i\varphi/2} \\ \cos\left(\frac{\theta}{2}\right) e^{i\varphi/2} \\ \frac{p}{E+m} \sin\left(\frac{\theta}{2}\right) e^{-i\varphi/2} \\ -\frac{p}{E+m} \cos\left(\frac{\theta}{2}\right) e^{i\varphi/2} \end{pmatrix}$$

$$q^\mu = p_c^\mu - p_a^\mu = \begin{pmatrix} E \\ E \sin \theta \\ 0 \\ E \cos \theta \end{pmatrix} - \begin{pmatrix} E \\ 0 \\ 0 \\ E \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ E \sin \theta \\ 0 \\ E (\cos \theta - 1) \end{pmatrix}$$

$$q^2 = -E^2 \sin^2 \theta - E^2 (\cos \theta - 1)^2$$

$$= -E^2 (\sin^2 \theta + \cos^2 \theta + 1 - 2 \cos \theta)$$

$$= -2E^2 (1 - \cos \theta)$$

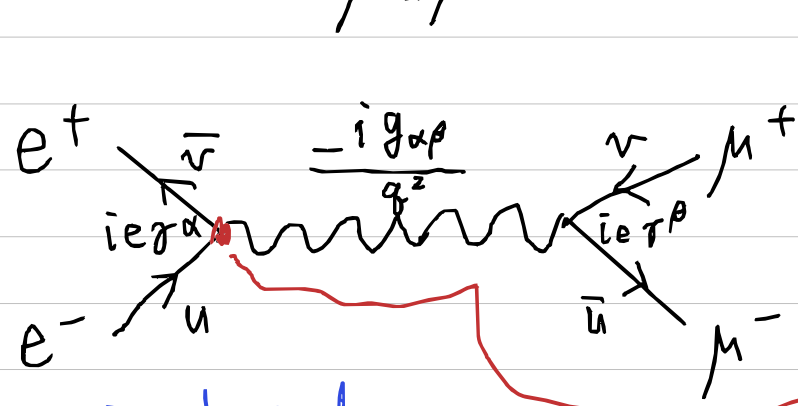
$$= -4E^2 \sin^2\left(\frac{\theta}{2}\right) = -s \sin^2\left(\frac{\theta}{2}\right)$$

$$|M_T|^2 = \frac{1}{4} \frac{8e^4 s^2 + 8e^4 s^2 \cos^4\left(\frac{\theta}{2}\right)}{s^2 \sin^4\left(\frac{\theta}{2}\right)} = 2e^4 \frac{1 + \cos^4\left(\frac{\theta}{2}\right)}{\sin^4\left(\frac{\theta}{2}\right)}$$

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{32\pi^2 s} \left(1 + \cos^4\left(\frac{\theta}{2}\right) \right) \frac{1}{\sin^4\left(\frac{\theta}{2}\right)} = \frac{\alpha^2}{2s} \frac{(1 + \cos^4\left(\frac{\theta}{2}\right))}{\sin^4\left(\frac{\theta}{2}\right)}$$

$$\alpha = \frac{e^2}{4\pi} = \frac{1}{137}$$

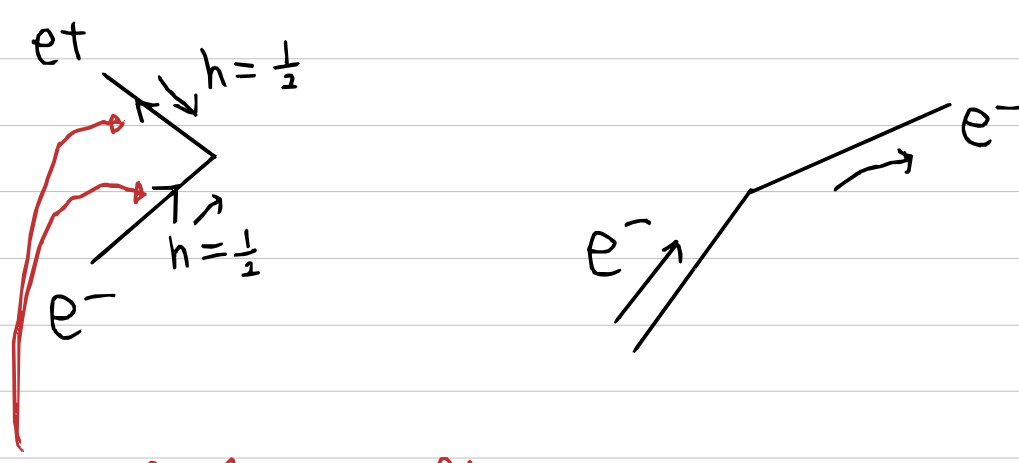
$$e_b^+ e_a^- \rightarrow \mu_d^+ \mu_c^-$$



$$q_\gamma^\mu = p_{e^+}^\mu + p_{e^-}^\mu$$

4-mtm of the photon = sum of the 4-mtm of the incoming particles.

$$-iM = \bar{u}_c i\epsilon\gamma^\beta v_d \left(-\frac{i g_{\alpha\beta}}{q^2} \right) \bar{v}_b i\epsilon\gamma^\alpha u_a$$



Dir. of fermion flow, not mtm. For antiparticles, the dir. is backwards.

$$J_{++} = J_{--} = 0$$

4 combinations of currents.

1 of which → give us

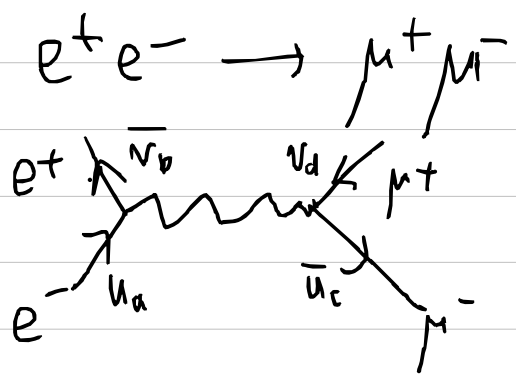
$$J_{\uparrow\downarrow}^\alpha J_{\alpha}^{\uparrow\downarrow} = ie^2 s (1 + \cos \theta)$$

$$J_{\uparrow\downarrow}^\alpha J_{\alpha}^{\downarrow\uparrow} = ie^2 s (1 - \cos \theta)$$

$$q^\mu = p_a^\mu + p_b^\mu = \begin{pmatrix} E \\ 0 \\ 0 \\ E \end{pmatrix} + \begin{pmatrix} E \\ 0 \\ 0 \\ -E \end{pmatrix} = \begin{pmatrix} 2E \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$q^2 = 4E^2 = s$$

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4s} (1 + \cos^2 \theta)$$

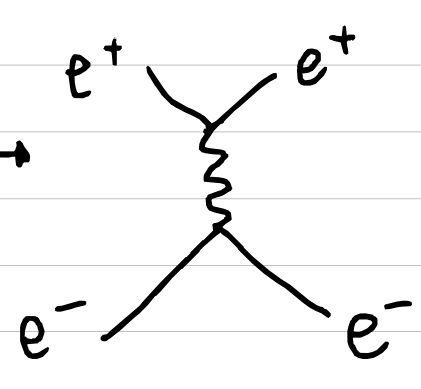


$e^+e^- \rightarrow \tau^+\tau^- \rightarrow$ similar.

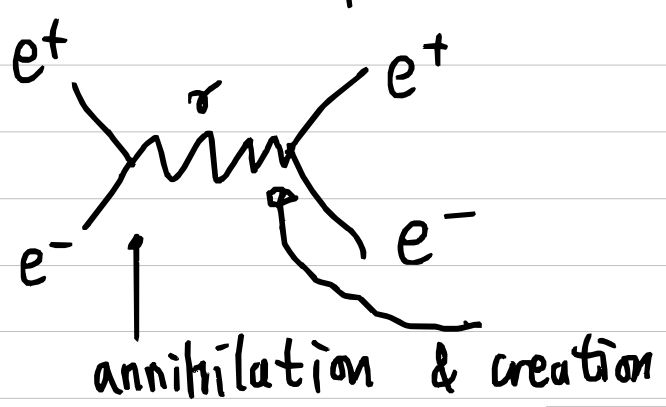
(Bhabha scattering)

$e^+e^- \rightarrow e^+e^- \rightarrow$ 2 diagrams

(1) t-channel



s-channel

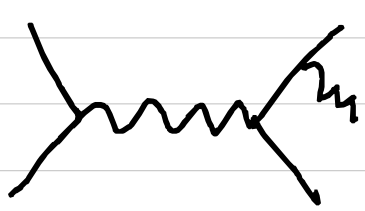


these are not distinguishable

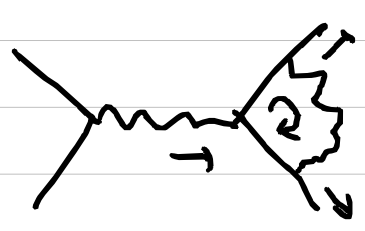
scattering via exchange of a virtual photon

§9.5. Higher-order effects

(EX)



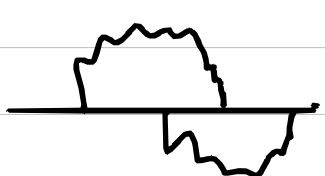
(EX)



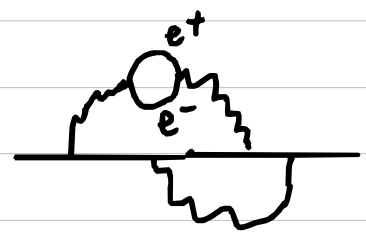
nastier

* Renormalization

to deal w/ such things.



loop correction



a particle emits a photon & reabsorbs it.

• Renormalization $\rightarrow$ subtracts off these infinities.

all of these loops end up being infinite

L10: Experiments & the Detection of Charged Particles.

§10.2.1. A source of particles

(1) Radioactivity — α particles (used in Rutherford scattering to discover the nucleus),

β particles

& γ particles

(2) Cosmic Rays

e^- 's from β^- decays

e^+ 's from β^+ decays

Cosmic Rays $\rightarrow \sim 100$ TeV

Pions $\rightarrow$ Muons

(3) Beams of protons or e^- 's (sometimes γ 's)

LN § 10.3,

§10.3.1. Interactions w/ atomic e^- s $p_{\text{DM}} \propto 1/\mu$

BLKT Lvl 5

→ Remains of
MICE

The COM En. that you get from a fixed target experim.
is much lower than the En. from a collision.

$$\sim EM$$

$$\sim 4E^2$$

Measure $\left\{ \begin{array}{l} m, m, p, \text{ En. } E, \\ \text{Spin } S \end{array} \right.$

In general, we cannot measure spin,
so, usually we measure m, m, p or En. E .

↳ We can measure m, m, p of a charged particle
very well.

↳ But, the measurement of En. E has been pretty
bad.

↳ Inaccurate.

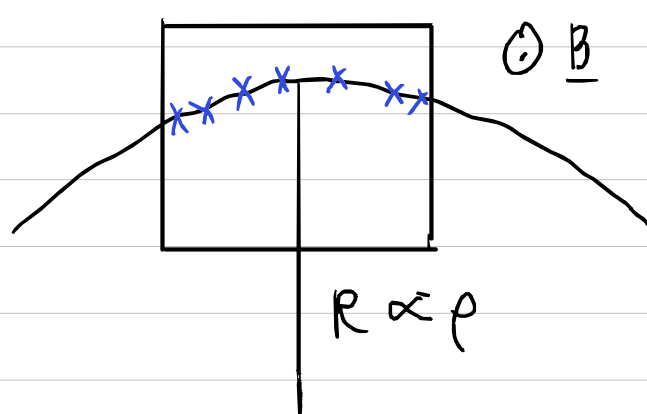
That's why

we measure

(p, m)

Then, compute E :

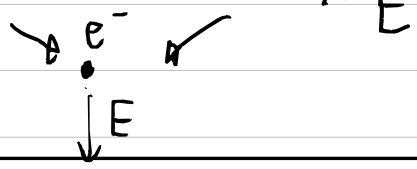
$$E^2 = p^2 + m^2$$



★ A very simple model:

The En. that this e^- picks up
a particle.

$$\sim E \sim \frac{p^2}{2m} \sim \frac{1}{\beta^2}$$



$$\frac{dp_e}{dt} = F$$

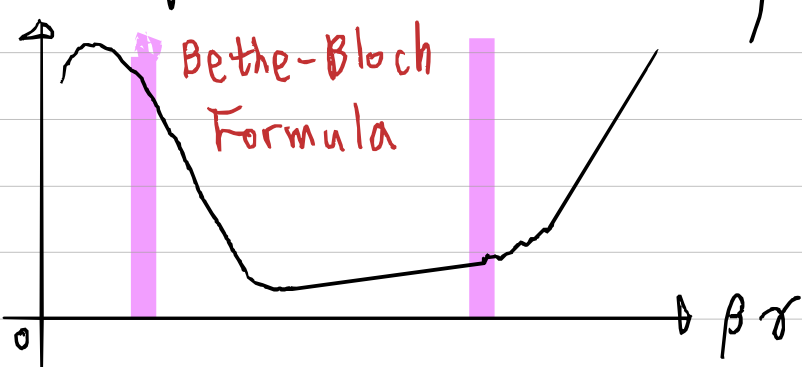
Force

$$\Delta p = \int F dt \propto \frac{1}{\beta} \quad \beta = \text{particle vel.}$$

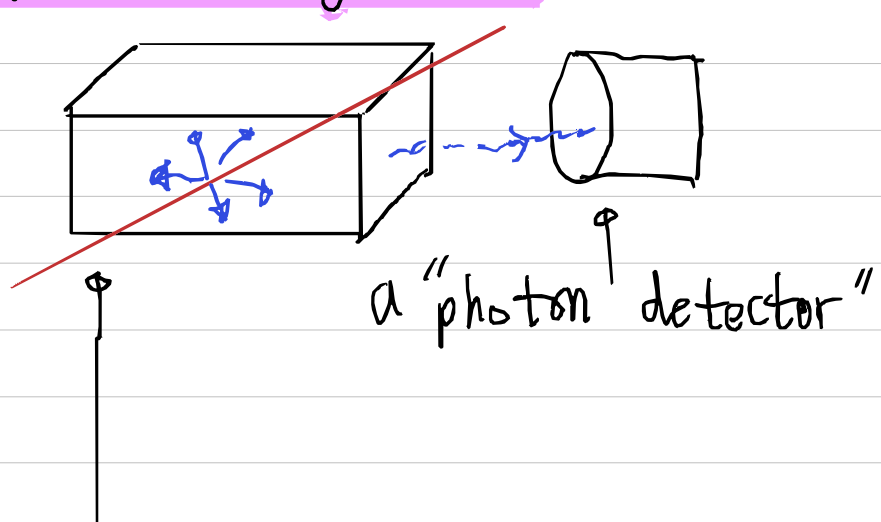
★ Bethe-Bloch Formula → Fig. 10.2.

Stopping power

μ^+ on Cu



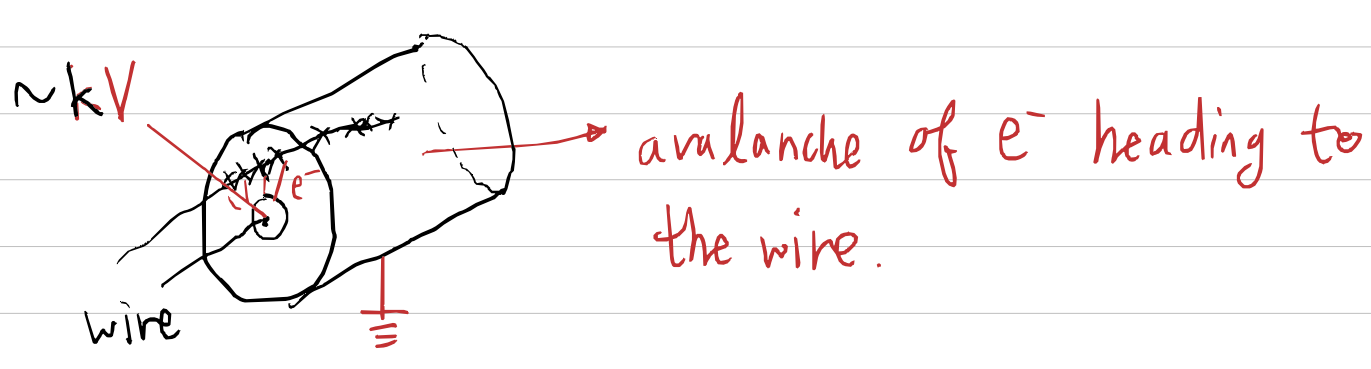
★ Scintillating Fibre



LHCb uses scintillating fibres instead of blocks.

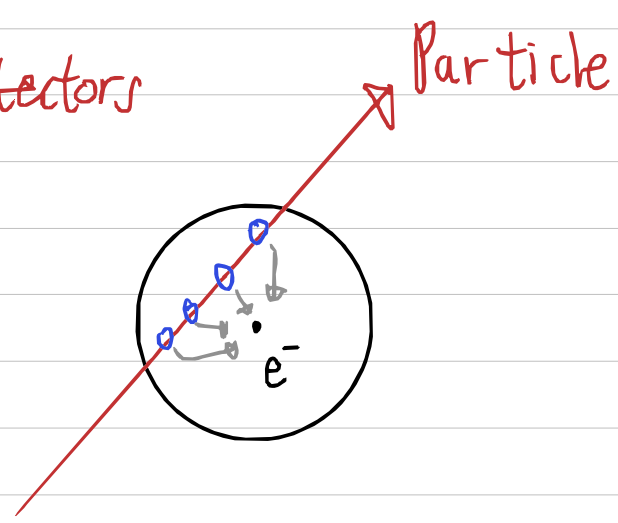
A block of material called a "scintillator",
↳ A lump of glass or plastic that contains some molecules that are particularly sensitive to being ionized; when they de-excite, they emit light.

★ Gas Detectors



ATLAS → pixel detector

Lvl 5 → Old gas detectors



① Multiwire Proportional Chamber

② Resistive Plate Chambers

③ Gas Electron Multiplier

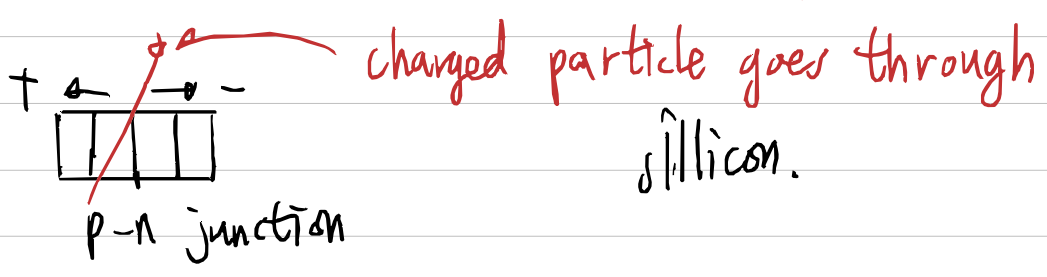
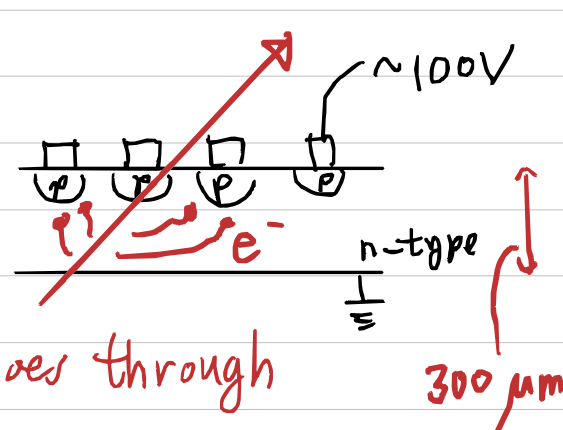
popular for experiments

↳ neutrinos or DM

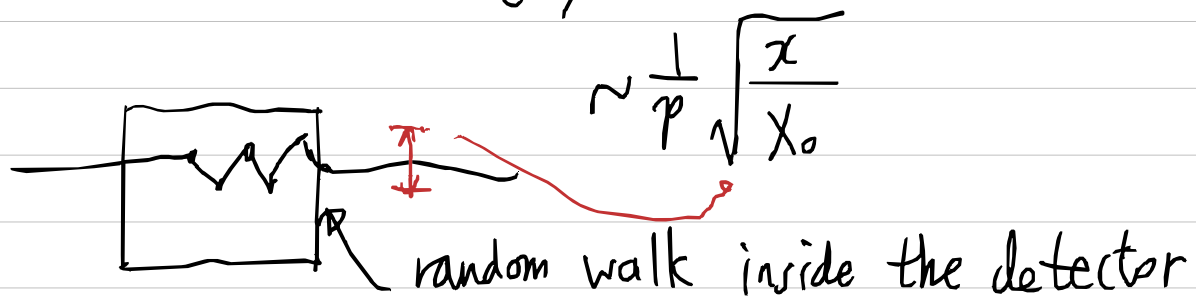
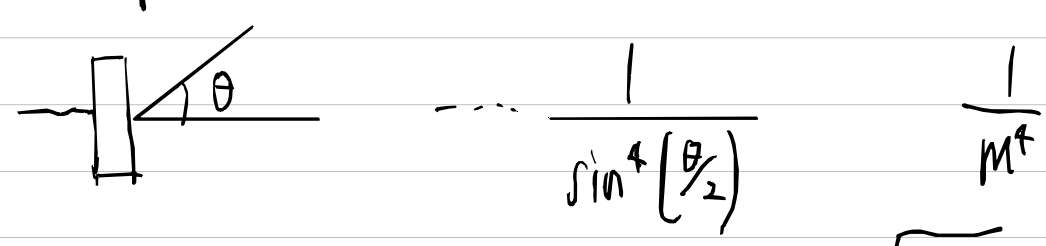
good for

④ TPC (time projection chamber) → low occupancy

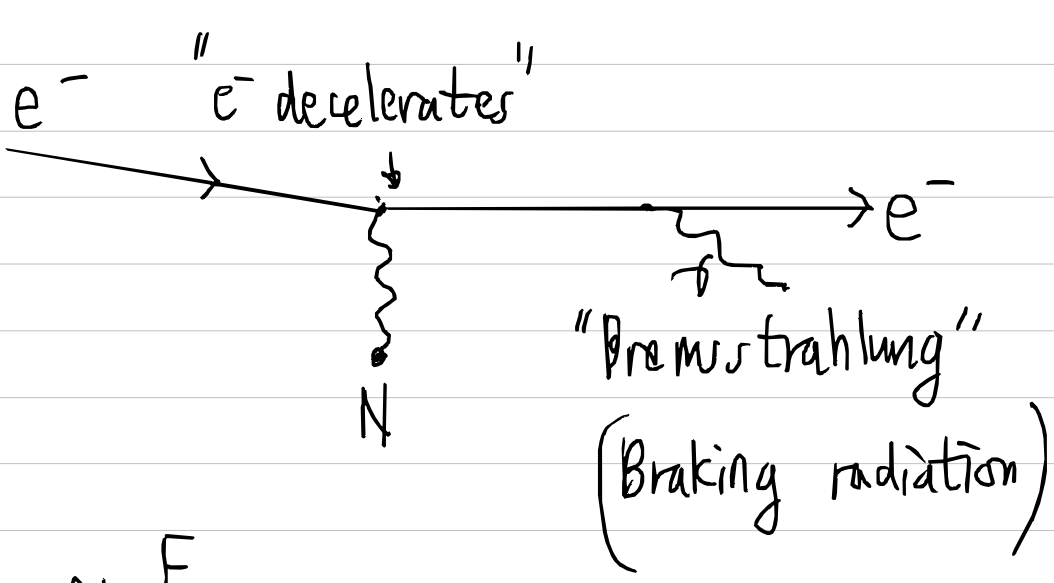
⑤ Silicon Strip Detectors



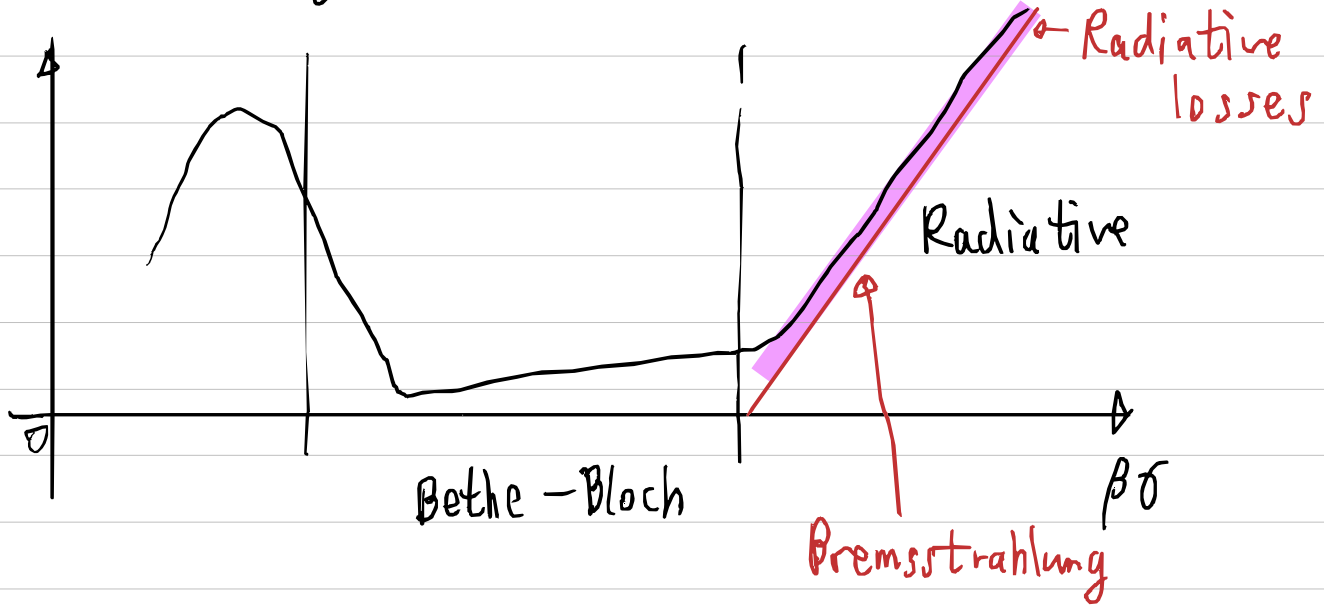
The particle can interact w/ nucleus.



X_0 : radiation leng.

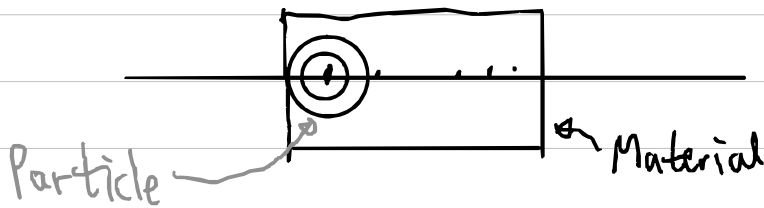


$$\frac{dE}{dx} \sim \frac{E}{X_0}$$

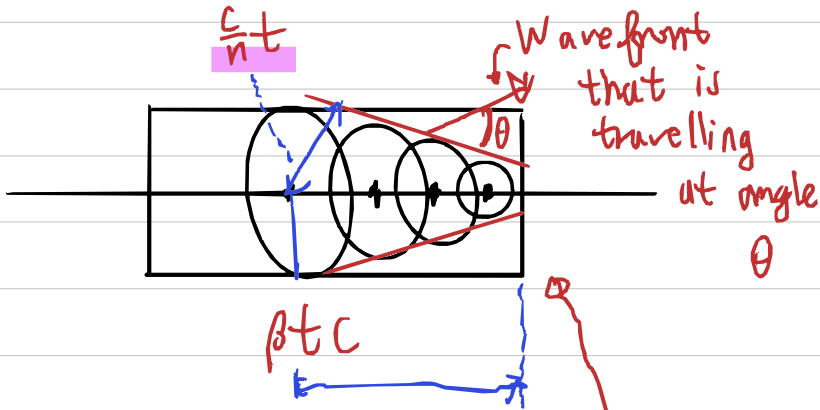


Cherenkov radiation

(4)



Its apparent color is the bluest blue.



$$\cos \theta = \frac{c}{n} \div \frac{1}{\beta ct} = \frac{1}{n\beta}$$

hypotenuse

$$\beta > \frac{1}{n}$$

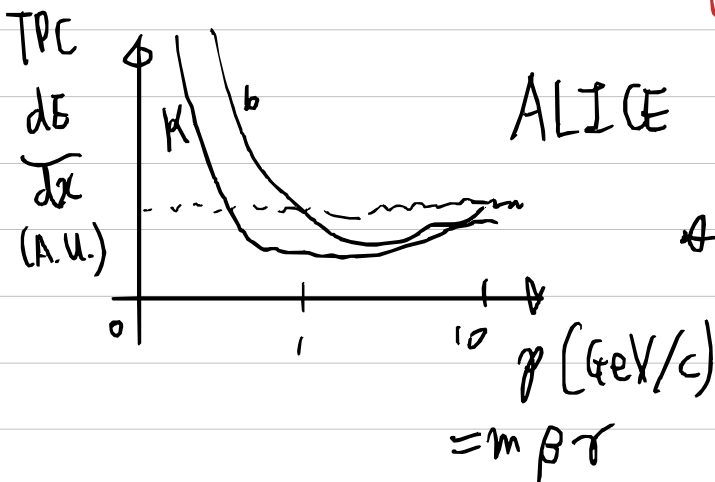
to get this emission.

Cherenkov effect.

* Particle ID

① TPC

TPC detector



As $\beta \rightarrow 1$,
(highly rel.)

$\cos \theta \rightarrow$ single number.

$$E \sim \frac{1}{\beta^2}$$

Particle ID

② Tof

Time-of-flight detectors (Tof detectors)

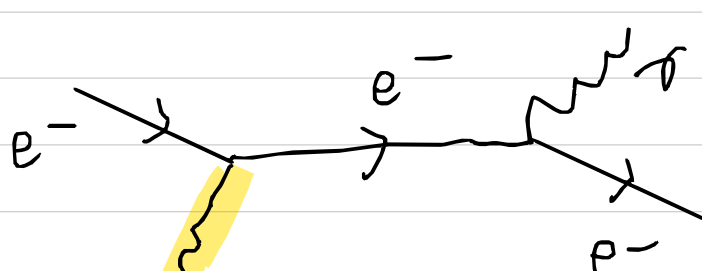
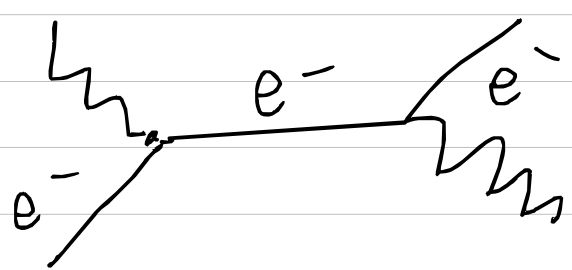
L11: Neutral Particles & Detector Systems.

We discuss photons & hadrons. (We don't discuss neutrinos here.)
In this lect..

Neutrinos experience only weak force.

→ special detectors for neutrinos.

→ Neutrinos have very small cross sections.



$$\frac{dE}{dx} = -\frac{E}{X_0}$$

"Bremsstrahlung"

Now, we draw this. (very similar)

$$\frac{dI}{dx} = -\frac{7}{9} \frac{I}{X_0}$$

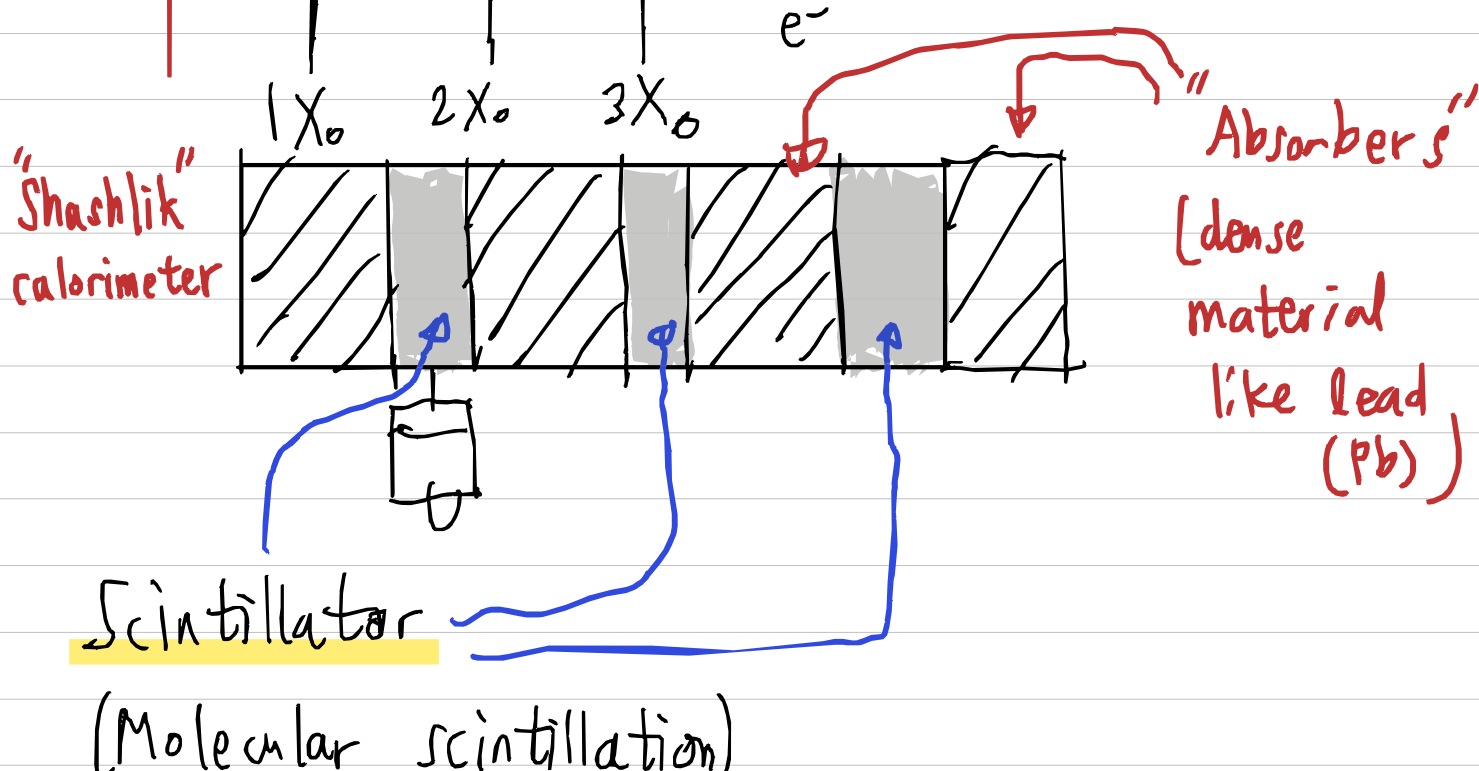
Pair-production

Bremsstrahlung

"EM Calorimeter"

"Electromagnetic Shower" effect.
(or cascade)

→ do the PS.



★ Another type of calorimeter — "Spaghetti" calorimeter.

★ (Ex)

CMS Calorimeter → both dense & scintillated
CMS → (lead tungstate)
PbWO₄

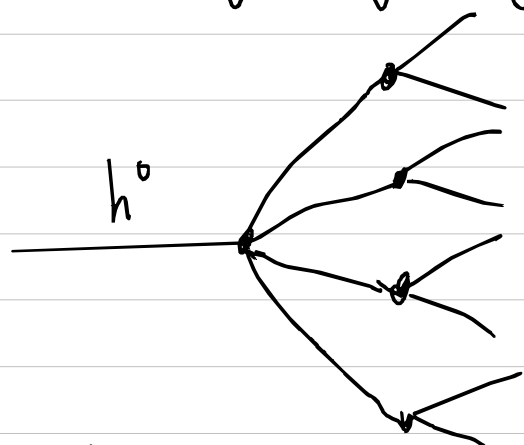
$M^4 = (\text{mass})^4$ → Bremsstrahlung
photon shower

§11.5. Hadronic Showers

★ Hadrons

Neutron (hadron)

↳ The only thing they interact → strong force



"Hadronic showers"

Hadronic calorimeter

B meson

$B^+ \rightarrow K^0 \pi^+$

★ Range of particles we're interested in:

Charged: $e^\pm, \mu^\pm, p, \bar{p}, \pi^\pm, K^\pm, (\tau^\pm)$

Some of 'em can be Vaporized

anti-proton

Kaons have strange quarks.

Neutral: $(\pi^0), \tau, K^0, n$

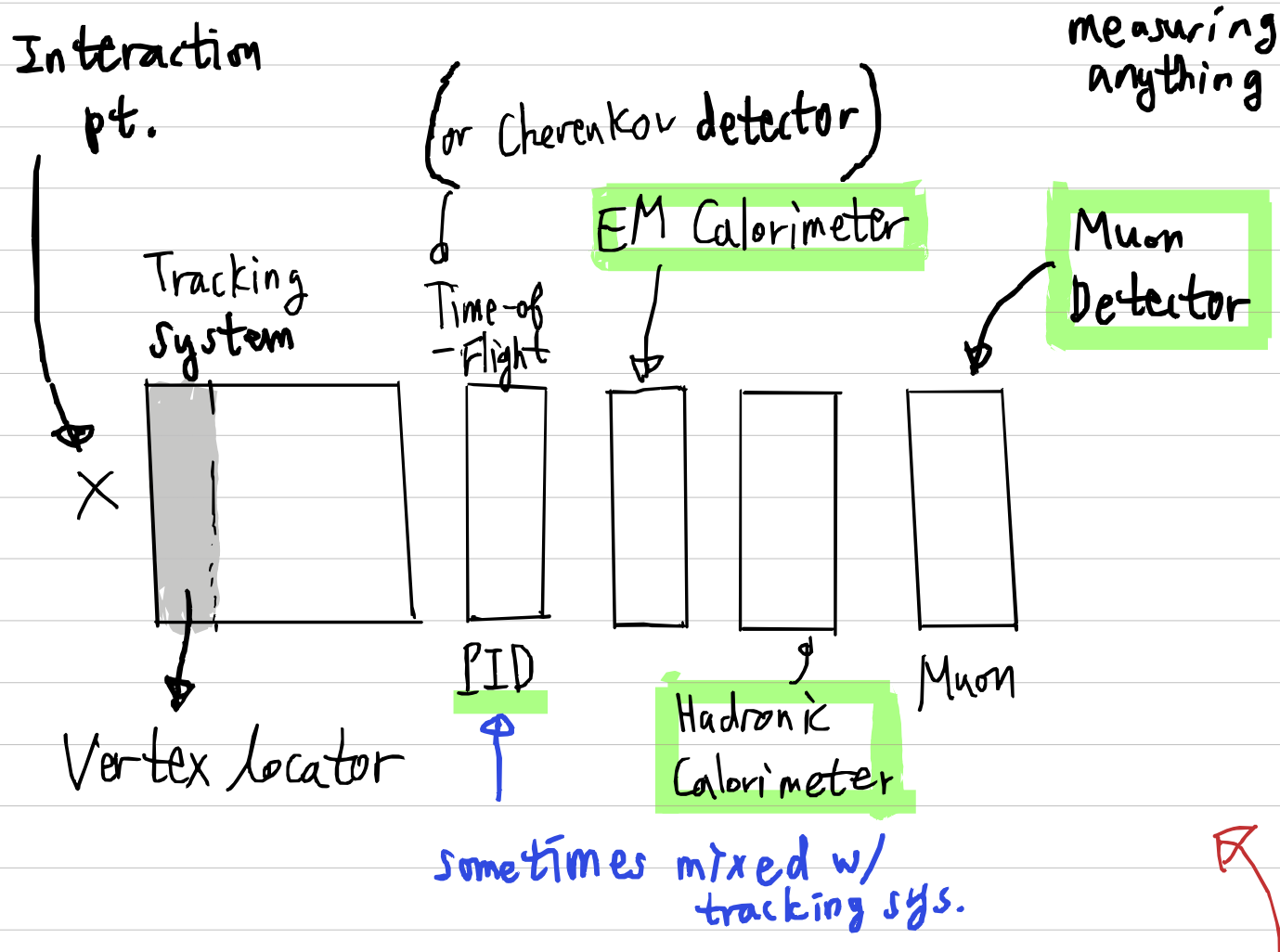
π^0 almost instantly decays into 2 photons.

$\tau^\pm$ → need very specialist detectors, but they're very short-lived.

§11.6. General Purpose Detectors "General" (e.g., CMS)

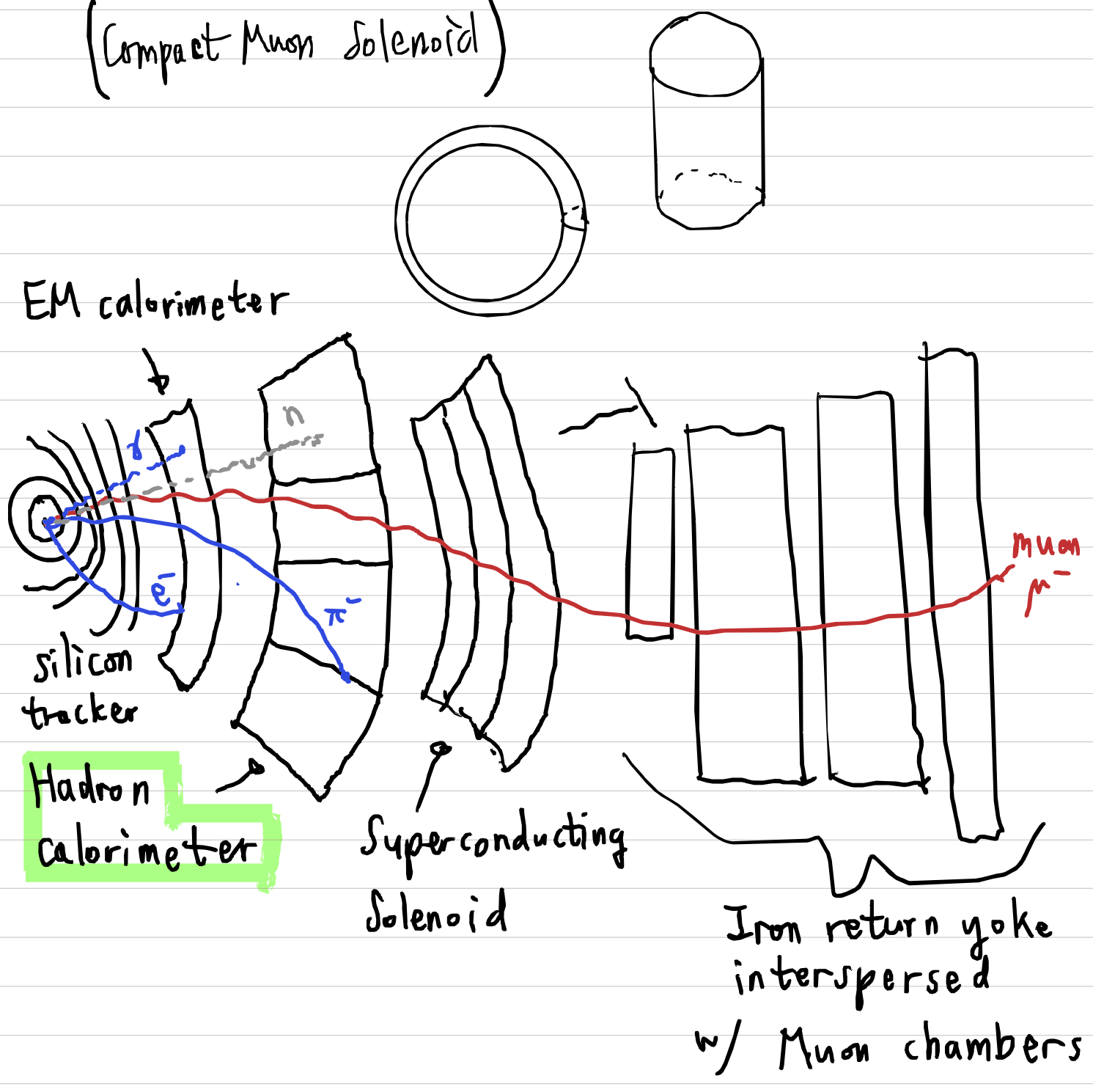
not exceptionally good at measuring anything.

EX.: CMS detector at the LHC.



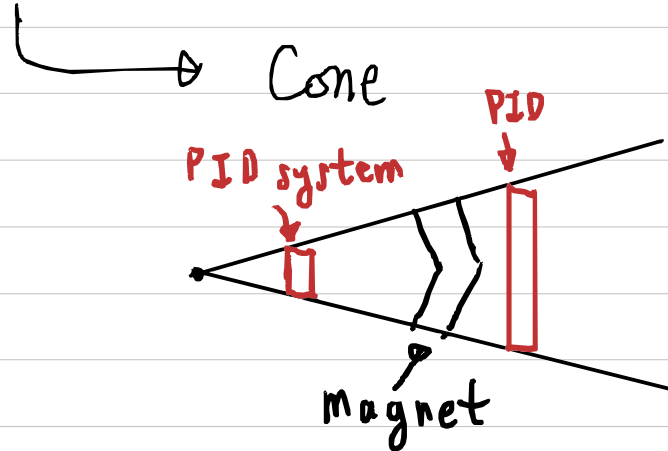
Almost all particle phys. experiments look like this.

(EX) CMS → The best detector (Compact Muon Solenoid)



3

Detections for
(EX) LHCb $\rightarrow$ B mesons



	ATLAS	CMS	LHCb
EM Calorimeter	Pb + liquid Ar	PbWO ₄ crystal	Pb + plastic scintillator
Hadron Calorimeter	Steel + scintillator	Brass + plastic scintillator	Fe + plastic scintillator

Economic choices

We'll explore in depth: Strong Force & Weak Force

Recap: QED

$$\mathcal{L} = i \bar{\psi} \partial_\mu \psi - m \bar{\psi} \psi$$

$$\psi \rightarrow \psi' = e^{i\alpha} \psi$$

$$\text{Noether: } \partial_\mu (\bar{\psi} \gamma^\mu \psi) = 0$$

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + i q A \quad (\text{local } \alpha(x^\mu))$$

$$A \rightarrow A + \partial \Lambda, \quad \Lambda = e\alpha$$

→ QED

In EM, the phase doesn't matter.

→ Need some new symmetry involving multiple fermions.

$$\mathcal{L} = i \bar{\psi}_r \partial_\mu \psi_r + i \bar{\psi}_b \partial_\mu \psi_b + i \bar{\psi}_g \partial_\mu \psi_g$$

$$- m [\bar{\psi}_r \psi_r + \bar{\psi}_b \psi_b + \bar{\psi}_g \psi_g]$$

looks like a length of a vec.

$$\Psi = \begin{pmatrix} \psi_r \\ \psi_b \\ \psi_g \end{pmatrix}$$

$$\hookrightarrow \mathcal{L} = i \bar{\Psi} \partial_\mu \Psi + m \bar{\Psi} \Psi$$

$$\Psi \rightarrow R \Psi$$

$$\Psi' = R \Psi$$

$$\text{Rotation matrix: } R^T = R^{-1}$$

$$\text{Unitary matrix: } U^{T*} = U^\dagger = U^{-1}$$

$$U = e^{-iH} \quad : \text{ power expansion}$$

$$e^M = 1 + M + \frac{M^2}{2!} + \frac{M^3}{3!} + \dots$$

$$U = e^{-i\alpha_i \lambda_i}$$

$\alpha_i = \text{rotation angle}$

λ_i : Gell-Mann matrices

→ generators of a group in Group theory.

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

G-M matrices → extension of Pauli matrices

$$U(3), \quad SU(3)$$

8 generators

(ie, 8 G-M matrices)

$$\rightarrow i=1, \dots, 8$$

$$U = e^{-i\alpha_i \lambda_i}$$

(2)

$$\Psi' = e^{-i\alpha_i \lambda_i} \Psi$$

$$J_i^\mu = \bar{\Psi} \gamma^\mu \lambda_i \Psi$$

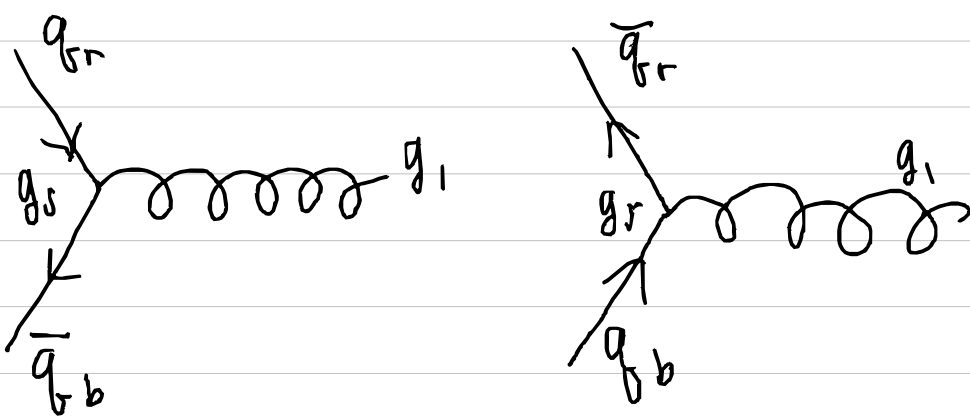
$$J_1^\mu = \bar{\Psi}_r \gamma^\mu \psi_b + \bar{\Psi}_b \gamma^\mu \psi_r$$

↑ blue-type qrk ↑ red-type qrk

$$\partial_\mu \Psi' = \partial_\mu (U \Psi) = U (\partial_\mu \Psi) + (\partial_\mu U) \Psi$$

$$\mathcal{L} \rightarrow \mathcal{L} + i \bar{\Psi} \gamma^\mu (\partial_\mu U) \Psi$$

$$-e A_\mu \bar{\Psi} \gamma^\mu \Psi \rightarrow -e A_\mu \bar{\Psi} \gamma^\mu \Psi - e (\partial_\mu \Lambda) \bar{\Psi} \gamma^\mu \Psi$$



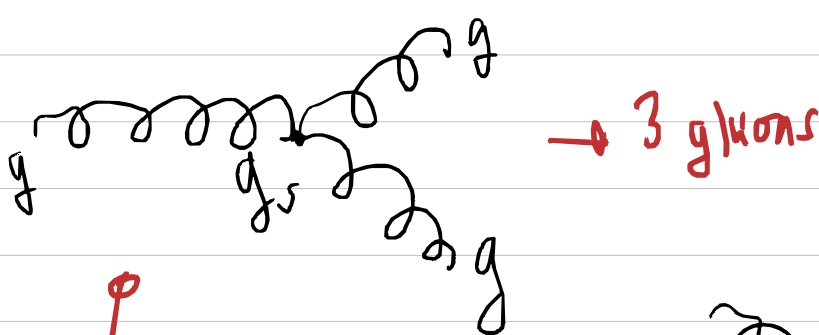
$$\mathcal{L}_{\text{QED}} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \quad \& \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} F_i^{\mu\nu} F_{\mu\nu}^i$$

$$F_i^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu - 2 g_s f_{ijk} A_j^\mu A_k^\nu + g_s^2 A^4$$

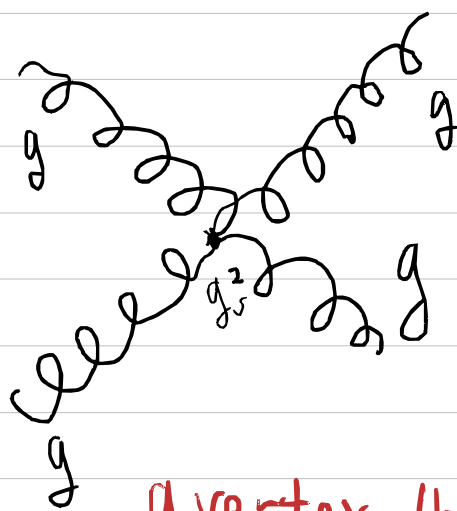
treat them as 2 separate terms
 kinetic part

$$\mathcal{L} \sim (\partial A)^2, \quad g_s (\partial A) A^2, \quad g_s^2 A^4$$



A gluon that couples to other 2 gluons.

A vertex that only couples "gluons".



A vertex that only couples "gluons".

In QED,

Photons do NOT interact w/ themselves.

To make photons interact w/ e.o., in QED, we need to put in some fermion loops.

Conversely, in QCD,

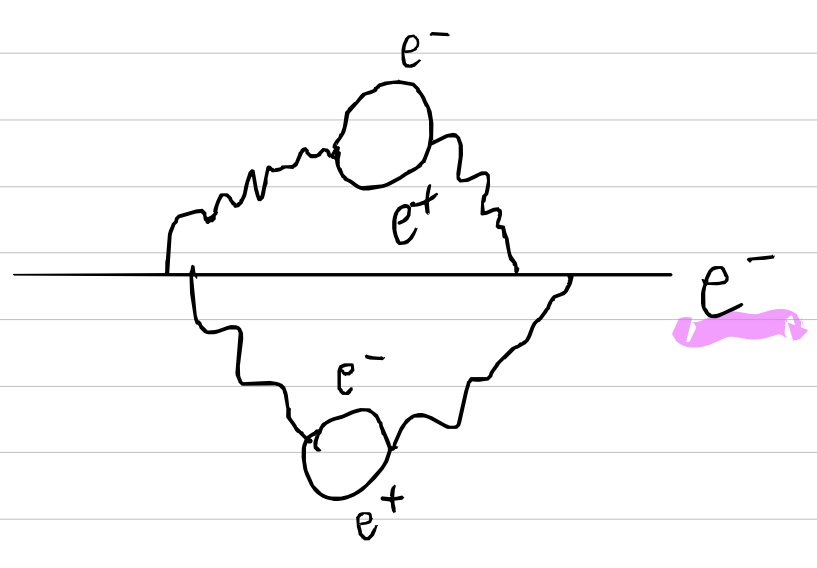
Gluons do interact w/ themselves.

These gluons have color charge. $\rightarrow$

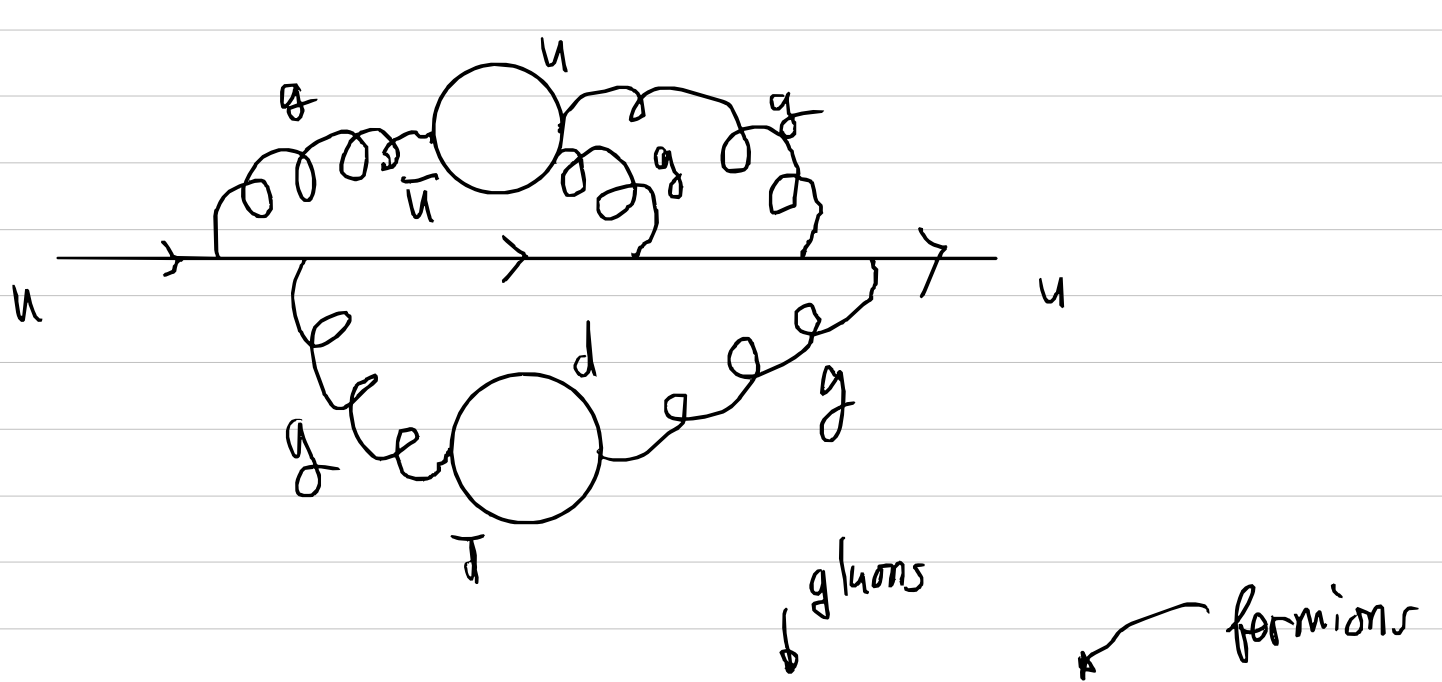
Use Gell-Mann matrices.

§ 12.5. The running coupling const.

In EM, for an e^- :



In QCD, for a quark, like an e^- , will be surrounded by QCD fld.s:



The critical number: $11n_c - 2n_f$
 $\uparrow$
 color

So, the effect of gluons wins out the effect of fermions.

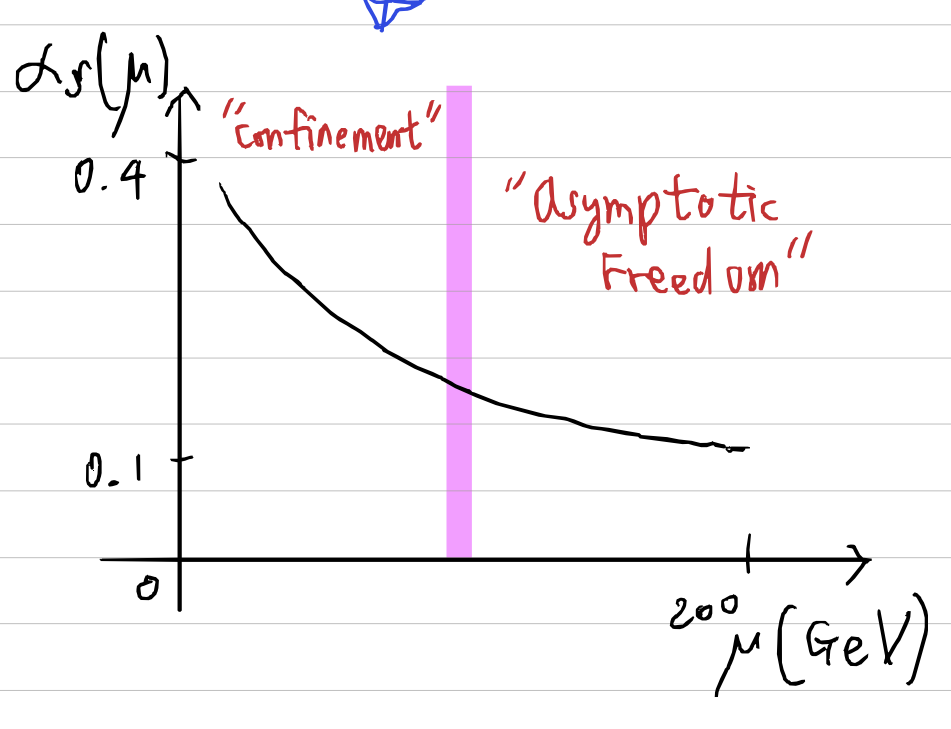
In EM, $\alpha = \frac{e^2}{4\pi}$ & $\alpha_s = \frac{g_s^2}{4\pi}$.

2 regions

① "Confinement"

At small En.s, α_s is very strong.

② "Asymptotic Freedom"



"glueballs" $\rightarrow$ observed? 50/50 chance.
 $\downarrow$
 not exactly certain.

APP L13:

QED Lagr.

$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - ie\bar{\psi}\gamma^\mu\psi A_\mu - \frac{1}{4}F^{\mu\nu}F_{\mu\nu}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

QCD Lagr.

$$\mathcal{L} = i\bar{\Psi}\gamma^\mu\partial_\mu\Psi - ig_s\bar{\Psi}\gamma^\mu\left(\frac{\lambda^a}{2}\right)\Psi A_\mu^a - \frac{1}{4}F_a^{\mu\nu}F_{\mu\nu}^a$$

$$\Psi = \begin{pmatrix} \Psi_u \\ \Psi_d \\ \Psi_s \end{pmatrix}$$

$$F_a^{\mu\nu} = \partial^\mu A_\nu^a - \partial^\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c$$



SU(2) equivalent

$$W^\pm, Z^\pm$$

W.F.

↑ ↓ spin up, down

4 ways 2 combine spins

$$\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$$

* S=1 :

$$\uparrow\uparrow, \frac{1}{\sqrt{2}}(\downarrow\uparrow + \uparrow\downarrow), \downarrow\downarrow \quad \text{: QM}$$

* S=0 :

$$\frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$$

* Colorless St.s C=0

$$\frac{1}{\sqrt{3}}(r\bar{r} + g\bar{g} + b\bar{b})$$

red green blue

state that Mesons exist in

$$\frac{1}{\sqrt{6}}(rbg - rgb + gbr - gbr + bgr - brg)$$

state that Baryons exist in

§13.3. Mesons

Restrict

$$(u, d, s)$$

$$q \text{ comb's of } q\bar{q} \begin{cases} u\bar{u}, u\bar{d}, u\bar{s}, d\bar{u}, d\bar{d}, d\bar{s}, s\bar{u}, s\bar{d}, s\bar{s} \end{cases}$$

$u\bar{u}, d\bar{d}, s\bar{s}$ decay in S.F. & EM.

$\text{e.g. } \pi^+ = u\bar{d} \rightarrow \text{P \& C conv.}$

$u\bar{d}, u\bar{s}, d\bar{u}, d\bar{s}, s\bar{u}, s\bar{d}$: stable to EM & strong decays

* Parity

$$P_L = (-1)^L$$

spher. harmonics AM

L: orbital ang. mtn.

$$\text{Tot. parity : } P_{f\bar{f}} = P_f P_{\bar{f}} P_L$$

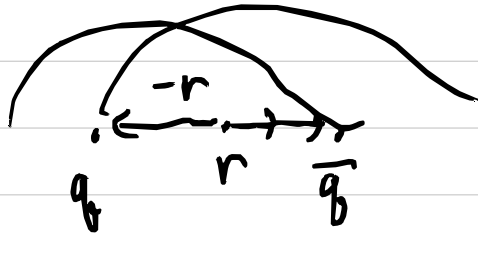
$$= P_f P_{\bar{f}} (-1)^L$$

f: fermion

$\bar{f}$: anti-fermion

$$= (+1)(-1)(-1)^L$$

$$= (-1)^{L+1}$$



$$\text{Way 1: } \leftarrow \frac{-r}{\text{com}} \frac{r}{\text{quark}} \rightarrow$$

r = dist. to quark

or $\frac{-r}{\bar{q}} \frac{r}{q}$ (swap)

Way 2: swap

$$C = (-1)^{L+1} (-1)^{S+1} = (-1)^{L+S}$$

$$= C_{f\bar{f}}$$

C=+1 for ground st.

C=-1 for excited st.

For J=0, → S=0, L=0 or S=1, L=1

§13.4. Meson mass spectrum

(2)

$$m_{\text{bound st.}} = m_q + m_{\bar{q}} - E_{\text{binding En.}}$$

1st excited st. $\rightarrow J^P = 1^-$ $\xrightarrow{\text{due to}}$ $J=1, L=0$

• "Hyperfine" interaction

$$\Delta E = \underline{\mu}_N \cdot \underline{B}_e = \underline{\mu}_e \cdot \underline{B}_N$$

[from e^- 's perspective]

$$\underbrace{\approx \underline{M}_1 \cdot \underline{M}_2}_{\text{Masses}} \approx \underbrace{\underline{S}_1 \cdot \underline{S}_2}_{\text{spins}} \frac{1}{M_1 M_2}$$

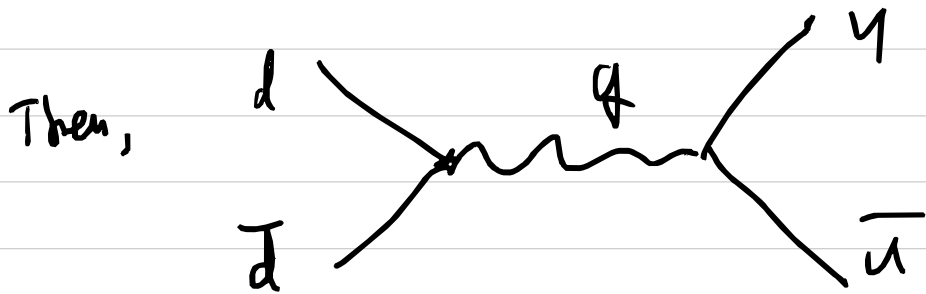
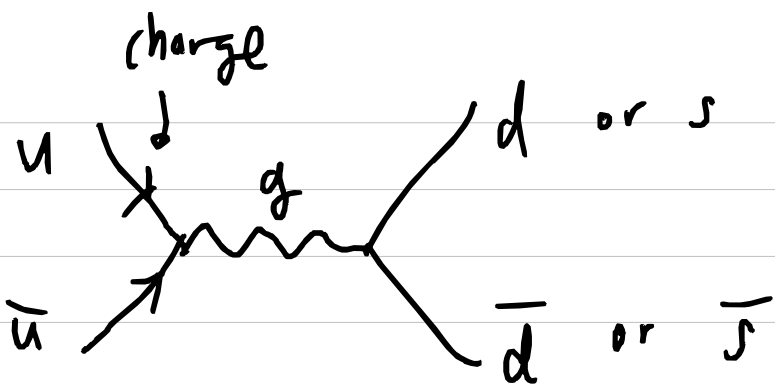
$$\therefore m = m_q + m_{\bar{q}} + K \frac{\langle \underline{S}_q \cdot \underline{S}_{\bar{q}} \rangle}{M_q M_{\bar{q}}}$$

$$\underline{J} = \underline{S}_q + \underline{S}_{\bar{q}} \rightarrow J^2 = S_q^2 + S_{\bar{q}}^2 + 2 \underline{S}_q \cdot \underline{S}_{\bar{q}}$$

$$\langle \underline{S}_q \cdot \underline{S}_{\bar{q}} \rangle = \frac{1}{2} [J(J+1) - S_q(S_q+1) - S_{\bar{q}}(S_{\bar{q}}+1)]$$

Expectation
val.

(3)



$\pi^0, \eta, \eta' \rightarrow$ mixtures of $u\bar{u}, d\bar{d}, s\bar{s}$

but π^0 has very little $s\bar{s}$

$$m_u \cong m_d$$

§ 13.5 Baryons

$uud, udd, \dots$
(proton) (neutron)

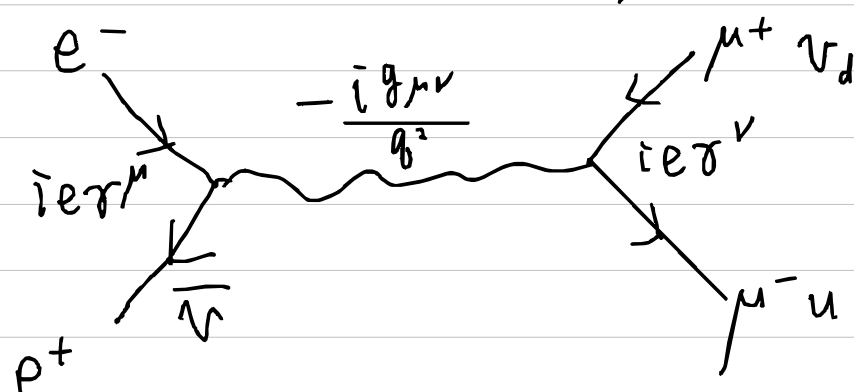
Uncolored $\rightarrow$ anti-symm. :

$$C=0 : \frac{1}{\sqrt{6}} (rgb - rbg + grb - gbr + bgr - brg)$$

§ 13.6. Exotic hadrons

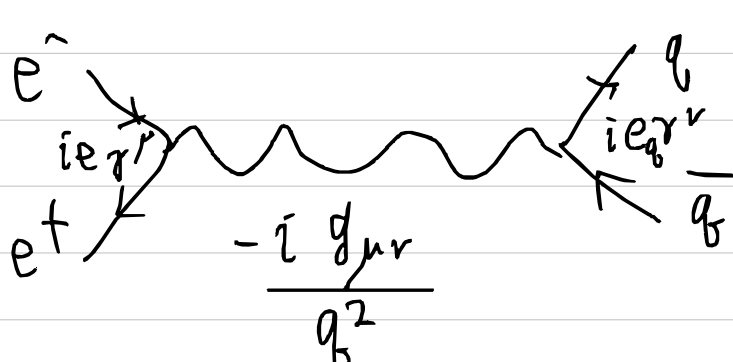
$(q\bar{q})(q\bar{q})$ } tetraquarks
 $(qqq)(q\bar{q})$ } pentaquarks

recently discovered.

L14: e^-e^+ annihilation to hadrons.Recap: $e^+e^- \rightarrow \mu^+\mu^-$ 

$$\rightarrow \frac{d\sigma_{\mu\mu}}{d\Omega} = \frac{\alpha^2}{4s} (1 + \cos^2\theta)$$

$$\sigma_{\mu\mu} = \frac{4\pi\alpha^2}{3s}$$

 $e^+e^- \rightarrow q\bar{q}$ 

$$M_{\mu^+\mu^-} = e^2 \quad \& \quad M_{q\bar{q}} = e e_q$$

$$|M|_{q\bar{q}}^2 = \left(\frac{e_q}{e}\right)^2 M_{\mu\mu}^2$$

$$\sigma_{q\bar{q}} = \left(\frac{e_q}{e}\right)^2 \sigma_{\mu\mu} = \frac{4\pi}{3s} \left(\frac{e_q}{e}\right)^2 \alpha^2$$

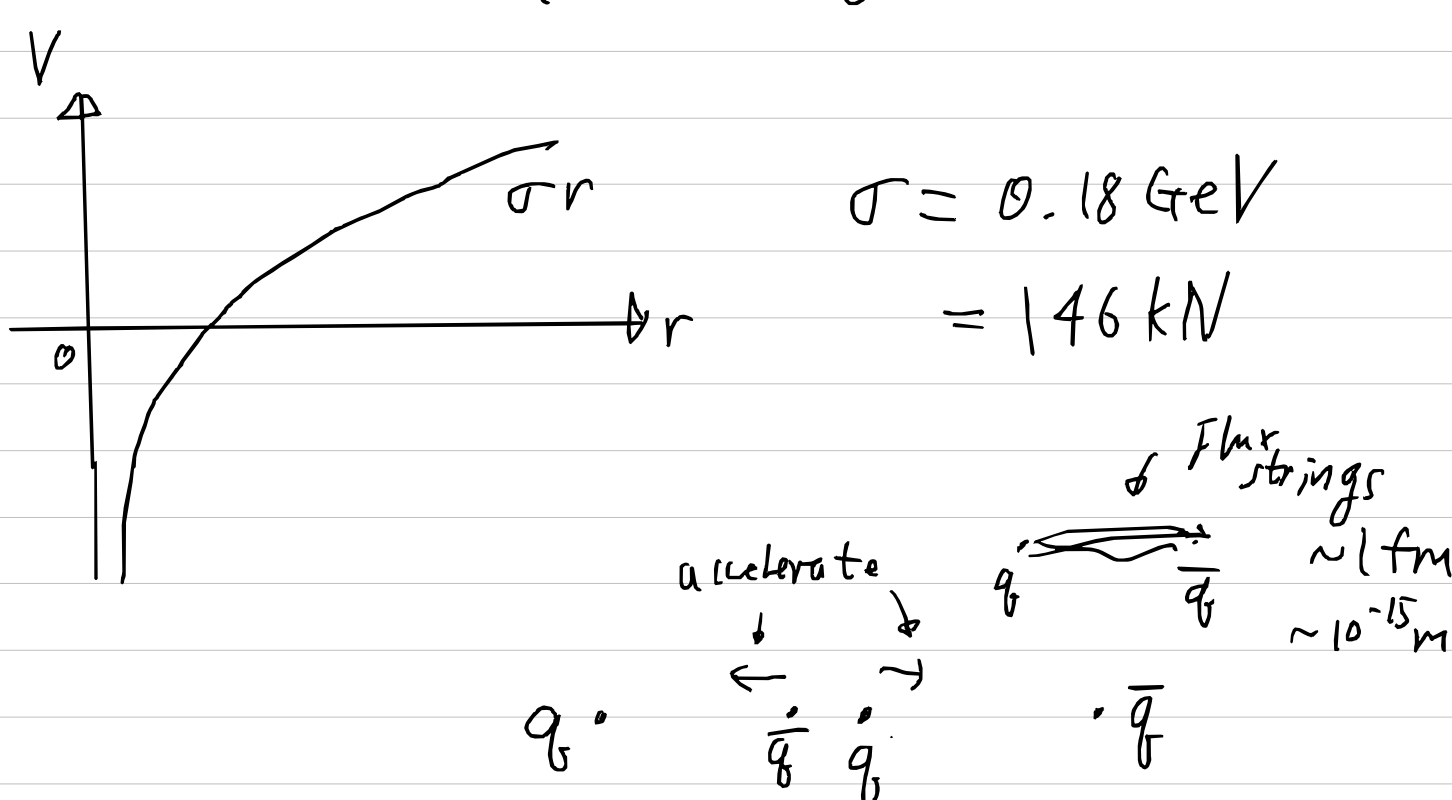
$$\sigma_{\text{Hadron}} = \frac{4\pi\alpha^2}{3s} \sum_q \left(\frac{e_q}{e}\right)^2 \times 3$$

color r, b, g

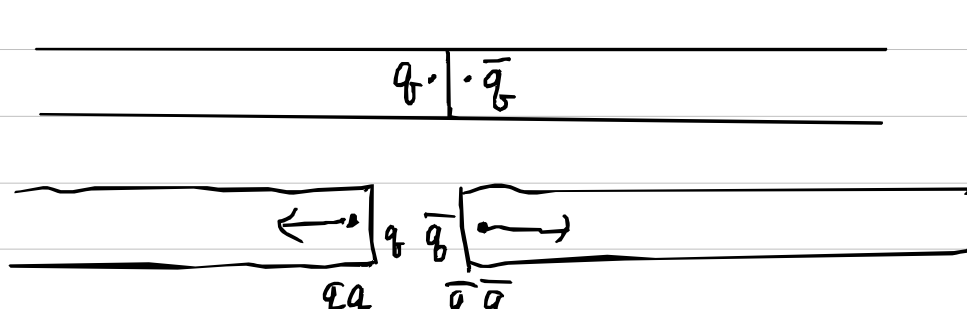
$$R = \frac{\sigma_{\text{Had}}}{\sigma_{\mu\mu}} = \sum_q \left(\frac{e_q}{e}\right)^2 \times 3$$

$$R(7\text{ GeV}) = \left(\left(-\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 \right) \times 3$$

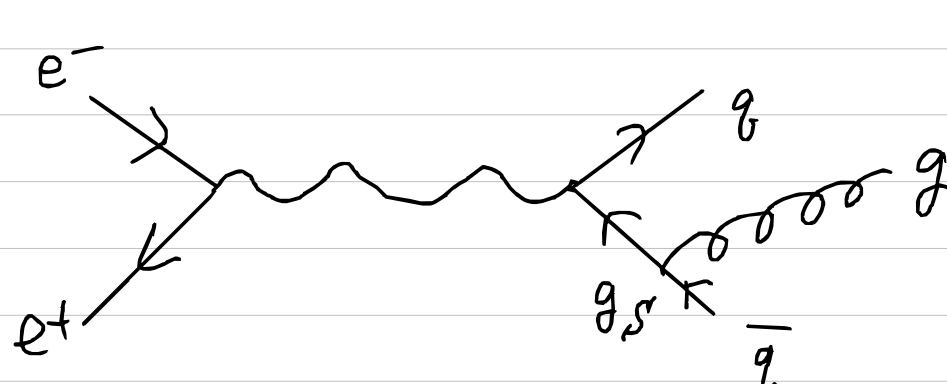
$$= \frac{10}{9} \times 3 = \frac{10}{3}$$



"Hadronization"

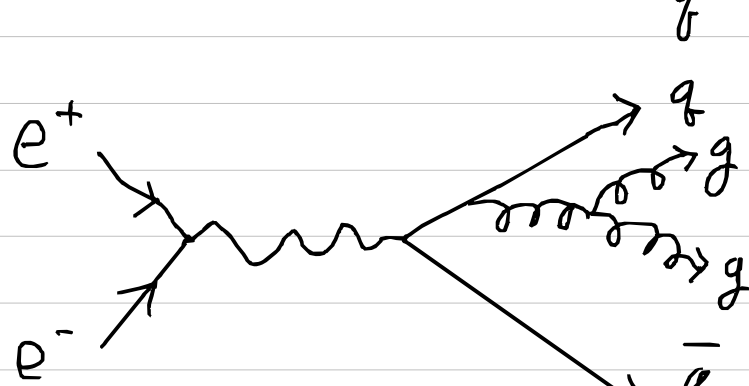
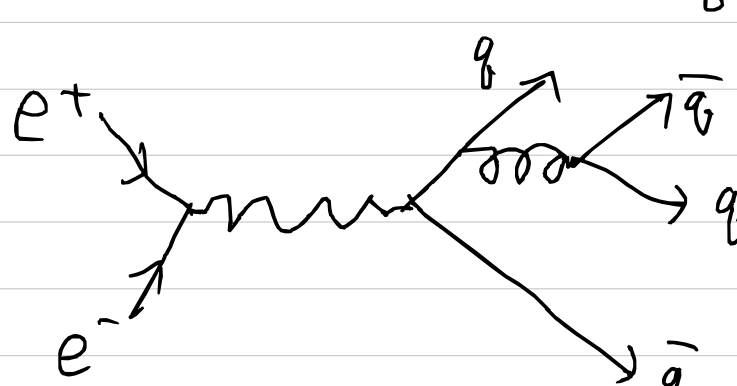
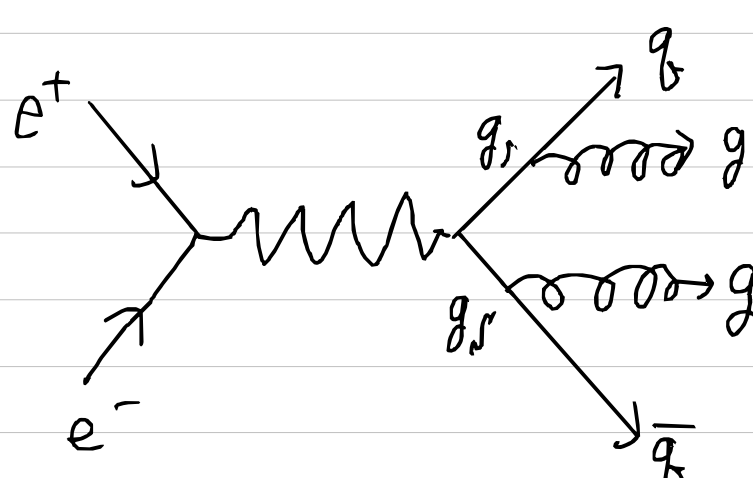
Define x : $x = \frac{p_h}{p_q} \rightarrow$ "scaling"

§14.4. Higher Order Effects

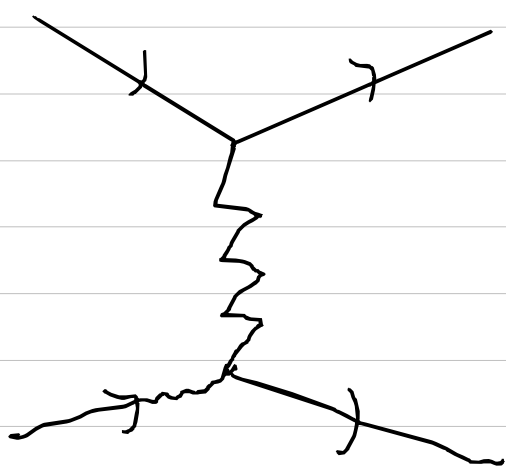


$$\sigma_H = \sigma_{H0} \left(1 + \frac{\alpha_s}{\pi}\right)$$

§14.5. Even Higher Orders



$$e^- \mu^- \rightarrow e^- \mu^-$$



Cross Sec.

$$\frac{d\sigma}{d(\cos\theta)} = \frac{\pi\alpha^2}{s} \frac{1 + \cos^4(\theta/2)}{\sin^4(\theta/2)}$$

 $e-p$ sys.

$$e^- p \rightarrow e^+ p$$

$$e^- p \rightarrow e^- X$$

$$p_T = \begin{pmatrix} p \\ 0 \\ 0 \\ p \end{pmatrix} + \begin{pmatrix} p \\ 0 \\ 0 \\ -p \end{pmatrix}$$

$$1/1\text{fm}$$

$$= \begin{pmatrix} 2p \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$s = p_T^2 = 4p^2$$

The tot. $e-q$ sys. En. :

$$x = \frac{|p_q|}{|p_p|}$$

$$E_{eq} = p + xp = (1+x)p$$

$$0 < x < 1$$

$$p_T = \begin{pmatrix} p \\ 0 \\ 0 \\ p \end{pmatrix} + \begin{pmatrix} xp \\ 0 \\ 0 \\ -xp \end{pmatrix}$$

lab frm.

$$p_T^2 = \hat{s} = (p+xp)^2 - (p-xp)^2 = 4xp^2 = x\hat{s}$$

$$= \begin{pmatrix} p+xp, 0, 0, p-xp \end{pmatrix} \begin{pmatrix} p+xp \\ 0 \\ 0 \\ p-xp \end{pmatrix}$$

$$q^2 = -\hat{s} \sin^2\left(\frac{\theta}{2}\right) = -\underbrace{Q^2}_{Q^2 > 0}$$

$$\sin^2\left(\frac{\theta}{2}\right) = \frac{Q^2}{\hat{s}}$$

$$\cos^2\left(\frac{\theta}{2}\right) = 1 - \sin^2\left(\frac{\theta}{2}\right) = 1 - \frac{Q^2}{\hat{s}}$$

$$\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = 1 - \frac{2Q^2}{\hat{s}}$$

The Weak Force - Part I

§16.1. Parity $\hat{P} \quad \underline{r} \rightarrow -\underline{r}$

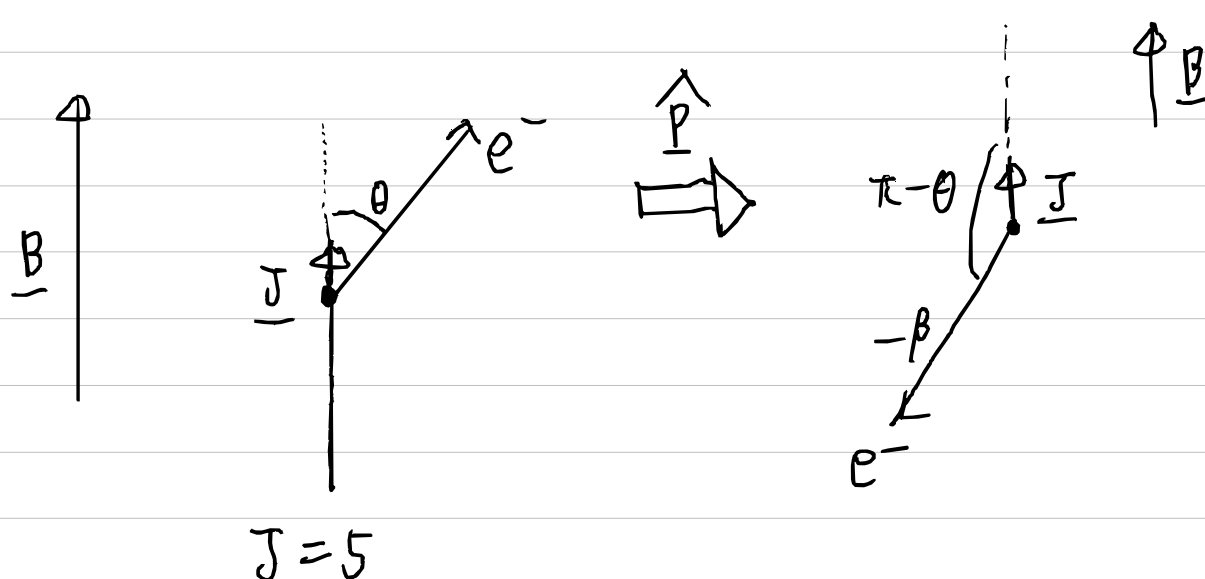
$$\hat{P}^2 \psi = \pm \psi$$

$$u \rightleftharpoons v$$

Charge conj'g. $f \rightarrow \bar{f}$

$$\hat{C} \psi = \pm \psi$$

$$\pi^0 \rightarrow \gamma\gamma, \quad \pi^0 \rightarrow \gamma\gamma\gamma$$



$$1 - \beta \cos(\pi - \theta)$$

$$\underline{\beta} \cdot \underline{J} = 1 + \beta \cos \theta \neq 1 - \beta \cos \theta$$

$$e^+e^- \rightarrow \mu^+\mu^- \sim 1 + \cos^2 \theta$$

$$\sim \underline{\beta} \cdot \underline{J}$$

Parity is massively violated.

(P $\rightarrow$ not conserved $\rightarrow$ "CP violation".(PT $\rightarrow$ together, conserved§16.5. SU(2) gauge invariance
(symmetry)

$$\Psi = \begin{pmatrix} \psi_\nu \\ \psi_e \end{pmatrix}$$

 $\rightarrow$ Need unitary rotation

$$U = e^{-i\alpha_j \sigma_j}$$

$$j=1, 3.$$

$$J_i^\mu = \bar{\Psi} \gamma^\mu \sigma_i \Psi$$

converts
neutrino
to e^- .converts
 e^- to
neutrino.

$$J_1^\mu = \bar{\Psi} \gamma^\mu \sigma_1 \Psi = \bar{\psi}_e \gamma^\mu \psi_\nu + \bar{\psi}_\nu \gamma^\mu \psi_e$$

$$J_2^\mu = \bar{\Psi} \gamma^\mu \sigma_2 \Psi = i\bar{\psi}_e \gamma^\mu \psi_\nu - i\bar{\psi}_\nu \gamma^\mu \psi_e$$

$$J_3^\mu = \bar{\Psi} \gamma^\mu \sigma_3 \Psi = \bar{\psi}_\nu \gamma^\mu \psi_\nu - \bar{\psi}_e \gamma^\mu \psi_e$$

$$\mathcal{L}_{int} = -g_w A_{i\mu} \bar{\Psi} \gamma^\mu \sigma_i \Psi$$

 A_i^μ flds are self-interacting

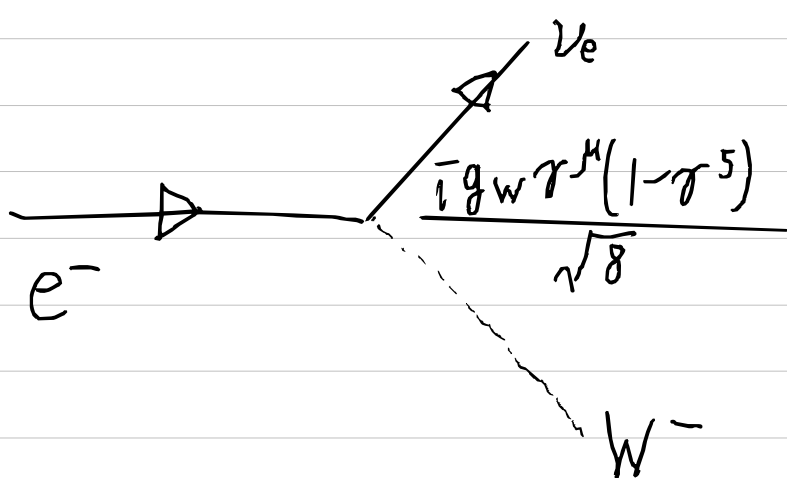
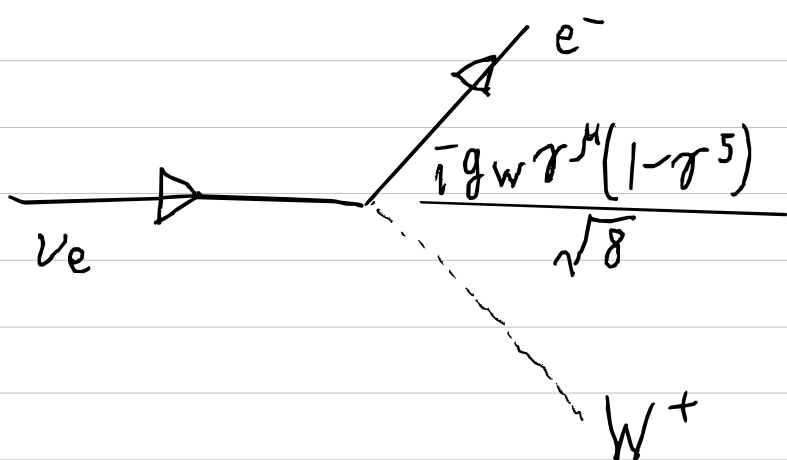
"Non-Abelian Theory"

"mass term" $\frac{m^2 A_i^\mu A_{i\mu}}{2}$ is not allowed.

(1)

★ Charged Current

$$\mathcal{L} = -\frac{g_w}{\sqrt{2}} \left[W_\mu^+ \bar{\psi}_e \gamma^\mu \frac{1}{2}(1-\gamma^5) \psi_\nu + W_\mu^- \bar{\psi}_\nu \gamma^\mu \frac{1}{2}(1-\gamma^5) \psi_e \right]$$



Propagator $\frac{-i g_{\mu\nu}}{q^2} \rightarrow -i \frac{g_{\mu\nu} - \frac{q_\mu q_\nu}{M_W^2}}{q^2 - M_W^2}$

For Weak Interaction $M_W = 0$

For $q^2 \ll M_W^2$, the propagator is

$$i \frac{g^{\mu\nu}}{M_W^2}$$

Fermi const.

$$\frac{G_F}{\sqrt{2}} = \frac{1}{8} \left(\frac{g_w}{M_W} \right)^2$$

$$\sim 10^{-5} \text{ GeV}^{-2}$$

so small $\rightarrow$ hence the nomenclature "weak"

$$g_w \approx 0.65$$

$$\alpha_w = \frac{g_w^2}{4\pi} \sim \frac{1}{29}$$

APP L17

L17 The Weak Force - Part II

Charged current $\rightarrow$ Weak Interactions

$$\mathcal{L} = -\frac{g_w}{2} \left[W_\mu^+ \bar{\psi}_e \gamma^\mu (1-\gamma^5) \psi_\nu + W_\mu^- \bar{\psi}_\nu \gamma^\mu (1-\gamma^5) \psi_e \right]$$

$$\bar{\psi} \gamma^\mu (1-\gamma^5) \psi$$

$$= \bar{\psi} \gamma^\mu \psi - \bar{\psi} \gamma^\mu \gamma^5 \psi$$

$$\hookrightarrow \partial_\mu \bar{\psi} \gamma^\mu \psi = 0 \rightarrow \partial_\mu \bar{\psi} \gamma^\mu \gamma^5 \psi = 2im \bar{\psi} \gamma^5 \psi$$

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\frac{1}{2} \gamma^5 \psi = \pm \frac{1}{2} \psi \quad \rightarrow \text{Chirality "op".}$$

$+\frac{1}{2} \psi_R$, $-\frac{1}{2} \psi_L$

Right-handed, Left-handed

+ve angular term
-ve angular term.

Spinor $\rightarrow$ can be decomposed into RH & LH.

For $\beta \rightarrow 1$, $m \rightarrow 0$:

$$\psi_R = \psi_{h=+\frac{1}{2}}, \quad \psi_L = \psi_{h=-\frac{1}{2}}$$

$$\chi = \psi_R + \psi_L \quad \frac{1}{2} \gamma^5 \psi_R = -\frac{1}{2} \psi_R$$

$$\frac{1}{2} (1-\gamma^5) \chi = \frac{1}{2} (1-\gamma^5) \psi_R + \frac{1}{2} (1-\gamma^5) \psi_L$$

$$= \cancel{\frac{1}{2} \psi_R} - \cancel{\frac{1}{2} \psi_R} + \frac{1}{2} \psi_L + \frac{1}{2} \psi_L$$

$$= \psi_L$$

$-\frac{1}{2} \gamma^5 \psi_L = +\frac{1}{2} \psi_L$

$$P_R = \frac{1}{2} (1+\gamma^5), \quad P_L = \frac{1}{2} (1-\gamma^5)$$

$$\frac{1}{2} (1-\gamma^5) \frac{1}{2} (1-\gamma^5) \chi = \psi_L$$

$$P_L P_L \psi = P_L \psi = \psi_L$$

$$\bar{\psi} \gamma^\mu \frac{1}{2} (1-\gamma^5) \frac{1}{2} (1-\gamma^5) \psi$$

$$= \bar{\psi} \gamma^\mu \frac{1}{2} (1-\gamma^5) \psi_L$$

$$= \bar{\psi} \frac{1}{2} (1+\gamma^5) \gamma^\mu \psi_L$$

$$= \psi^\dagger \gamma^0 \frac{1}{2} (1+\gamma^5) \gamma^\mu \psi_L$$

$$= \psi^\dagger \frac{1}{2} (1-\gamma^5) \gamma^0 \gamma^\mu \psi_L$$

$$= \left[\frac{1}{2} (1-\gamma^5) \psi \right]^\dagger \gamma^0 \gamma^\mu \psi_L$$

$\hookrightarrow \psi_L$

$$= \psi_L^\dagger \gamma^0 \gamma^\mu \psi_L$$

$$= \bar{\psi}_L \gamma^\mu \psi_L$$

"Weak Interaction" $\rightarrow$ p.s.

$$\bar{\psi} = \psi^\dagger \gamma^0$$

$$\gamma^{5\dagger} = \gamma^5$$

Do it again

$\psi_{h=\frac{1}{2}} \rightarrow \text{mostly } \psi_R$

(2)

$$\psi_{h=\frac{1}{2}} = \sqrt{\frac{1+\beta}{2}} \psi_R - \sqrt{\frac{1-\beta}{2}} \psi_L$$

$$\psi_{h=-\frac{1}{2}} = \sqrt{\frac{1-\beta}{2}} \psi_R + \sqrt{\frac{1+\beta}{2}} \psi_L$$

$\psi_{h=-\frac{1}{2}} \rightarrow \text{mostly } \psi_L$

ϕ

Weak interactions for neutrinos.

Helicity $\rightarrow$ projection of spin at the mfm.

We have the maximum parity violation.

Neutrinos will couple w/
LH particle str

$$U_{R,L} = \frac{1}{2} (1 \pm \sigma^5) U$$

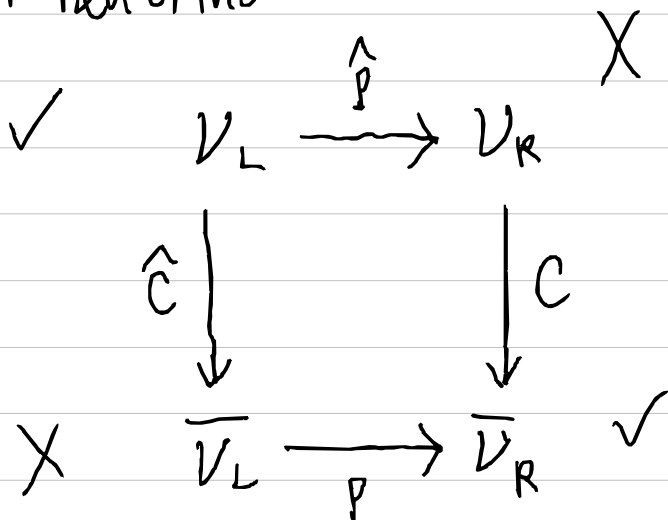
$$V_{R,L} = \frac{1}{2} (1 \mp \sigma^5) V$$

OR

RH anti-particle
strs

MAXIMUM
CP violation

* LH neutrino



$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}, \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}$$

Applying
Global $U(1)$ symmetry to EM

$\rightarrow$ we get conservation laws

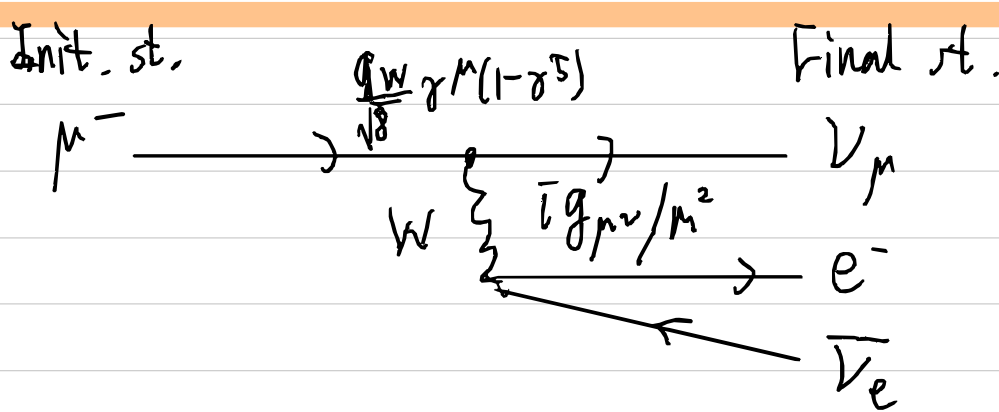
electronious, muonious, tauonious

e^- number, μ^- num., lepton num.

$1 = e^-$, $1 = \text{particle}$, $1 = \tau^-$
 $-1 = e^+$, $-1 = \text{anti-p.}$, $-1 = \tau^+$

μ^- can only couple to ν_μ

e^-



$$\Gamma = \frac{G_F^2 m_\mu^5}{24(2\pi)^3} \sim 3 \times 10^{-19} \text{ GeV}$$

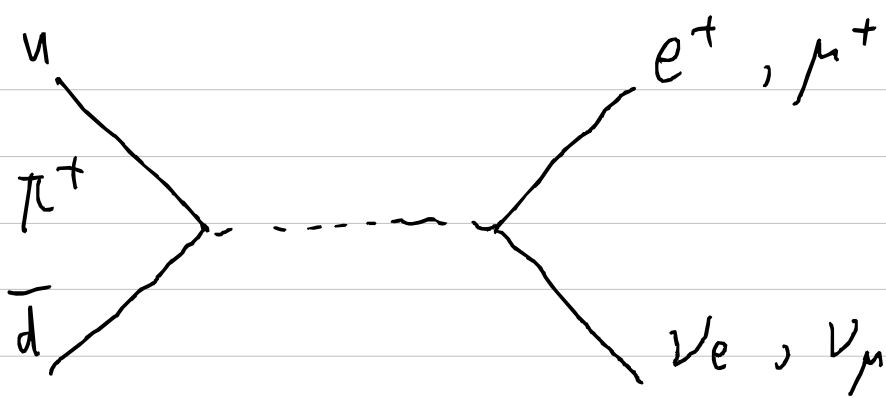
L.T. of muon $\approx 2.2 \text{ ns}$

$$\therefore \frac{d\Gamma}{d\Omega} = 1 - \frac{1}{3} \cos \theta$$

$$\Gamma_\tau = \Gamma_\mu \left(\frac{m_\tau}{m_\mu} \right)^5 (1 + 1 + 3) \approx 20 \times 10^{-13} \text{ GeV}$$

Width of τ τ^- mode μ^- mode quark mode

3



$$\underline{\underline{M}} \propto J_\mu^\pi J_L^\mu$$

$$J_\pi^\mu = f_\pi P_\pi^\mu$$

$$J_\pi^\mu \propto P_\pi^\mu$$

Ang. Mtm.

Mtm

$$|\underline{\underline{M}}|^2 = 2 f_\pi^2 m_e^2 (m_\pi^2 - m_e^2)$$

$$\frac{d\Gamma}{d\Omega} = \frac{1}{32\pi^2} \frac{m_\pi^2 - m_e^2}{m_\pi^2}$$

$$\therefore \Gamma \propto f_\pi^2 \frac{m_e^2 (m_\pi^2 - m_e^2)^2}{m_\pi^2}$$

$$\frac{\Gamma_e}{\Gamma_\mu} = \frac{m_e^2 (m_\pi^2 - m_e^2)^2}{m_\mu^2 (m_\pi^2 - m_\mu^2)^2} \sim 1.3 \times 10^{-4}$$

"e⁻ - to - μ⁻ ratio"



The Weak Force - P III

§18.1. Kaon decays

$$K^- (s\bar{u}) \rightarrow \mu^- \bar{\nu}_\mu$$

$$\pi^- (d\bar{u}) \rightarrow \mu^- \bar{\nu}_\mu$$



§18.2. Cabibbo (quark) mixing

$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos \theta_c & \sin \theta_c \\ -\sin \theta_c & \cos \theta_c \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}$$

$$\left(\begin{array}{l} \theta_c : \text{Cabibbo angle} \approx 13^\circ \\ \\ = \begin{pmatrix} 0.975 & 0.222 \\ -0.222 & 0.975 \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix} \end{array} \right)$$

$$\text{Lagr.} : \mathcal{L} = \frac{g_W}{\sqrt{2}} W_\mu \left[\bar{\psi}_u \gamma^\mu \frac{1}{2} (1 - \gamma^5) \psi_d + \bar{\psi}_c \gamma^\mu \frac{1}{2} (1 - \gamma^5) \psi_s \right]$$

$$\mathcal{L} = \frac{g_W}{\sqrt{2}} W_\mu (\bar{\psi}_u, \bar{\psi}_c) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \gamma^\mu \frac{1}{2} (1 - \gamma^5) \begin{pmatrix} \psi_d \\ \psi_s \end{pmatrix}$$

$$u \rightarrow d \text{ coupling} : g_W \cos \theta_c$$

$$c \rightarrow s \text{ coupling} : g_W \cos \theta_c$$

$$u \rightarrow s \text{ coupling} : +g_W \sin \theta_c$$

$$c \rightarrow d \text{ coupling} : -g_W \sin \theta_c$$

↳ a 4-th quark is needed for consistency,

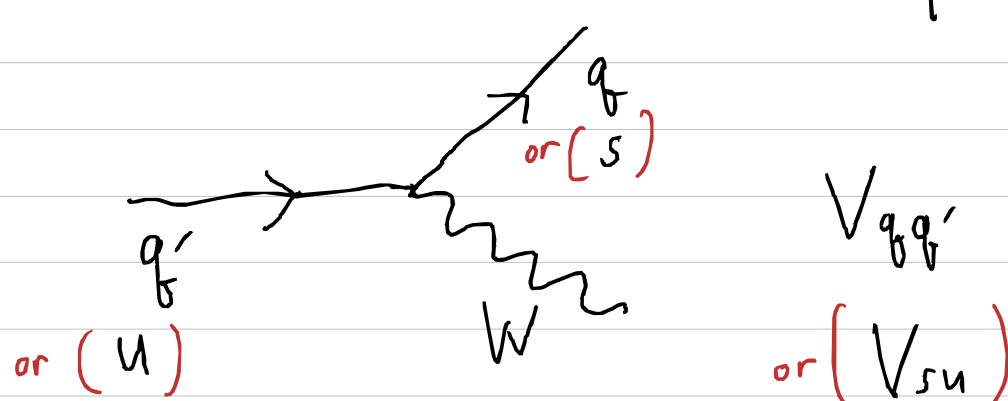
§18.3. The CKM (Cabibbo-Kobayashi-Maskawa) matrix

3x3 matrix

$$\begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} = V_{\text{CKM}}$$

The reason they expanded $2 \times 2 \rightarrow 3 \times 3$ was to explain CP violation.

$$\text{Then, } \mathcal{L} = \frac{g_W}{\sqrt{2}} W_\mu (\bar{\psi}_u \quad \bar{\psi}_c \quad \bar{\psi}_t) \begin{pmatrix} V_{\text{CKM}} \end{pmatrix} \gamma^\mu \frac{1}{2} (1 - \gamma^5) \begin{pmatrix} \psi_d \\ \psi_s \\ \psi_b \end{pmatrix}$$



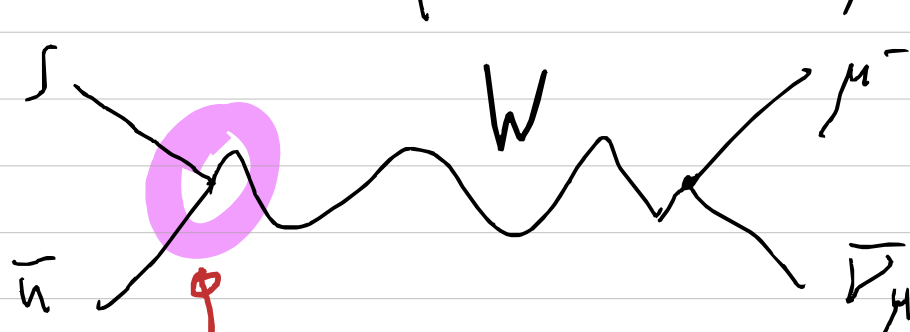
$$V_{ij} = V_{ji}^*$$

Unitary Matrix

$$V_{su} = V_{us}^*$$

Write down the outgoing one 1st & the incoming 2nd.

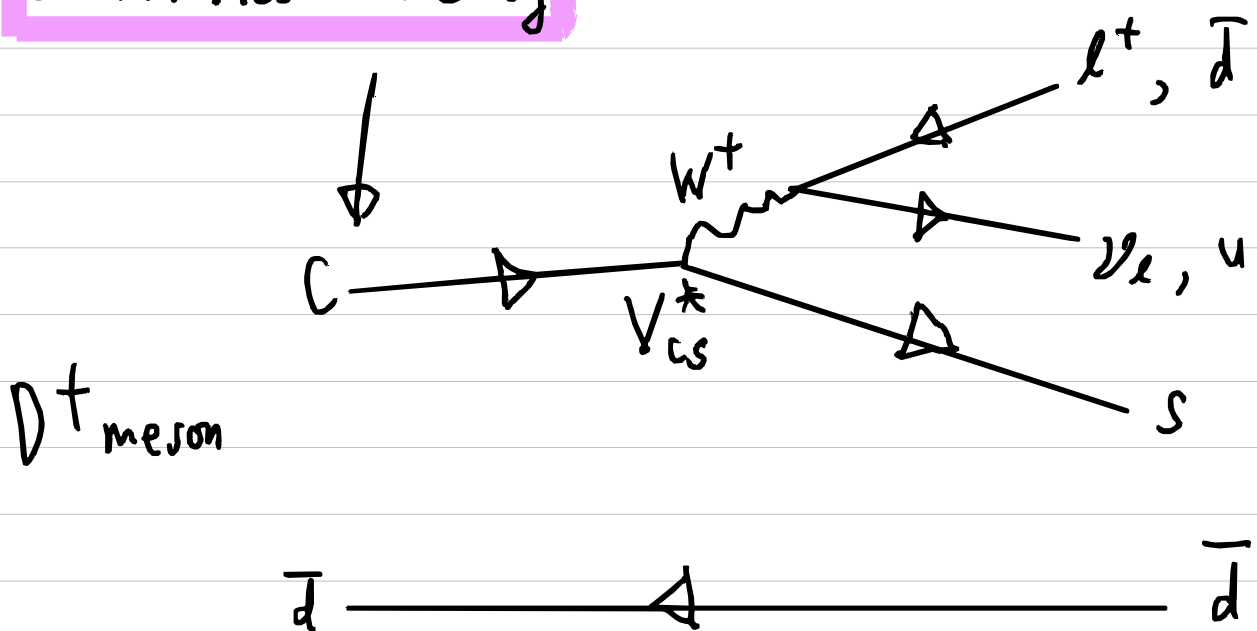
$$|V_{\text{CKM}}| \approx \begin{pmatrix} 0.975 & 0.223 & 0.003 \\ 0.222 & 0.974 & 0.04 \\ 0.009 & 0.039 & 0.999 \end{pmatrix} \approx \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



su coupling happens here.

§18.4. Charged D decay

Charm meson decay

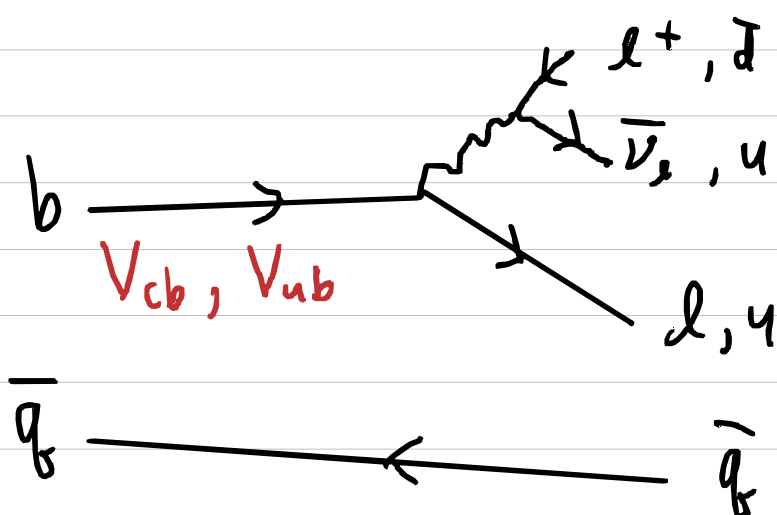


"Spectator" quarks have influence.

$$\text{Width: } \Gamma_D \sim \left(\frac{m_c}{m_\tau} \right)^5 \Gamma_\tau$$

$$\text{Lifetime: } \tau_D \sim 1.5 \tau_\tau \sim \sum [\text{final}]$$

$$\tau_c \sim \left(\frac{m_\tau}{m_c} \right)^5 \tau_\tau$$

§18.5. b & t hadrons

$$V_{cb} \sim 0.04$$

$$V_{ub} \sim 0.003$$

$$\tau_b \sim \left(\frac{m_\tau}{m_b} \right)^5 \frac{1}{|V_{cb}|^2 + |V_{ub}|^2} \tau_\tau$$

$$\sim 1.8 \times 10^{-12} \text{ s}$$

$$\text{for } m_b = 4.5 \text{ GeV}$$

$$\tau_t \sim \left(\frac{m_\tau}{m_t} \right)^5 \tau_\tau \sim 3 \times 10^{-23} \text{ s}$$

$$\text{for } m_t = 175 \text{ GeV}$$

top qrk $\rightarrow$ heavier than W bosons.

Incorporating W production on mass shell

$$\tau_t \sim 4 \times 10^{-25} \text{ s}$$

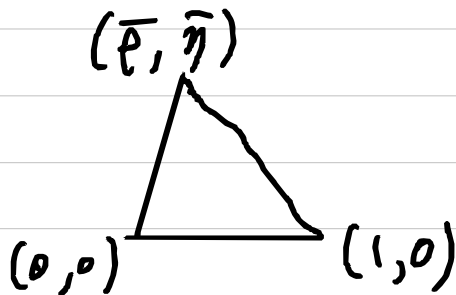
Recently, CMS & ATLAS $\rightarrow$ top qrks

(3)

Prob. 6. Unitary Triangles

$$V_{CKM}^\dagger V_{CKM} = I$$

$$V_{ud} V_{ub}^* + V_{cd} V_{cb}^* + V_{td} V_{tb}^* = 0$$



As of now,

$$V_{CKM} \approx \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ & & 1 \end{pmatrix}$$

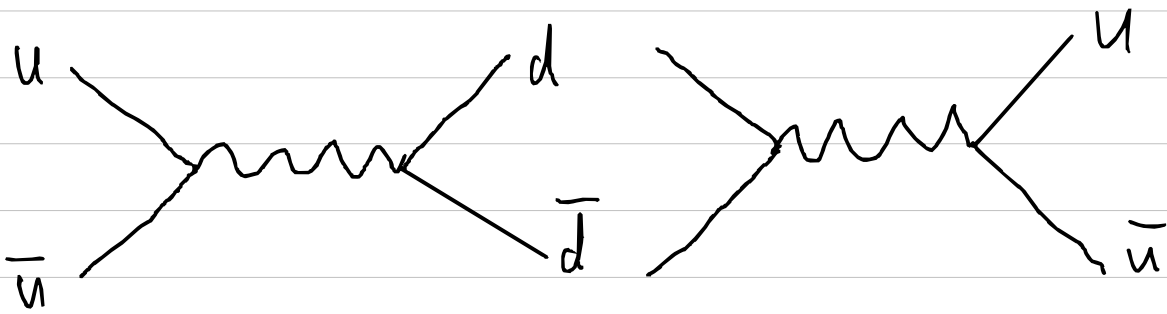
9
estimated

WE
 Haven't seen any $\mu^- - e^-$ couplings
 or $\mu^- - \tau^-$ couplings.

APP L19

The Weak Force - Part IV

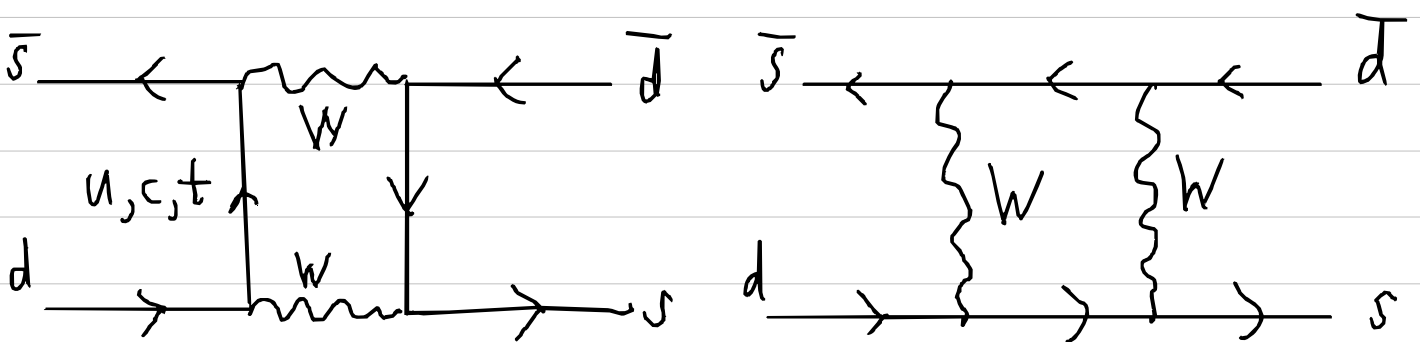
§19.1. The CP violation



$$\rightarrow \pi^0 = \frac{1}{\sqrt{2}} (u\bar{u} + d\bar{d})$$

Direct CP : $\Gamma(x \rightarrow y) \neq \Gamma(\text{CP}x \rightarrow \text{CP}y)$

Mixing $K^0 \rightarrow \bar{K}^0$



K_S, K_L

↑
short-lived

↑
long-lived

$J^P = 0^-$

$$\hat{P} |K^0\rangle = -|K^0\rangle$$

$$\hat{P} |\bar{K}^0\rangle = -|\bar{K}^0\rangle$$

$$\hat{C} |K^0\rangle = -|\bar{K}^0\rangle$$

* Apply C twice:

$$CC |K^0\rangle = C^2 |K^0\rangle = |K^0\rangle$$

$$C|\bar{K}^0\rangle = -|K^0\rangle$$

(2)

$$\hat{C}\hat{P}|K^0\rangle = |\bar{K}^0\rangle$$

$$\hat{C}\hat{P}|\bar{K}^0\rangle = |K^0\rangle$$

$$|K_1\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle)$$

$$|K_2\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle)$$

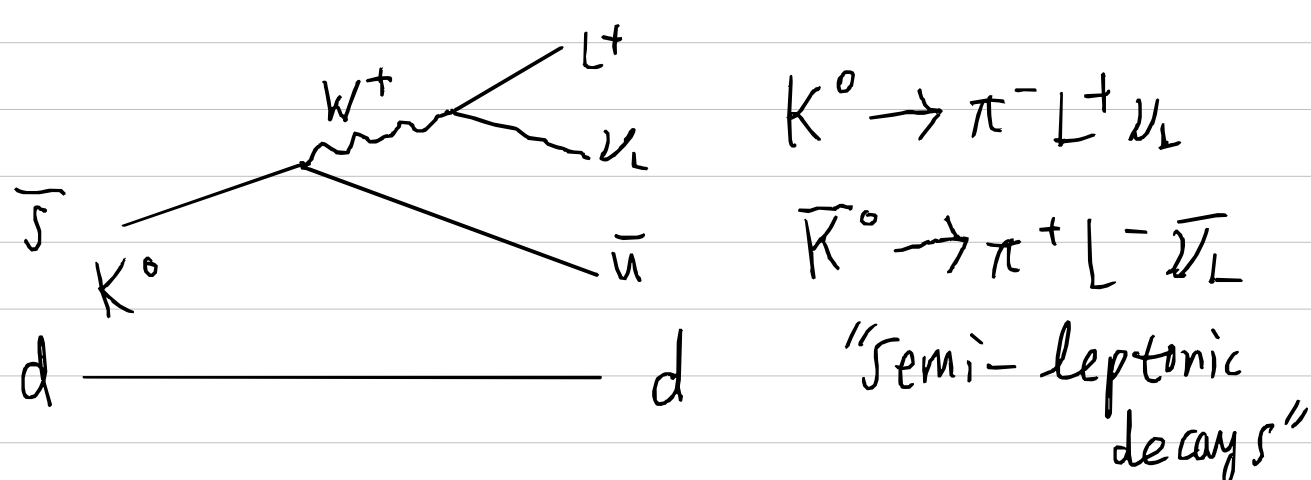
$$CP|K_1\rangle = \frac{1}{\sqrt{2}}(|\bar{K}^0\rangle + |K^0\rangle) = |K_1\rangle$$

$$CP|K_2\rangle = \frac{1}{\sqrt{2}}(|\bar{K}^0\rangle - |K^0\rangle) = -|K_2\rangle$$

$$|K^0\rangle = \frac{1}{\sqrt{2}}(|K_1\rangle + |K_2\rangle)$$

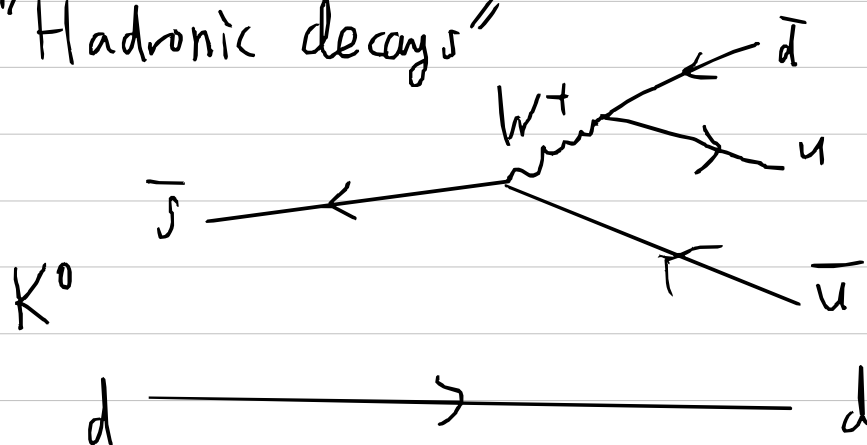
$$|\bar{K}^0\rangle = \frac{1}{\sqrt{2}}(|K_1\rangle - |K_2\rangle)$$

§19.3. Neutral Kaon Decays



$$K^0 \rightarrow \pi\pi, \pi\pi\pi$$

"Hadronic decays"



$$CP(|\pi\pi\rangle) = |\pi\pi\rangle$$

$$CP(|\pi\pi\pi\rangle) = -|\pi\pi\pi\rangle$$

$$K_1 \rightarrow \pi\pi, CP = +1$$

$$K_2 \rightarrow \pi\pi\pi, CP = -1$$

$$3m_\pi \approx m_{K^0}$$

$$K_S \rightarrow \pi^+\pi^-$$

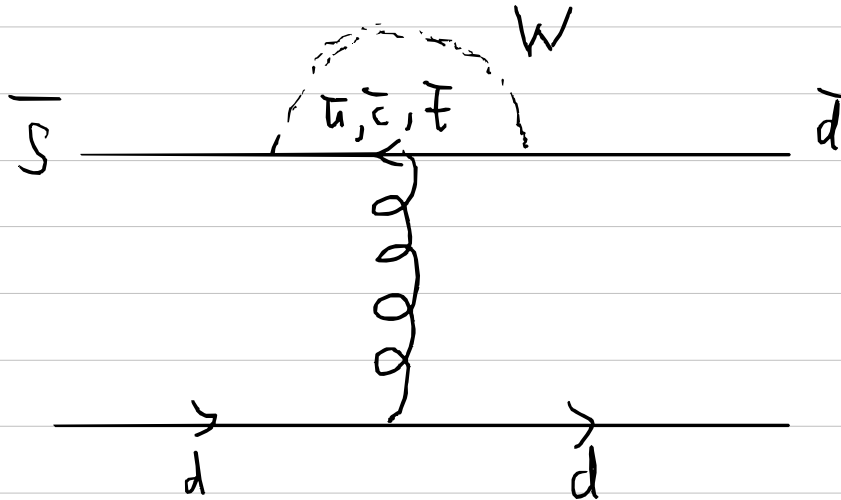
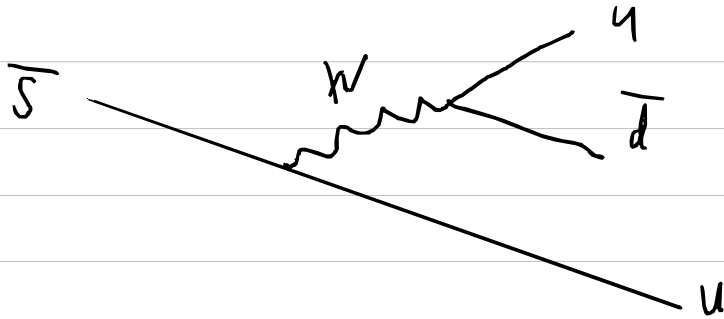
$$K_L \rightarrow \pi^+\pi^-\pi^0$$

$$|K_S\rangle = N(|K_1\rangle + \epsilon|K_2\rangle)$$

$$|K_L\rangle = N(|K_2\rangle + \epsilon|K_1\rangle)$$

$$\text{Normalization factor} : N = \frac{1}{\sqrt{1+|\epsilon|^2}}$$

3



$K^0(\bar{s}d)$

Penguin decay.

L20: Electroweak Unification

* Is weak F. $SU(2)$?

$$\Psi = \begin{pmatrix} \psi_u \\ \psi_L \end{pmatrix}_L \quad \& \quad \begin{pmatrix} \psi'_u \\ \psi'_d \end{pmatrix}_L$$

$$J^\mu = \bar{\Psi} \gamma^\mu \Psi$$

$$J_1^\mu = \bar{\psi}_e \gamma^\mu \psi_\nu + \bar{\psi}_\nu \gamma^\mu \psi_e$$

$$J_2^\mu = i \bar{\psi}_e \gamma^\mu \psi_\nu - i \bar{\psi}_\nu \gamma^\mu \psi_e$$

$$J_3^\mu = \bar{\psi}_{\nu L} \gamma^\mu \psi_{\nu L} - \bar{\psi}_{eL} \gamma^\mu \psi_{eL}$$

$$W^\pm = \frac{1}{\sqrt{2}} (W_1 \mp i W_2)$$

$$W_3^\mu \neq Z^\mu$$

$$\mathcal{L} = g_w Z^\mu J_{3\mu}$$

$$\mathcal{L} = g_z Z^\mu \bar{\Psi} \gamma^\mu (C_V - C_A) \Psi$$

$$\mathcal{L} = g_z Z^\mu [C_L \bar{\psi}_L \gamma^\mu \psi_L + C_R \bar{\psi}_R \gamma^\mu \psi_R]$$

$$C_L = C_V + C_A, \quad C_R = C_V - C_A$$

§20.3. GWS Model

$$Z^\mu = (\cos \theta_w) W_3^\mu - (\sin \theta_w) Y^\mu$$

$$A^\mu = (\cos \theta_w) Y^\mu + (\sin \theta_w) W_3^\mu$$

$$\mathcal{L} = g_Y Y_\mu J_Y^\mu \xrightarrow{\text{then}} \sin \theta_w = \frac{e}{g_w}$$

$$\& \quad g_Y = \frac{g_w e}{\sqrt{g_w^2 - e^2}}$$

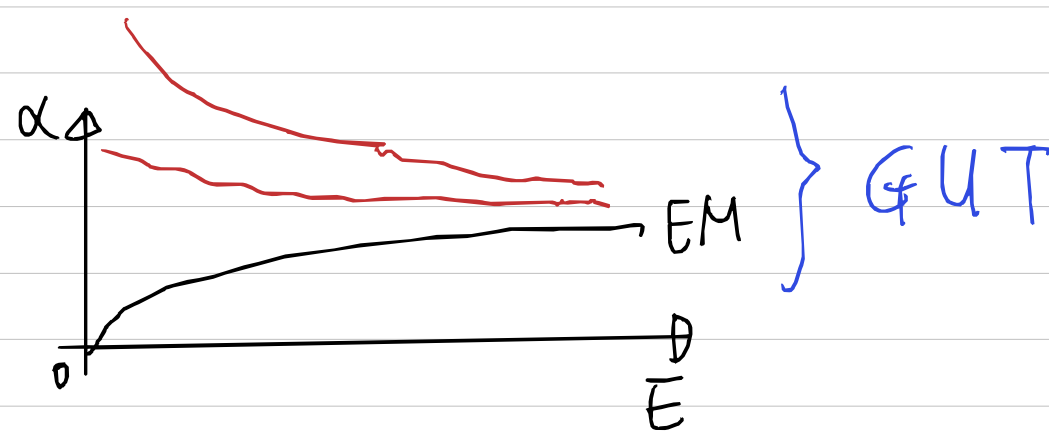
• The current that must couple to the hypercharge

$$J_Y^\mu = -\bar{\psi}_{\nu L} \gamma^\mu \psi_{\nu L} - \bar{\psi}_{eL} \gamma^\mu \psi_{eL} - 2 \bar{\psi}_{eR} \gamma^\mu \psi_{eR}$$

$$\text{Mixing angle: } \sin \theta_w = \frac{e}{g_w}$$

$$\sin^2 \theta_w \approx 0.23 \rightarrow \theta_w \sim 0.5 \text{ rad} = 28.7^\circ$$

$\sim 30^\circ$
(Nobel Prize worthy)
"WHY?"



§20.4. other pts



$$\sim e \underline{E} \cdot \underline{B}$$

"Anomaly" condition

$$\sum q = 0 \quad -1 + \frac{2}{3} - \frac{1}{3} = -\frac{2}{3}$$

ν, e^-

Include color too

$$\sum q = 0 \quad -1 + 3 \times \left(\frac{2}{3}\right) - 3 \times \left(\frac{1}{3}\right) = 0$$

neutral

* CKM matrix equivalent for Z ,

$$\begin{pmatrix} \psi_u \\ \psi_c \\ \psi_t \end{pmatrix} = U_u \begin{pmatrix} \psi'_u \\ \psi'_c \\ \psi'_t \end{pmatrix}$$

up-type qarks

$$\begin{pmatrix} \psi_d \\ \psi_s \\ \psi_b \end{pmatrix} = U_d \begin{pmatrix} \psi'_d \\ \psi'_s \\ \psi'_b \end{pmatrix}$$

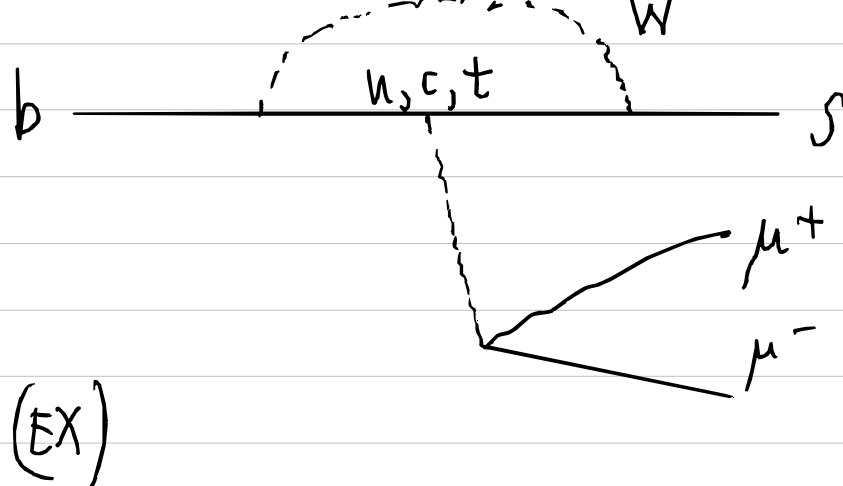
down-type qarks

$$\therefore \bar{\psi}'_u \psi'_d = \bar{\psi}_u \underbrace{U_u U_d^\dagger}_{V_{CKM}} \psi_d$$

$$V_{CKM} = U_u U_d^\dagger$$

$$(\bar{\psi}'_u, \bar{\psi}'_c, \bar{\psi}'_t) \begin{pmatrix} \psi'_d \\ \psi'_s \\ \psi'_b \end{pmatrix}$$

$$\bar{\psi}'_u \psi_u = \bar{\psi}_u \cancel{U_u U_u^\dagger} \psi_u = \bar{\psi}_u \psi_u$$



(ex)

$$K^- \rightarrow \pi^- e \nu$$



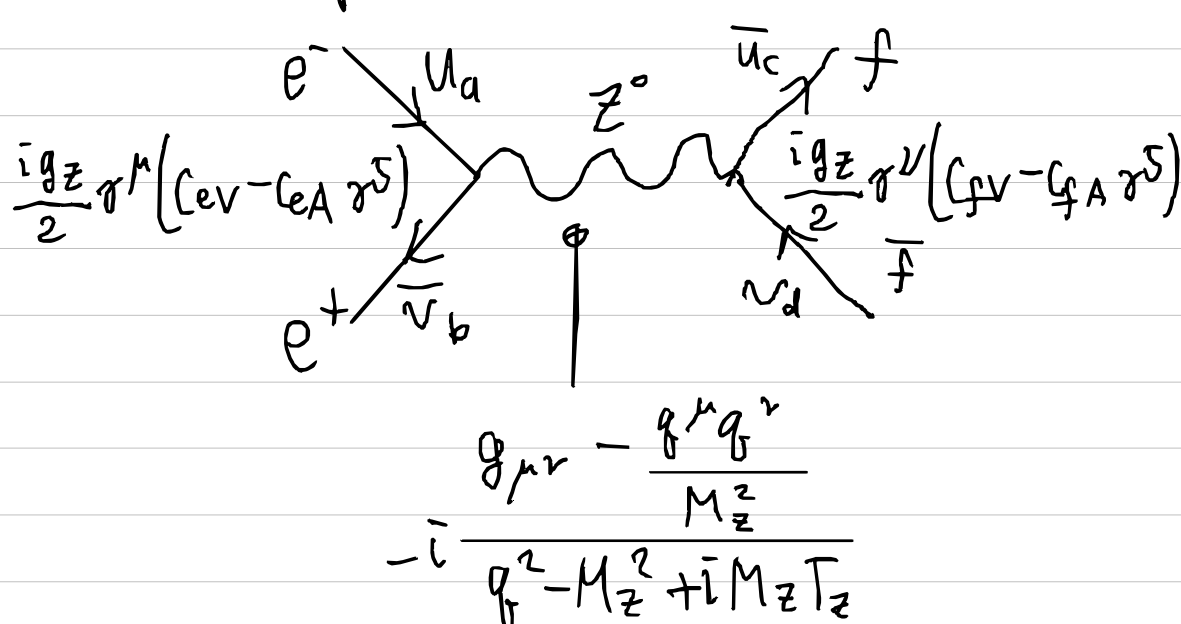
$$K^- \rightarrow \pi^- \nu \bar{\nu}$$

$e^+e^- \rightarrow Z \rightarrow f\bar{f}$ at COM energies close to m_Z .

$$-i \frac{g^{\mu\nu} - q^\mu q^\nu \frac{1}{M_Z^2}}{q^2 - M_Z^2 + iM_Z \Gamma_Z} \quad ; \text{ propagator}$$

↳ for a spin-1 boson

§21.3. Z production



cross-section: $\sigma \propto \frac{s}{(s - M_Z^2)^2 + M_Z^2 \Gamma_Z^2} \propto \frac{1}{\Gamma_Z^2}$

$$A^2 \propto \frac{1}{(\omega^2 - \omega_0^2)^2 + \omega^2 \Gamma^2}$$

$$\left. \begin{array}{l} m_Z \sim 91 \text{ GeV} \\ \Gamma \sim 2.45 \text{ GeV} \end{array} \right\} \rightarrow \Gamma_{\text{tot}} = \underbrace{3\Gamma_e + 3\Gamma_d + 3\Gamma_\mu}_{\text{"Total Width"}} + N\Gamma_\nu$$

↑
 $\Gamma_{\text{invisible}}$

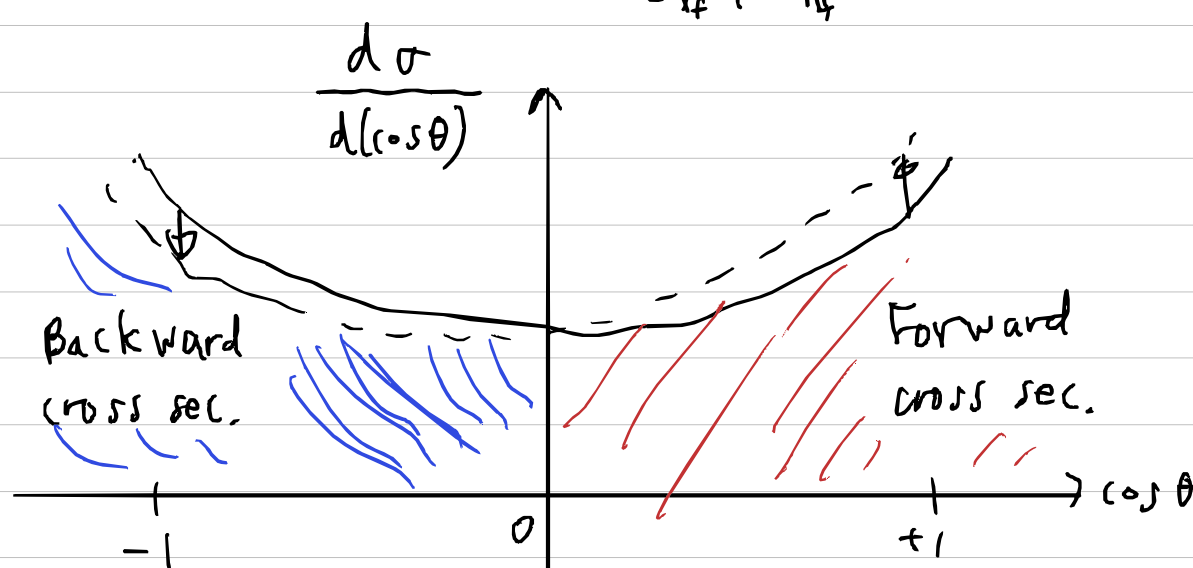
$$\begin{aligned} \Gamma_{\text{invisible}} &= \Gamma_{\text{tot}} - \Gamma_{\text{vis}} = 0.5 \text{ GeV} \\ &= N\Gamma_\nu \\ &= N(0.17 \text{ GeV}) \end{aligned}$$

$$N=3$$

§21.4. Fermion couplings

$$\frac{d\sigma}{d(\cos\theta)} \propto 1 + \cos^2\theta + 2A_e A_f \cos\theta$$

$$A_f = \frac{2C_{Vf}C_{Af}}{C_{Vf}^2 + C_{Af}^2}$$

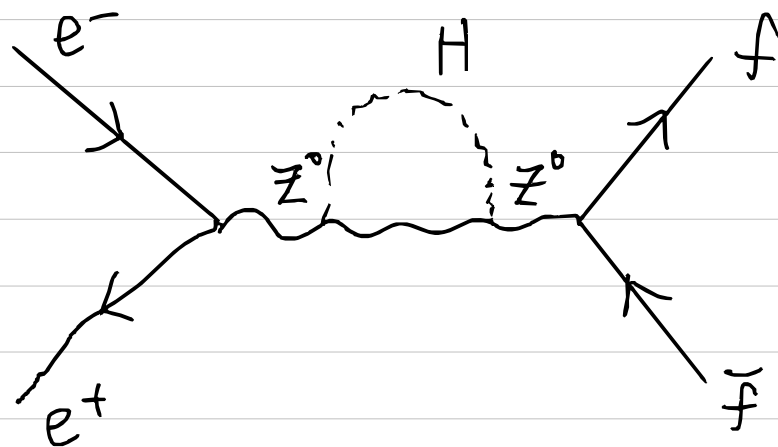
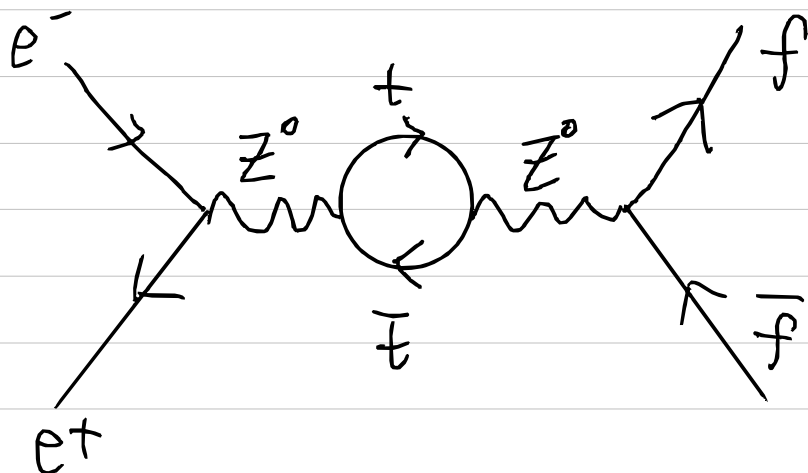
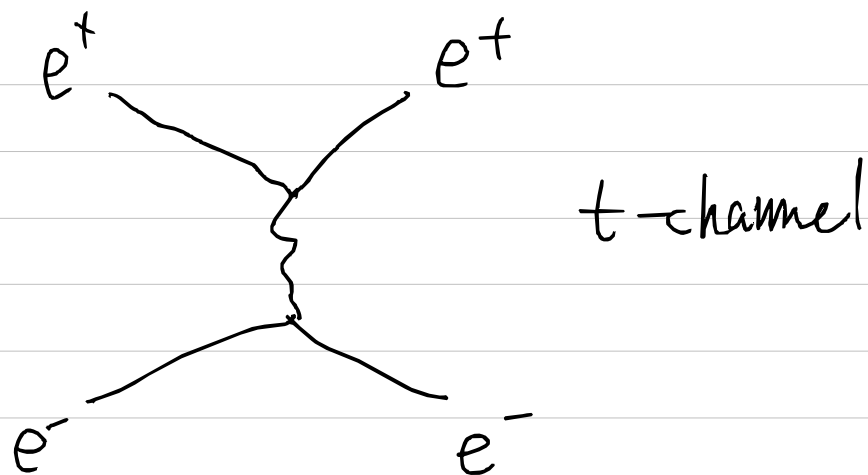


$$\sigma_F \propto \int_0^1 \frac{d\sigma}{d(\cos\theta)} d(\cos\theta) \propto \frac{4}{3} + A_e A_f$$

$$\sigma_B \propto \int_{-1}^0 \frac{d\sigma}{d(\cos\theta)} d(\cos\theta) \propto \frac{4}{3} - A_e A_f$$

$$A_{FB} = \frac{\sigma_F - \sigma_B}{\sigma_F + \sigma_B} = \frac{3}{4} A_e A_f$$

2



L22: The Higgs & mass generation

* Massive Lagrangians

• Scalar : $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$

• Fermion : $\mathcal{L} = i \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi$

• Vector : $\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{1}{2} m^2 A^\mu A_\mu$

$$\mathcal{L}_{\text{mass}} = \frac{1}{2} m^2 (A^\mu + \partial^\mu \Lambda) (A_\mu + \partial_\mu \Lambda)$$

$$= \frac{1}{2} m^2 (A^\mu A_\mu + \partial^\mu \Lambda A_\mu + A^\mu \partial_\mu \Lambda + \partial^\mu \Lambda \partial_\mu \Lambda)$$

$$\phi \rightarrow e^{-i\alpha} \phi$$

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi$$

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + i q A_\mu$$

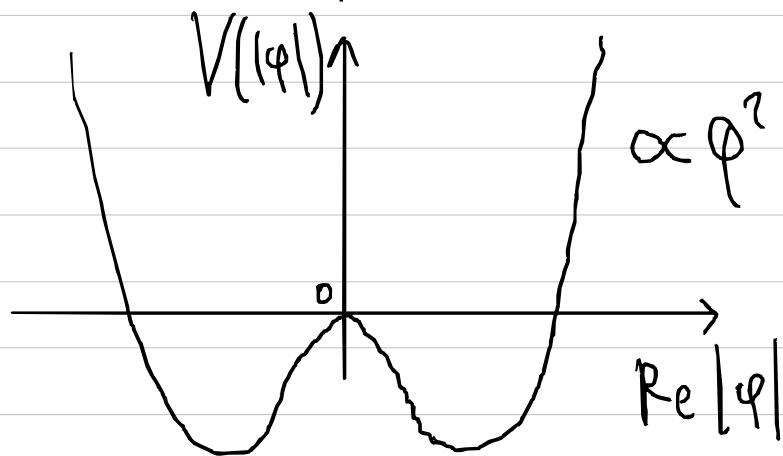
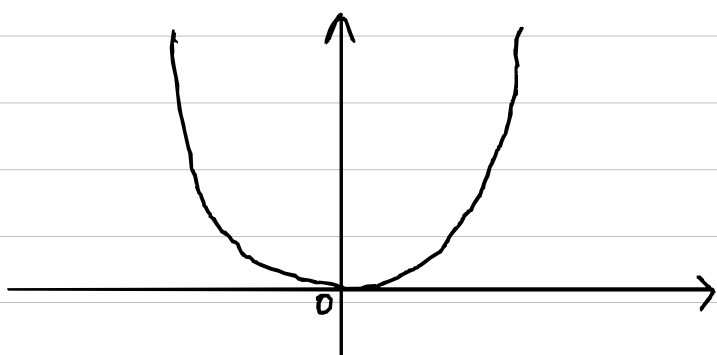
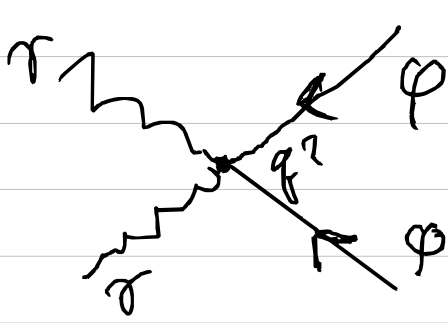
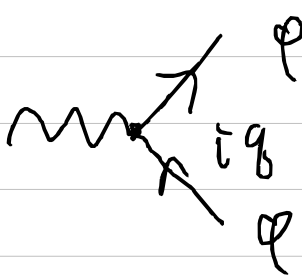
$$\mathcal{L} = (\partial_\mu \phi + i q A_\mu \phi)^* (\partial^\mu \phi + i q A^\mu \phi) - m \phi^* \phi$$

$$= \partial_\mu \phi^* \partial^\mu \phi + i q A_\mu (\partial_\mu \phi^*) \phi - i q A_\mu \phi^* \partial^\mu \phi$$

$$+ q^2 A^\mu A_\mu \phi^* \phi$$

$$= \partial_\mu \phi^* \partial^\mu \phi - i q A_\mu (\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*)$$

$$+ q^2 A^\mu A_\mu \phi^* \phi - m \phi^* \phi$$



$$V = -\mu^2 |\phi|^2$$

$$+ \frac{\lambda}{2} |\phi|^4$$

ϕ

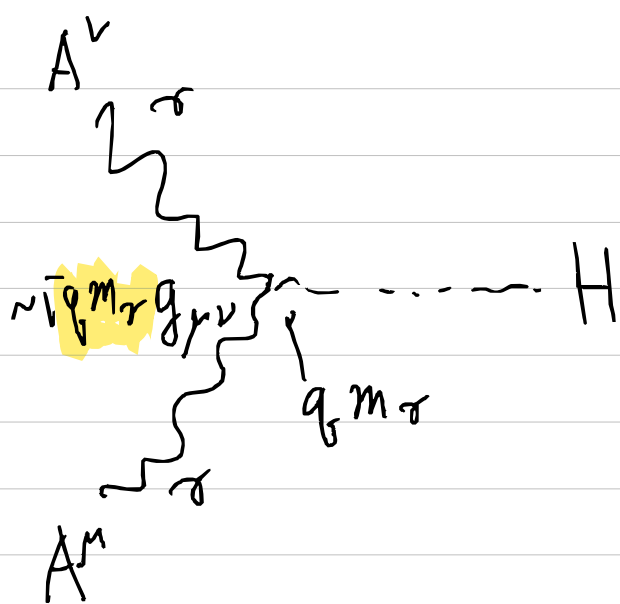
" ϕ^4 theory"

$$|\phi| = \frac{v}{\sqrt{2}} = \sqrt{\frac{\mu^2}{\lambda}} \quad \& \quad |\phi| = \frac{v+H}{\sqrt{2}}$$

$$\frac{1}{2} q^2 A^\mu A_\mu (v+H)^2 = \frac{1}{2} q^2 v^2 A^\mu A_\mu + \frac{1}{2} q^2 H^2 A^\mu A_\mu + v q^2 H A^\mu A_\mu$$

$$m_\sigma = q v$$

②



$$A^\mu \partial_\mu H_\alpha$$

$$\sim \dots H_\alpha$$

$$A^\mu \rightarrow A^\mu - \partial^\mu \Delta$$

§ 22.5. The SM case for $SU(2)$

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \quad \Phi^\dagger \Phi$$

$$M_W = \frac{g_W}{2} v$$

$$\& \quad M_Z = \frac{g_Z}{2} v = \frac{g_W v}{2 \cos \theta_W}$$

$$\hookrightarrow \therefore M_W = M_Z \cos \theta_W$$

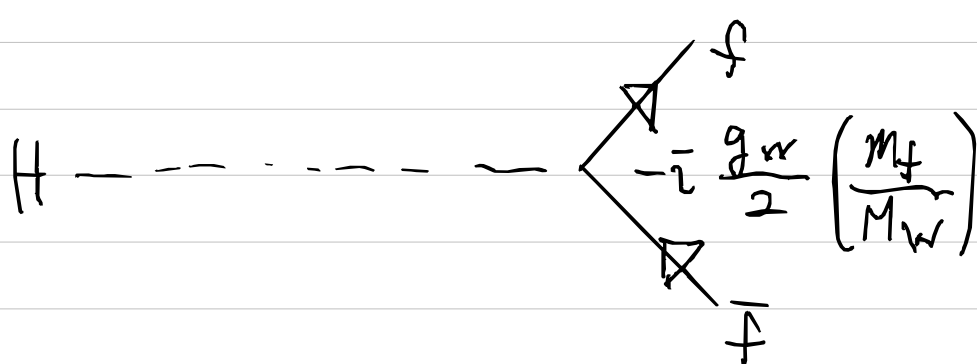
§ 22.6. Fermion masses

$$\mathcal{L} = g_f |\Phi| \bar{\psi} \psi$$

$$= g_f \frac{v + H}{\sqrt{2}} \bar{\psi} \psi$$

$$m_f = \frac{g_f v}{\sqrt{2}}$$

$$\frac{g_f}{\sqrt{2}} = \frac{m_f}{v} = \frac{g_W}{2} \frac{m_f}{M_W}$$



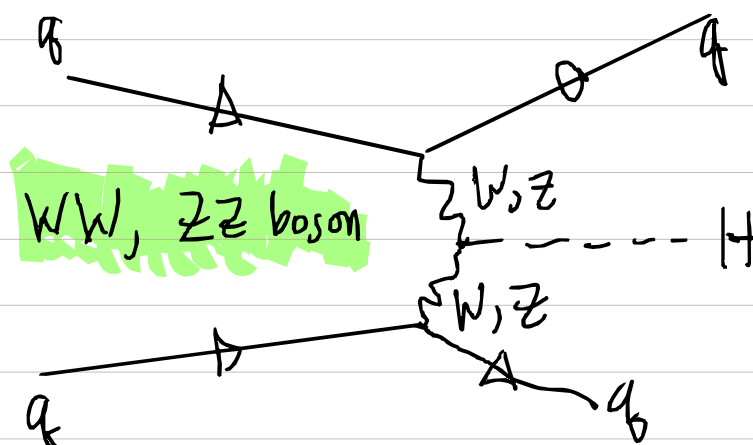
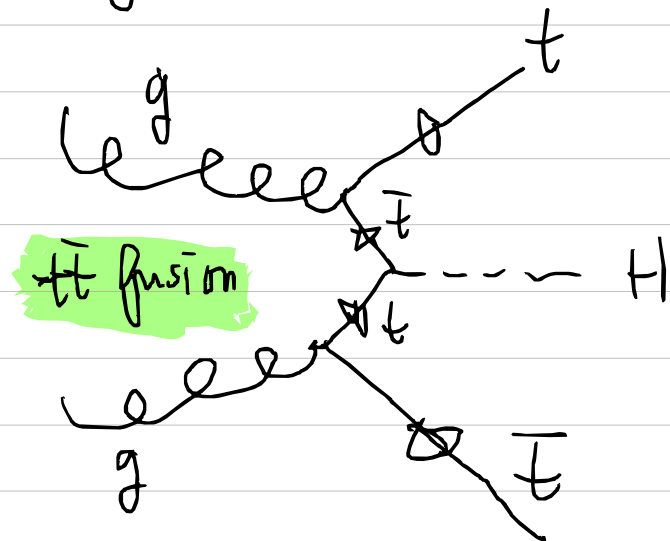
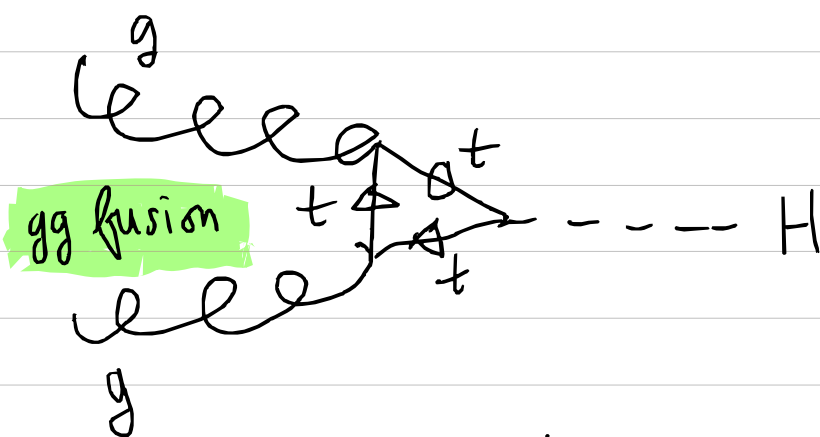
APP L23

Dr. M. McCann (1)
March 11, 2025

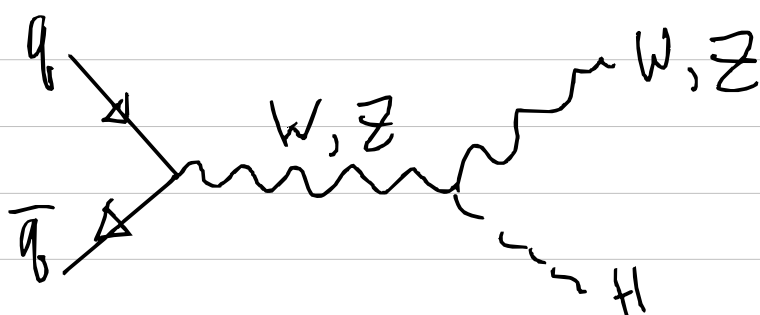
L23: Higgs discovery & properties

4 main production modes

- ggF
- VBF
- Higgsstrahlung
- $t\bar{t}$ fusion



: VBF (vector boson fusion)
(Electroweak boson fusion)



L24: Neutrinos

$$\nu_e, \nu_\mu, \nu_\tau, \quad \nu_1, \nu_2, \nu_3$$

$$\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_{12} & \sin \theta_{12} \\ -\sin \theta_{12} & \cos \theta_{12} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix}$$

$$|\nu_e\rangle = c_{12} |\nu_1\rangle + s_{12} |\nu_2\rangle$$

$$|\nu_1(t)\rangle = |\nu_1(0)\rangle e^{-i p_1^\mu x_\mu}$$

$$|\nu_2(t)\rangle = c_{12} |\nu_1\rangle e^{-i p_1^\mu x_\mu} + s_{12} |\nu_2\rangle e^{-i p_2^\mu x_\mu}$$

$$|\nu_1\rangle = c_{12} |\nu_e\rangle - s_{12} |\nu_\mu\rangle$$

$$|\nu_2\rangle = c_{12} |\nu_\mu\rangle + s_{12} |\nu_e\rangle$$

$$|\nu_e(t)\rangle = \cancel{|\nu_e\rangle} \left(c_{12}^2 e^{-i p_1^\mu x_\mu} + s_{12}^2 e^{-i p_2^\mu x_\mu} \right) + \underbrace{|\nu_\mu\rangle}_{\substack{|| \\ |}} s_{12} c_{12} \left(e^{-i p_1^\mu x_\mu} + s_{12}^2 e^{-i p_2^\mu x_\mu} \right)$$

$$P(\nu_e \rightarrow \nu_\mu) = 2 s_{12}^2 c_{12}^2 \left[1 - \cos \left((p_1 x - E_1 t) - (p_2 x - E_2 t) \right) \right]$$

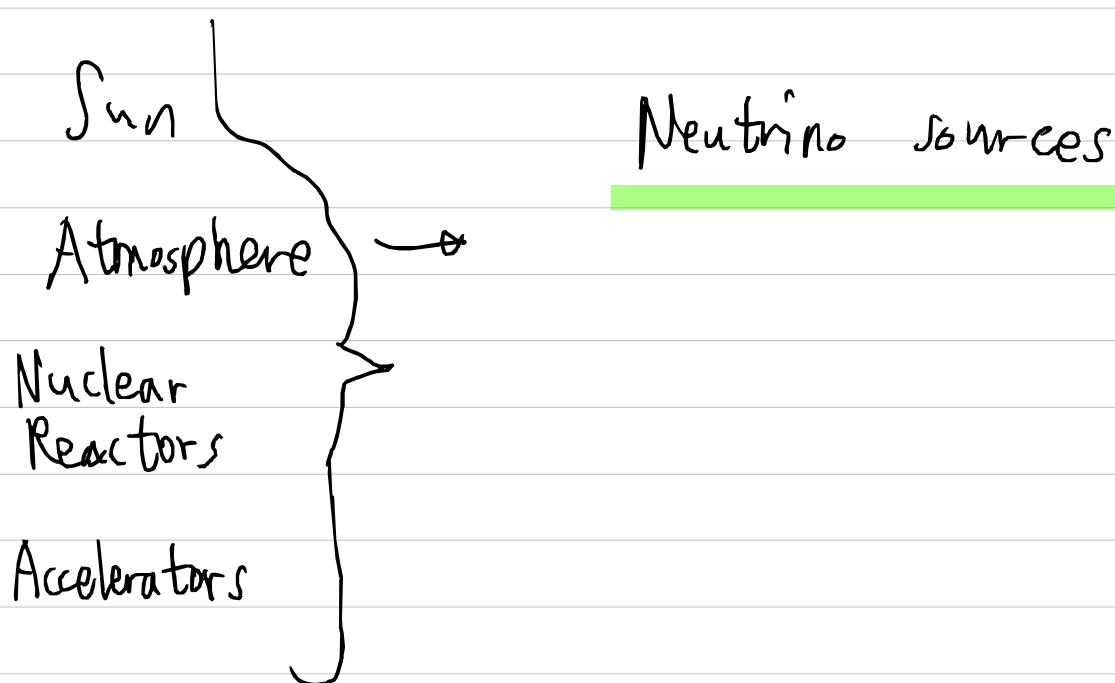
$$(p_1 - E_1) x$$

$$E = \left(p^2 + m^2 \right)^{\frac{1}{2}} \approx p + \frac{m^2}{2p}$$

②

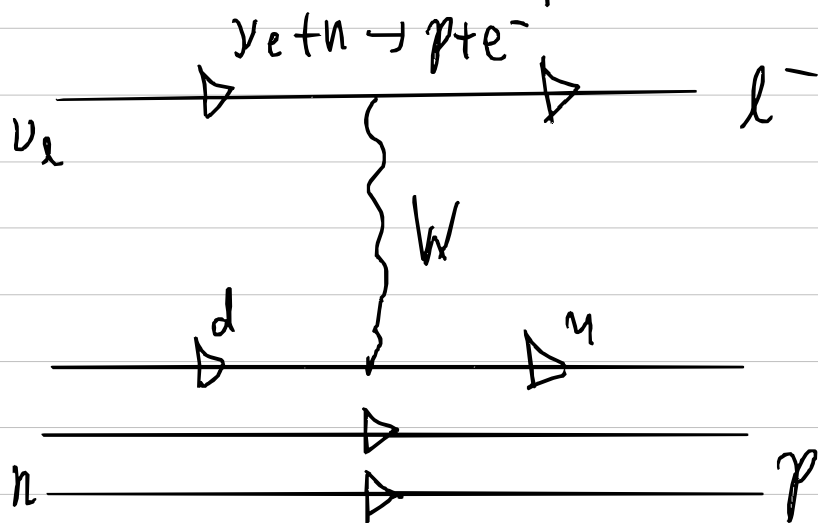
§24.3. The PMNS mixing matrix

$$V_{\text{PMNS}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{-i\delta} & 0 & c_{13} \end{pmatrix} \times \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

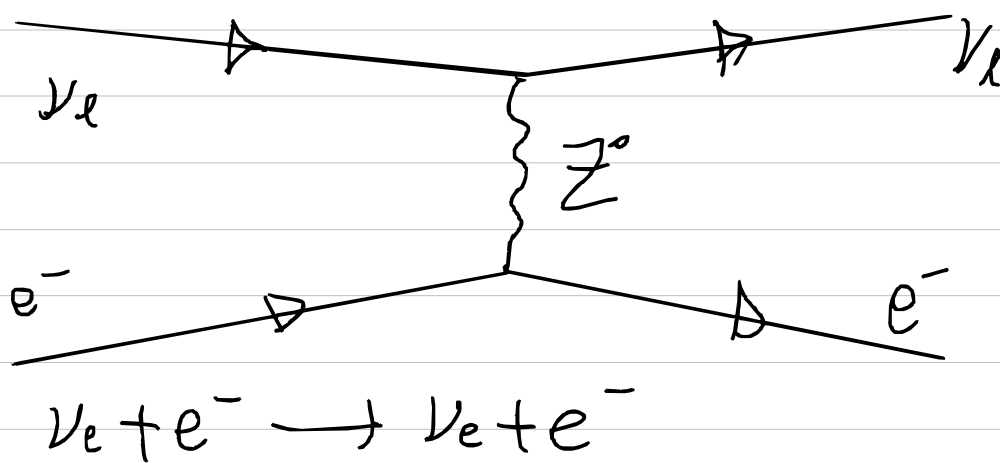


★ Detecting Neutrinos

§24.4. Neutrino Experiments

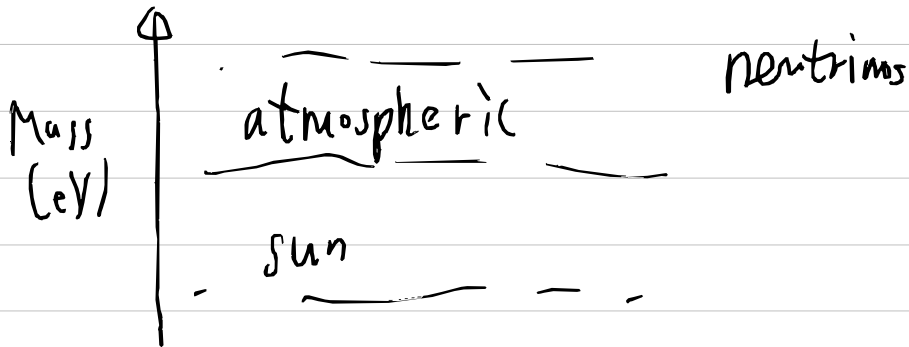


$W^\pm$ boson interactions



Z^0 boson interactions

§24.5. MSW Effect



§24.6. Accommodating ν masses

Dirac Mass $m = -m_D$ $\psi = \psi_L + \psi_R$

$$\begin{aligned}
 -m_D \bar{\psi} \psi &= -m_D (\bar{\psi}_R + \bar{\psi}_L) (\psi_R + \psi_L) \\
 &= -m_D (\bar{\psi}_R \psi_L + \bar{\psi}_L \psi_R)
 \end{aligned}$$

* Majorana-type mass $\rightarrow$ couples Majorana terms together.

$$\frac{m_M}{2} \overline{(\psi_L^M)^c} \psi_L^M \rightarrow \text{BSM physics}$$

"Why $m_\nu \sim 0$?" $\rightarrow$ "Seesaw mechanism"

Rewatch!

L25: Beyond the SM (BSM)

The SM $\rightarrow SU(3)_c \times SU(2)_L \times U(1)_Y$
 $\downarrow$ gauge theory

Very successful $\rightarrow$ [Theory $\leftrightarrow$ Measurement]
 Very accurate $\rightarrow$ Very good agreement

The difference is only 1 part in a trillion
 $= 10^{-12}$.

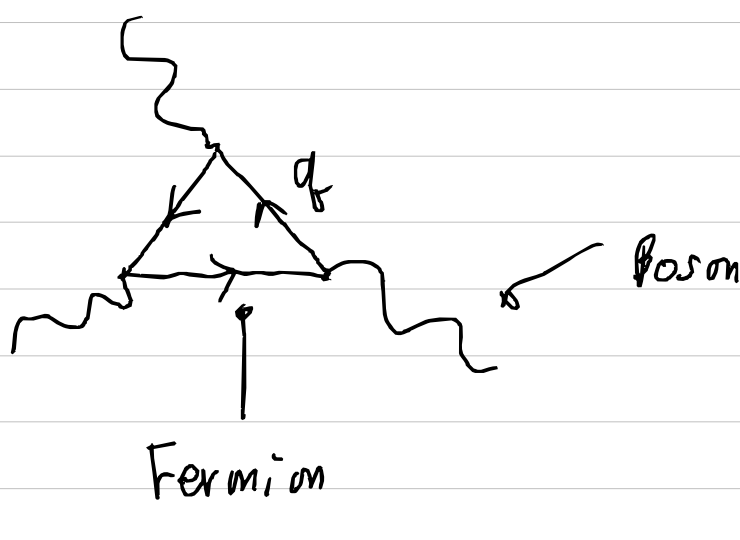
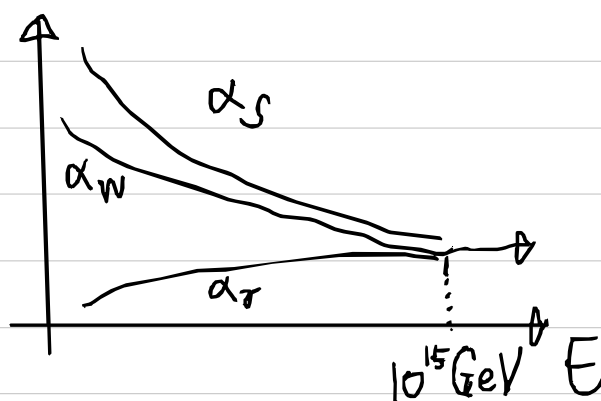
$$F^{\mu\nu} F_{\mu\nu}$$

$$- \epsilon^{\alpha\beta\mu\nu} F^{\alpha\beta} F_{\mu\nu}$$

$\uparrow$
anti-symm. tensor

The SM has 26 parameters.

★ GUT



We need the strong force in order for the EW force to make any sense.

At some point, the SF & EW force are merged together, & they split up later on.

$$★ \text{GUT} : G \supset SU(3) \times SU(2)_L \times U(1)$$

$\downarrow$
 The En. scale at which this unification occurs
 $= \Lambda_{\text{GUT}}$ (or M_{GUT}) = GUT scale.

(2)

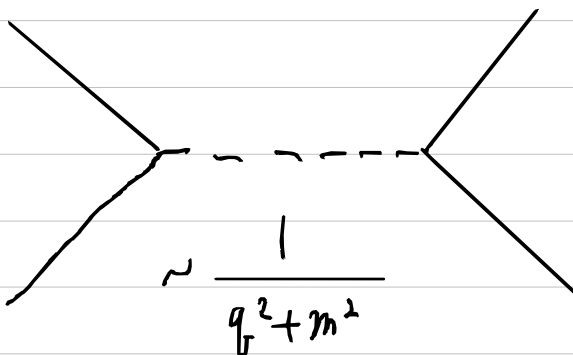
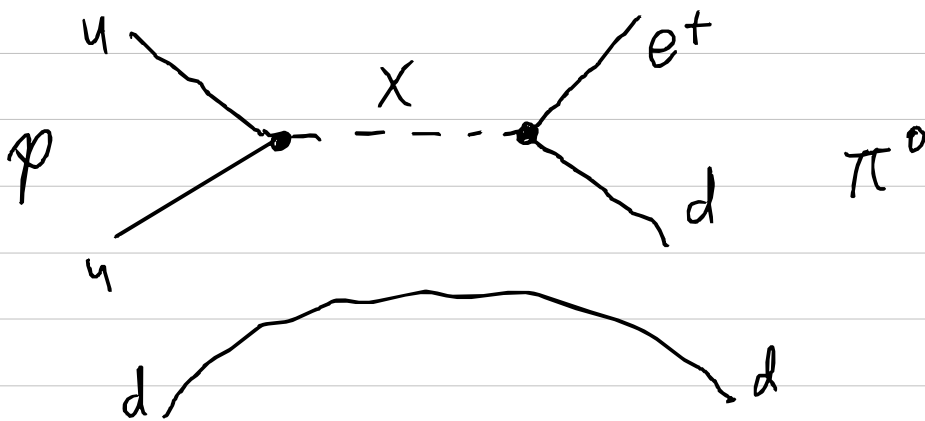
★ $SU(5) \rightarrow$ the smallest grp that contains
 $SU(3) \times SU(2) \times U(1)$.

could be decomposed into $SU(3) \times SU(2) \times U(1)$.

Multiplet

$$SU(N) \rightarrow N^2 - 1$$

★ Leptoquark : proton decay



$$\sim \left| \frac{g^2}{m^2} \right|$$

APP L26

Dr. M. McCann ①
March 14, 2025

L26: BSM II "Dark Matter"

- Direct Detection Measurements: "LZ experiment"
- Indirect Detection Measurements: "Fermi-LAT"

Brookhaven result 

 Fermilab
result


SM
prediction


Experiment
Avg.

$a_\mu \times 10^9 - 1165900$

"Anomalous magnetic moment of the muon (a_μ)"

$$\hbar = c = 1$$

z -axis $\rightarrow$ particle's motion aligned.

$$\#1. (E, \underline{p}) \longleftrightarrow (t, \underline{r})$$

$\rightarrow$ means the L.T.s for $(E, \underline{p})$ are equivalent to those for $(t, \underline{r})$.

For a particle at rest, the tot. En. is due to the mass:

$$E = m$$

(a) particle w/ spd. β

$$E_{\text{tot.}} = E = \gamma m$$

$$m_{\text{tm.}} = p = \gamma m \beta$$

$$\text{where } \gamma = \frac{1}{\sqrt{1-\beta^2}}$$

S(a) L.T.s along the z -ax. for t & z :

$$t' = \gamma(t - \beta z) \quad \& \quad z' = \gamma(z - \beta t)$$

$\hookrightarrow$ the same for E & p_z

$$E' = \gamma(E - \beta p_z) \quad \& \quad p'_z = \gamma(p_z - \beta E)$$

$$t \rightarrow E \quad \& \quad z \rightarrow p_z$$

In the rest frm. (R), the val.s are $E_R = m$

$$\& \quad p_{R,x} = p_{R,y} = p_{R,z} = 0.$$

Boosting the observer by $-\beta$ so that the particle appears to be moving by $+\beta$ along the z -axis, then the boosted val.s are given:

$$E = \gamma(E_R + \beta p_{R,z}) = \gamma m$$

$$\& \quad p_z = \gamma(p_{R,z} + \beta E_R) = \gamma \beta m.$$

The motion in the x & y dir.s $\rightarrow$ not affected by a L.T. along the z -dir., so the boosted mtm in those dir.s are $p_x = p_y = 0$.

$\therefore$ The magnitude of the mtm: $p = \gamma m \beta$.

(2)

(b) Invert these eqns to give β & γ in terms of $m, p,$ & E .

S(b) Noting: $\gamma = \gamma m \beta$ gives $p = E \beta$,

then inverting these eqs simply gives

$$\gamma = \frac{E}{m} \quad \& \quad \beta = \frac{p}{E}.$$

$$E = \gamma m \quad \& \quad p = E \beta.$$

(c) Show: the quantity $E^2 - p^2$ is an invariant under a L.T..

S(c) The init. val. of $E_R^2 - p_R^2$:

$$E^2 - p^2 = m^2 - 0^2 = m^2.$$

The boosted val.:

$$E^2 - p^2 = \gamma^2 m^2 - \gamma^2 m^2 \beta^2 = \gamma^2 m^2 (1 - \beta^2).$$

Since $\gamma^2 = \frac{1}{1 - \beta^2}$, then $E^2 - p^2 = m^2$.

↳ So, this combination is invariant.

(3)

(d) To what physical quantity does

 $E^2 - p^2$ correspond?(5)(d) Physically, $E^2 - p^2$ corresponds to the square of the particle mass.

#2. [The calculation of the kinematics for a 2-body decay.]

Consider the decay of a particle X to 2 particles c & d in the rest frame of X .

(a) Using only En. & mom conservation, show that

$$E_c = \frac{m_X^2 + m_c^2 - m_d^2}{2m_X} \quad \text{\& similarly for } E_d.$$

(b) Check En. is conserved by this formulae.

(c) [The val. of the En. if the final state particles are the same (or antiparticles of each other)]

(5) #2. ^(a) Use 4-vec.s, in the rest frame of X (the CM frame), we have:

$$X^\mu = (E_X, \underline{p}_X) = (m_X, 0) \quad \& \quad c^\mu = (E_c, \underline{p}_c) \\ \& \quad d^\mu = (E_d, \underline{p}_d)$$

4-mom. cons. (i.e., E cons. & p cons.) implies:

$$X^\mu = c^\mu + d^\mu, \text{ i.e., } \begin{cases} m_X = E_c + E_d \\ \& \underline{p}_d = -\underline{p}_c. \end{cases}$$

$$\therefore d^\mu = X^\mu - c^\mu$$

& Taking the 4-vec. dot products w/ itself

(squaring each side)

$$\begin{aligned} d^\mu d_\mu &= (X^\mu - c^\mu)(X_\mu - c_\mu) \\ &= X^\mu X_\mu + c^\mu c_\mu - 2X^\mu c_\mu \\ &= m_X^2 + m_c^2 - 2m_X E_c \end{aligned}$$

$$\text{Recall: } d^\mu d_\mu = E_d^2 - p_d^2 = m_d^2 \quad \text{\& "on-shell"}$$

= LHS

$$\& \quad m_d^2 = m_X^2 + m_c^2 - 2m_X E_c$$

$$\hookrightarrow \text{Rearranging: } E_c = \frac{m_X^2 + m_c^2 - m_d^2}{2m_X}$$

Similarly, using $c^\mu = X^\mu - d^\mu$

& taking the 4-vec. I.P.s of each side:

$$c^\mu c_\mu = (X^\mu - d^\mu)(X_\mu - d_\mu) = X^\mu X_\mu + d^\mu d_\mu - 2X^\mu d_\mu$$

 $\downarrow$

$$m_c^2 = m_X^2 + m_d^2 - 2m_X E_d$$

$$E_d = \frac{m_X^2 + m_d^2 - m_c^2}{2m_X} \quad \text{\& what we'd expect from } c \leftrightarrow d. \text{ (arbitrary)}$$

(5)(b) Check En. cons. by summing these two, $E_d + E_c$:

$$\begin{aligned} E_c + E_d &= \frac{m_X^2 + m_c^2 - m_d^2}{2m_X} + \frac{m_X^2 + m_d^2 - m_c^2}{2m_X} \\ &= \frac{2m_X^2}{2m_X} = m_X \end{aligned}$$

(5)(c) If the final particles $\rightarrow$ the same, or antiparticles of e.o., then $m_c = m_d$:

$$E_c = \frac{m_X^2}{2m_X} = \frac{m_X}{2} = E_d.$$

 $\hookrightarrow$ Each particle has the available En., as expected by symmetry.

#2.

④

(d) The general expression for the magnitude of the mtm of c & d $\rightarrow$ more complicated.

Show:

$$p_c = \frac{\sqrt{m_x^4 + m_c^4 + m_d^4 - 2m_x^2 m_c^2 - 2m_x^2 m_d^2 - 2m_c^2 m_d^2}}{2m_x}$$

(e) [The val. of p_d .]

(f) [The val. of the mtm if the final particle st.s are the same (or anti-ptcl.s of e.o.).]

(g) Show: if 1 of the final st. particles is massless, e.g., $m_d = 0$, then the expression for the mtm simplifies to

$$p_c = \frac{m_x^2 - m_c^2}{2m_x}$$

S(d)

= magnitude of mtm of c.

$$p_c = \sqrt{E_c^2 - m_c^2}$$

$$= \sqrt{\frac{(m_x^2 + m_c^2 - m_d^2)^2}{4m_x^2} - m_c^2} \quad \rightarrow \text{becomes}$$

$$p_c = \frac{\sqrt{m_x^4 + m_c^4 + m_d^4 - 2m_c^2 m_d^2 + 2m_x^2 m_c^2 - 2m_x^2 m_d^2 - 4m_x^2 m_c^2}}{2m_x}$$

$$= \frac{\sqrt{m_x^4 + m_c^4 + m_d^4 - 2m_c^2 m_d^2 - 2m_x^2 m_c^2 - 2m_x^2 m_d^2}}{2m_x}$$

S(e)

symmetric for c & d

$$\rightarrow p_c = p_d$$

necessary as the magnitudes of momenta are by defn. the same in the COM.

(5)

S(f) If $m_c = m_d$, then

$$p_c = \frac{\sqrt{m_x^4 - 4m_x^2 m_c^2}}{2m_x} = \sqrt{\frac{m_x^2}{4} - m_c^2}$$

S(g) If $m_d = 0$, then

$$p_c = \frac{\sqrt{m_x^4 + m_c^4 - 2m_x^2 m_c^2}}{2m_x} = \frac{\sqrt{(m_x^2 - m_c^2)^2}}{2m_x}$$

#3. The decaying particle is rarely at rest.

Looking at decays from a moving frm., for some special cases.

Consider the decay : $X \rightarrow c + d$.

For simplicity, consider the $m_c = m_d$ case,

$X \xrightarrow{\quad}$
Let the z -ax. to be along the dir. of travel of X in our lab frm.

(a) Assume: partcl $c \rightarrow$ emitted directly along the z -ax.

By doing an explicit L.T. into a frm. moving along the z -ax. w/ spd. $-\beta$, find the p_c in the frm. where the prtcl X is moving w/ spd. β (the lab frm.).

(S)(a) Using the L.T. for the mtn.:

$$p'_c = \gamma(p_c + \beta E_c) = \gamma \left[\sqrt{\frac{m_x^2}{4} - m_c^2} + \beta \frac{m_x}{2} \right]$$

We've taken the expressions for E_c & p_c in the rest frm. from 2(c) & 2(f), resp..

(b) The cond. under which the boosted mtn of c bigger than that of X . Give the cond. in terms of β & the vel. of particle c in particle X 's rest frm β_c .

(S)(b) We want to find the cond. s.t. $p'_c > p'_x$,
 p'_x : the mtn of X in the lab frm.

Taking p'_x from #1(a):

$$p'_c > p'_x \Rightarrow p'_c > \gamma \beta m_x$$



$$\gamma \left[\sqrt{\frac{m_x^2}{4} - m_c^2} + \beta \frac{m_x}{2} \right] > \gamma \beta m_x$$

$$\cancel{\gamma} \sqrt{\frac{m_x^2}{4} - m_c^2} > \left(\cancel{\gamma} \beta m_x - \cancel{\gamma} \beta \frac{m_x}{2} \right) = \frac{\cancel{\gamma} \beta m_x}{2}$$

$$\therefore \sqrt{\frac{m_x^2}{4} - m_c^2} > \frac{\beta m_x}{2}$$

(6)

In terms of the rest frame mfm & E_c , E_c & p_c ,

this is

$$p_c > \beta E_c \quad \& \quad \frac{p_c}{E_c} = \beta_c > \beta$$

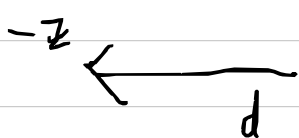
∴ The Requirement :

$$\left(\begin{array}{l} \text{spd. of particle c} \\ \text{in the rest frame of X} \end{array} \right) > \left(\begin{array}{l} \text{spd of X in the} \\ \text{moving frm., i.e.,} \\ \text{the boost vel.} \end{array} \right)$$

$$\left\{ \begin{array}{l} \beta_c > \beta \\ v_c > v \end{array} \right\}$$

↳ Can be understood using p-cons. (i.e., $p_x = p_c + p_d$ & $p'_x = p'_c + p'_d$).

When $p'_c > p'_x$, then particle d must be moving in the oppo. dir. ($p'_d < 0$) to balance mfm.



In the boosted frm., d must have a vel. along the z -ax.,

$$\beta_d > \beta$$

↑
 $\beta_d = \beta_c =$ its spd in the R.F.

(7)

Conversely, if $p'_c < p'_x$, then

the boost vel. $\beta > \beta_d$, it's going in the same
z-dir. as c.

$$\frac{d}{dz} \quad \rightarrow \quad \left(\text{if } p'_c < p'_x \right)$$

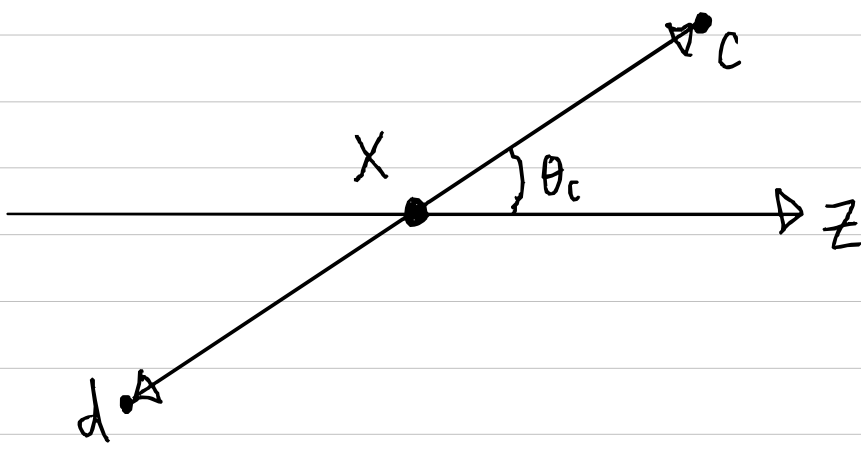
#3(c). We'll restrict the following to the simpler case where

$m_c = m_d = 0$. We now look at the case when

particle c is emitted at an angle θ_c to the z-ax.

in the R.F. of X. By doing an explicit L.T., show:

$$E'_c = \gamma \frac{m_x}{2} (1 + \beta \cos \theta) \quad : \text{En. of c in the boosted frm.}$$



For $m_c = m_d = 0$, then $p_c = E_c = \frac{m_x}{2}$.

The compo. of the mtm. along the boost dir. :

$$p_{c,z} = \frac{m_x}{2} \cos \theta_c,$$

Now : L.T. the R.F. En. & mtm.s to find the En.

in the boosted frm. :

$$\begin{aligned} E'_c &= \gamma \left(E_c + \beta p_{c,z} \right) = \gamma \left[\frac{m_x}{2} + \beta \frac{m_x}{2} \cos \theta_c \right] \\ &= \gamma \frac{m_x}{2} (1 + \beta \cos \theta_c) \end{aligned}$$

#3(d) For an unpolarized decay, there's no preferred dir. in space. In the R.F., the particles are emitted equally thruout the solid angle.

i.e. $\hookrightarrow$ they have a probab. distribution in $\cos \theta_c$ of
 $P(\cos \theta_c) d(\cos \theta_c) = \frac{1}{2} d(\cos \theta_c)$

[The distrib. of En. of particle c in the boosted frm.]

$$\rightarrow P(E'_c) dE'_c$$

S(d) $P(E'_c) dE'_c = P(\cos \theta_c) d(\cos \theta_c) = \frac{1}{2} d(\cos \theta_c)$

Can find the infinitesimal ang. elem. $d(\cos \theta_c)$

in terms of the infinitesimal En. elem.

$$dE'_c = \frac{\gamma \beta m_x}{2} d(\cos \theta_c)$$

$$\hookrightarrow P(E'_c) = \frac{1}{\gamma \beta m_x}$$

$$\therefore P(E'_c) dE'_c = \frac{1}{\gamma \beta m_x} dE'_c,$$

$$E'_{c, \max/\min} = \frac{\gamma m_x}{2} (1 \pm \beta) = \frac{E'_x}{2} (1 \pm \beta)$$

#4. T.D.

Cosmic ray muons are produced high in the atmos.
 (at 8000m, say) & travel toward Earth at very
 nearly the spd of lgt. ($0.998c$).

(a) Assuming that CM holds (i.e., no relativity),
 how far does the muon travel b/4 decaying?

$$\tau_{\mu} = 2.2 \times 10^{-6} \text{ s (muon's lifetime).}$$

Would the muons reach the ground level?

S(a)

dist. = velocity $\times$ time

APP P52

#1. Deep Inelastic Scattering (DIS)

An e^- scatters from a nucleon,

$$e^-(k) + N(P) \rightarrow e^-(k') + X(P')$$

(a) For ep DIS, in the R.F. of the proton

$$s \equiv (k + P)^2 = 2M_p E$$

E = En. of the incoming e^-
 θ = e^- scattering angle

$$q^2 \equiv (k' - k)^2 = -2EE'(1 - \cos \theta)$$

$$\nu \equiv \frac{P \cdot q}{M_p} = E - E'$$

E' = En. of the outgoing e^-

M_p = Proton Mass

(a) 4-mom:

Incoming e^- : $k = (E, \underline{k})$

Scattered e^- : $k' = (E', \underline{k}')$

Proton (at rest): $P = (M_p, \underline{0})$

Photon: $q = k - k' = (E - E', \underline{k} - \underline{k}')$

The ep COM En.:

$$\begin{aligned} s &= (P + k)^2 = P^2 + k^2 + 2P \cdot k = M_p^2 + 2M_p E \\ &= M_p (2E + M_p) \end{aligned}$$

Where the e^- mass $\rightarrow$ neglected.

In DIS, M_p alone can also be neglected.
 (proton mass)

$$\therefore s = 2M_p E$$

Then,
En. transfer :

proton (at rest) photon

$$\nu = \frac{P \cdot q}{M_p} = \frac{1}{M_p} (M_p(E - E'))$$

$$= E - E'$$

(2)

★ 4-mtm transfer squared:

$$q^2 = (k - k')^2 = k^2 + (k')^2 - 2k \cdot k'$$

$$= -2(E E' - \underline{k} \cdot \underline{k}') \quad (+ \dots)$$

$$= -2EE' + 2|\underline{k}| |\underline{k}'| \cos \theta \quad \text{I.P.}$$

$$= -2EE'(1 - \cos \theta)$$

$$|\underline{k}| |\underline{k}'| = EE'$$

(b) Consider the 4-mtm of the struck qrk before & after the interaction.

Show: the fraction of the proton's mtm carried by the struck quark : $x = -\frac{q^2}{2P \cdot q}$

5(b)(1) Let the target qrk carry a fraction ε of the proton mtm.

(2) the 4-mtm of the qrk b/4 & aft the coll:

b/4	aft
εP	$\varepsilon P + q$

The qarks $\rightarrow$ bound by QCD into hadrons.

For DIS (i.e., HE) $\rightarrow$ can assume

qarks $\rightarrow$ almost free
(asymptotic freedom)

$$m_q^2(b/4) \approx m_q^2(aft) \approx 0$$

$$\therefore (\cancel{\varepsilon P})^2 = (\varepsilon P + q)^2 = (\varepsilon P)^2 + 2\varepsilon P \cdot q + q^2$$

$$x = \frac{Q^2}{2P \cdot q} \rightarrow \downarrow = \cancel{(\varepsilon P)^2} + \varepsilon \frac{Q^2}{x} + q^2$$

$$\rightarrow \varepsilon \frac{Q^2}{x} + q^2 = 0$$

$$\rightarrow \varepsilon \frac{Q^2}{x} - Q^2 = 0$$

$$\hookrightarrow Q^2 \left(\frac{\varepsilon}{x} - 1 \right) = 0$$

$$\therefore \varepsilon = x \quad (\text{for } Q^2 \neq 0)$$

$\therefore$ the defn. of $x \rightarrow$ consistent w/ it being the fraction of the qrk mtm carried by the qrk.

(ep) $p \rightarrow 2u \& 1d$

(3)

(c) The $e^- - p$ DIS structure fn. F_2^{ep} :

$$\frac{1}{x} F_2^{ep} \equiv \sum_q \left(\frac{e_q}{e} \right)^2 q(x) = \frac{1}{q} (4u_v + d_v) + \frac{4}{3} S$$

$u_v(x)$ & $d_v(x)$: proton valence qrk distributions

$S(x)$: sea distribution

$$\frac{1}{x} F_2^{en} = \frac{1}{q} (u_v + 4d_v) + \frac{4}{3} S$$

Find: $\frac{1}{x} (F_2^{ep} - F_2^{en})$.

3d) From the defn. of DIS struc. fn.

$$\frac{1}{x} F_2^{ep} \equiv \sum_q \left(\frac{e_q}{e} \right)^2 q(x)$$

neglecting the heavy qarks (c, b, t)

$$\frac{1}{x} F_2^{ep} = \left(\frac{2}{3} \right)^2 [u(x) + \bar{u}(x)] + \left(\frac{1}{3} \right)^2 [d(x) + \bar{d}(x)] + \left(\frac{1}{3} \right)^2 [s(x) + \bar{s}(x)]$$

$e_u = 2e$

$e_s = e$

$e_d = e$

The proton has valence qarks $u_v(x)$ & $d_v(x)$

& also sea qarks $u_s(x)$, $\bar{u}_s(x)$, $d_s(x)$, $\bar{d}_s(x)$,

$s_s(x)$, $\bar{s}_s(x)$, w/

$$u(x) = u_v(x) + u_s(x)$$

$$d(x) = d_v(x) + d_s(x)$$

The sea qarks $\rightarrow$ produced as qrk-antigrk pairs

So :

$$u_s(x) = \bar{u}_s(x) = d_s(x) = \bar{d}_s(x) = s_s(x) = \bar{s}_s(x) = S(x)$$

$$\therefore \frac{1}{x} F_2^{ep} = \left(\frac{2}{3} \right)^2 [u_v + u_s + \bar{u}_s] + \left(\frac{1}{3} \right)^2 [d_v + d_s + \bar{d}_s] + \left(\frac{1}{3} \right)^2 [s_s + \bar{s}_s]$$

$$= \left(\frac{2}{3} \right)^2 [u_v + 2S] + \left(\frac{1}{3} \right)^2 [d_v + 2S] + \left(\frac{1}{3} \right)^2 [2S]$$

$$= \left(\frac{4}{9} + \frac{1}{9} + \frac{1}{9} \right) (2S) + \frac{1}{9} (4u_v + d_v)$$

$$= \frac{6^2}{9 \times 3} (2S) + //$$

$$= \frac{1}{9} (4u_v + d_v) + \frac{4}{3} S$$

$$n \rightarrow 2d \text{ \& \& } 1u$$

(4)

Similarly, $e^- - n$:

$$\frac{1}{x} F_2^{en} = \left(\frac{2}{3}\right)^2 [d + \bar{d}] + \left(\frac{1}{3}\right)^2 [u + \bar{u}] + \left(\frac{1}{3}\right)^2 [s + \bar{s}]$$

$$= \left(\frac{2}{3}\right)^2 [d_v + d_s + \bar{d}_s] + \left(\frac{1}{3}\right)^2 [u_v + u_s + \bar{u}_s] + \left(\frac{1}{3}\right)^2 [s_s + \bar{s}_s]$$

$$= \left(\frac{2}{3}\right)^2 [d_v + 2s'] + \left(\frac{1}{3}\right)^2 [u_v + 2s'] + \left(\frac{1}{3}\right)^2 [2s']$$

$$= \frac{1}{9} [4d_v + u_v] + 2s' \left[\frac{4+1+1}{9} \right]$$

$$= \frac{4s}{3} + \frac{1}{9} [4d_v + u_v] \quad \underbrace{\frac{6}{9}}_{\frac{2}{3}}$$

$$\therefore \frac{F_2^{ep}}{x} - \frac{F_2^{en}}{x} = \frac{4s}{3} + \frac{1}{9} [4u_v + d_v]$$

$$- \left(\frac{1}{9} [u_v + 4d_v] + \frac{4s}{3} \right)$$

$$= \frac{1}{9} \left[\underset{\sim}{4u_v} + \underset{\sim}{d_v} - \underset{\sim}{u_v} - \underset{\sim}{4d_v} \right]$$

$$= \frac{1}{3} (3u_v - 3d_v) = \frac{1}{3} (u_v - d_v)$$

(5)

(d) Integrating over experimental data

$$\left. \begin{aligned} \int_0^1 dx F_2^{ep}(x) &= 0.18 \\ \int_0^1 dx F_2^{en}(x) &= 0.12 \end{aligned} \right\} \rightarrow$$

Neglecting s & heavier qarks, use them to estimate the fraction of the nucleon's mfm carried by the qarks.

5(d) If we neglect the s & $\bar{s}$ qarks, then

$$F_2^{ep} = \frac{4}{9} [xu(x) + x\bar{u}(x)] + \frac{1}{9} [xd(x) + x\bar{d}(x)]$$

The mfm fraction carried by the $u + \bar{u}$ qarks:

$$f_u = \int_0^1 dx x(u + \bar{u})$$

& by the $d + \bar{d}$ qarks:

$$f_d = \int_0^1 dx x(d + \bar{d}).$$

So,

$$\int_0^1 dx F_2^{ep}(x) = \frac{4}{9} f_u + \frac{1}{9} f_d = 0.18$$

$$\int_0^1 dx F_2^{en}(x) = \frac{1}{9} f_u + \frac{4}{9} f_d = 0.12.$$

$$\therefore \left. \begin{aligned} \frac{5}{9} (f_u + f_d) &= 0.3 \\ \frac{3}{9} (f_u - f_d) &= 0.06 \end{aligned} \right\} \rightarrow \text{gives } \begin{aligned} f_u &= 0.36 \\ f_d &= 0.18 \end{aligned}$$

So, we estimate: the mfm fraction carried by all the qarks & anti-qarks:

$$f_u + f_d \approx 0.54.$$

leaves the gluons w/ mfm fraction

$$f_g = 1 - (f_u + f_d) \approx 0.46.$$

(6)

#2. In the general case of any op. w/ only 2 distinct e-val.s, projection op.s can be formed.

(a) Consider an op. $\hat{A}$

— several e-st.s ψ_i w/ e-val.s λ_i

$$\rightarrow \hat{A} \psi_i = \lambda_i \psi_i$$

The e-st.s form a complete set.

* Any st. can be decomposed into a sum of e-st.s :

$$\Psi = \sum_i \alpha_i \psi_i$$

Show : applying the combi. $(\lambda_1 - \hat{A})$ to the gen. st. Ψ removes the e-state ψ_1 from this sum.

S(a) Applying the op. combination to Ψ :

$$(\lambda_1 - \hat{A}) \Psi = \sum_i \alpha_i (\lambda_1 - \hat{A}) \psi_i$$

$$\downarrow$$

$$\hat{A} \psi_i = \lambda_i \psi_i$$

$$\text{so : } (\lambda_1 - \hat{A}) \Psi = \sum_i \alpha_i (\lambda_1 - \lambda_i) \psi_i$$

(b) Consider : the case where $\hat{A}$ has only 2 distinct e-val.s.

Show : the projection op.s

$$\hat{P}_1 = \frac{\lambda_2 - \hat{A}}{\lambda_2 - \lambda_1} \quad \& \quad \hat{P}_2 = \frac{\lambda_1 - \hat{A}}{\lambda_1 - \lambda_2} .$$

S(b) W/ only 2 e-val.s ,

$$(\lambda_1 - \hat{A}) \Psi = \sum_i \alpha_i (\lambda_1 - \lambda_i) \psi_i \quad \rightarrow \left(\begin{array}{l} i=2 \\ 2 \text{ e-val.s} \end{array} \right)$$

becomes

$$(\lambda_1 - \hat{A}) \Psi = \alpha_2 (\lambda_1 - \lambda_2) \psi_2$$

so, $\frac{\lambda_1 - \hat{A}}{\lambda_1 - \lambda_2} \Psi = P_2 \Psi = \alpha_2 \psi_2$

$\therefore P_2$ removes ψ_1 , & leaves the coeff. of ψ_2 unchanged.

Similarly, $\frac{\lambda_2 - \hat{A}}{\lambda_2 - \lambda_1} \Psi = P_1 \Psi = \alpha_1 \psi_1 .$

(c) Define a new op. from $\hat{A}$:

$$\hat{B} = \frac{2\hat{A} - (\lambda_1 + \lambda_2)}{(\lambda_1 - \lambda_2)}$$

Show: $\hat{B}$ has the same e-sts as $\hat{A}$,

but w/ e-val.s of ± 1 , i.e.,

$$\hat{B}\psi_1 = \psi_1 \quad \& \quad \hat{B}\psi_2 = -\psi_2.$$

S(c) ① apply $\hat{B}$ to ψ_1 .

$$\begin{aligned} \hat{B}\psi_1 &= \frac{2\hat{A}\psi_1 - (\lambda_1 + \lambda_2)\psi_1}{\lambda_1 - \lambda_2} = \frac{2\lambda_1\psi_1 - (\lambda_1 + \lambda_2)\psi_1}{\lambda_1 - \lambda_2} \\ &= \frac{\lambda_1\psi_1 - \lambda_2\psi_1}{\lambda_1 - \lambda_2} = \psi_1 \end{aligned}$$

② apply $\hat{B}$ to ψ_2 .

$$\begin{aligned} \hat{B}\psi_2 &= \frac{2\hat{A}\psi_2 - (\lambda_1 + \lambda_2)\psi_2}{\lambda_1 - \lambda_2} = \frac{2\lambda_2\psi_2 - (\lambda_1 + \lambda_2)\psi_2}{\lambda_1 - \lambda_2} \\ &= \frac{\lambda_2\psi_2 - \lambda_1\psi_2}{\lambda_1 - \lambda_2} = -\psi_2 \end{aligned}$$

$\therefore \hat{B}$ has the same e-sts as $\hat{A}$, but w/ e-val.s of ± 1 .

(9)

(d) Consider the general st.: $\Psi = \alpha_1 \psi_1 + \alpha_2 \psi_2$.

Show: $\hat{B}^2 = I$, i.e., $\hat{B}^2 \Psi = \Psi$.

$$\begin{aligned} \text{S(d)} \quad \hat{B}^2 \Psi &= \hat{B} \hat{B} (\alpha_1 \psi_1 + \alpha_2 \psi_2) = \hat{B} (\alpha_1 \psi_1 - \alpha_2 \psi_2) \\ &= \alpha_1 \psi_1 + \alpha_2 \psi_2 = \Psi. \end{aligned}$$

Since this holds for

$\forall \text{ st. } \Psi,$

then $\hat{B}^2 = I$.

We know, from (c), that
 $\hat{B} \psi_1 = \psi_1$ & $\hat{B} \psi_2 = -\psi_2$.

(e) Show: in terms of $\hat{B}$, the projec. ops:

$$P_1 = \frac{1}{2}(I + \hat{B}) \quad , \quad P_2 = \frac{1}{2}(I - \hat{B})$$

(e) Solving for $\hat{A}$ in terms of $\hat{B}$:

$$\hat{A} = \frac{1}{2} [(\lambda_1 + \lambda_2) + \hat{B}(\lambda_1 - \lambda_2)]$$

$$\text{from } \hat{B} = \frac{2\hat{A} - (\lambda_1 + \lambda_2)}{(\lambda_1 - \lambda_2)}$$

$\therefore$

$$\hat{P}_1 = \frac{\lambda_2 - \hat{A}}{\lambda_2 - \lambda_1} = \frac{\frac{1}{2}\lambda_2 - \frac{1}{2}\lambda_1 - \cancel{\frac{1}{2}\lambda_2} - \frac{1}{2}\hat{B}(\lambda_1 - \lambda_2)}{\lambda_2 - \lambda_1}$$

$$= \frac{(\lambda_2 - \lambda_1) - \hat{B}(\lambda_1 - \lambda_2)}{2(\lambda_2 - \lambda_1)} = \frac{1}{2}(I + \hat{B})$$

$$\hat{P}_2 = \frac{\lambda_1 - \hat{A}}{\lambda_1 - \lambda_2} = \frac{\frac{1}{2}\lambda_1 - \cancel{\frac{1}{2}\lambda_1} - \frac{1}{2}\lambda_2 - \frac{1}{2}\hat{B}(\lambda_1 - \lambda_2)}{\lambda_1 - \lambda_2}$$

$$= \frac{(\lambda_1 - \lambda_2) - \hat{B}(\lambda_1 - \lambda_2)}{2(\lambda_1 - \lambda_2)} = \frac{1}{2}(I - \hat{B})$$

Cross-check that these do not act as projec. ops:

$$P_1 \Psi = \frac{1}{2}(I + \hat{B})(\alpha_1 \psi_1 + \alpha_2 \psi_2)$$

$$= \frac{1}{2} [(\alpha_1 \psi_1 + \alpha_2 \psi_2) + (\alpha_1 \psi_1 - \alpha_2 \psi_2)] = \alpha_1 \psi_1$$

&

$$P_2 \Psi = \frac{1}{2}(I - \hat{B})(\alpha_1 \psi_1 + \alpha_2 \psi_2)$$

$$= \frac{1}{2} [(\alpha_1 \psi_1 + \alpha_2 \psi_2) - (\alpha_1 \psi_1 - \alpha_2 \psi_2)] = \alpha_2 \psi_2$$

$\rightarrow \therefore P_{1,2} \rightarrow \text{act as proj. ops.}$

(10)

(f)

Show: the $P_{1,2}$ have the other required properties for projec. op.s, i.e.

$$P_1^2 = P_1, \quad P_2^2 = P_2, \quad P_1 P_2 = P_2 P_1 = 0,$$

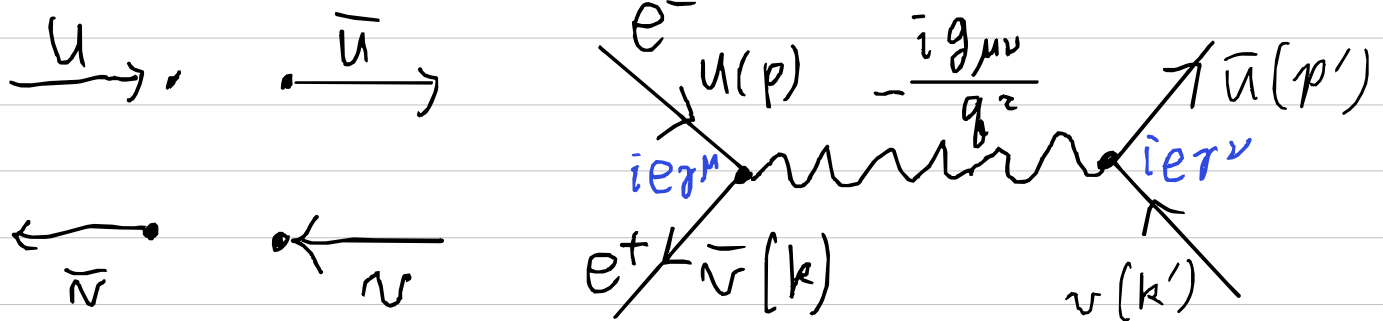
$$P_1 + P_2 = I.$$

APP RF4

Feb 4, 2025 ^①

#1.

$$e^-(p) e^+(k) \rightarrow \tau^-(p') \tau^+(k')$$



$$-iM = \left(\bar{u}_{p'} i e \gamma^\nu v_{k'} \right) \left(\bar{v}_k i e \gamma^\mu u_p \right)$$

Particle	Qrk content	Mass (MeV)	Width (MeV)
$\pi^\pm$	$u\bar{d}/\bar{u}d$	139.6	~ 0
ρ^0	$(u\bar{u}-d\bar{d})/\sqrt{2}$	775.3	149.1
$D^\pm$	$c\bar{d}/\bar{c}d$	1869.7	~ 0
J/ψ	$c\bar{c}$	3096.9	92.9×10^{-3}

#1. The ρ^0 is a $J^{PC} = 1^{--}$ state, which we will take as $u\bar{u}$

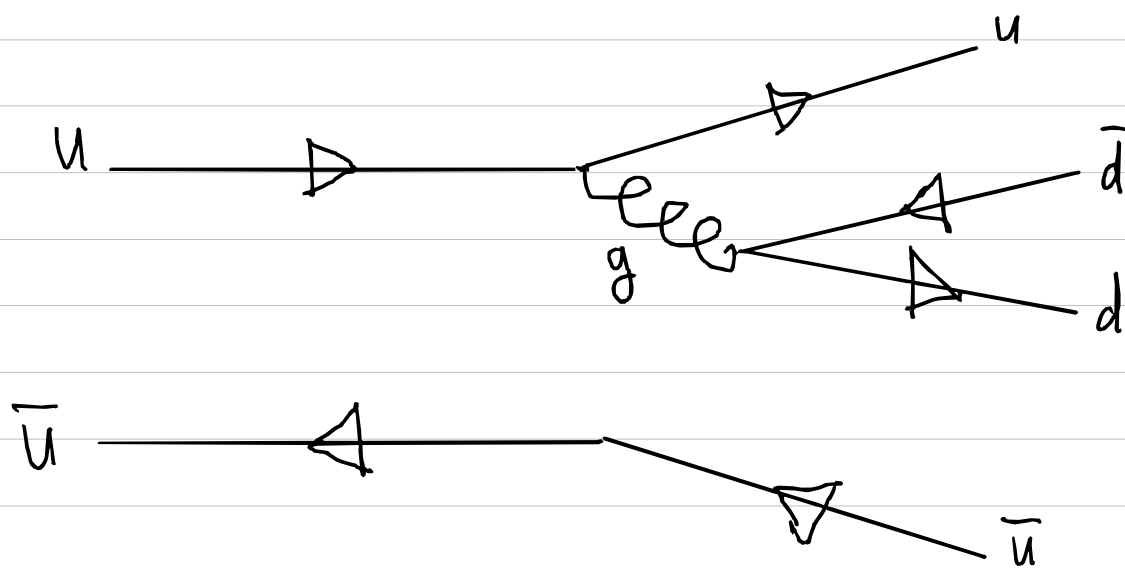
↓
Tot. A.M. = 1 & Parity = -1
& Charge conj. = -1.

[NB: it's really: $(u\bar{u}-d\bar{d})/\sqrt{2}$, but that doesn't change our argument as $m_u \approx m_d$].

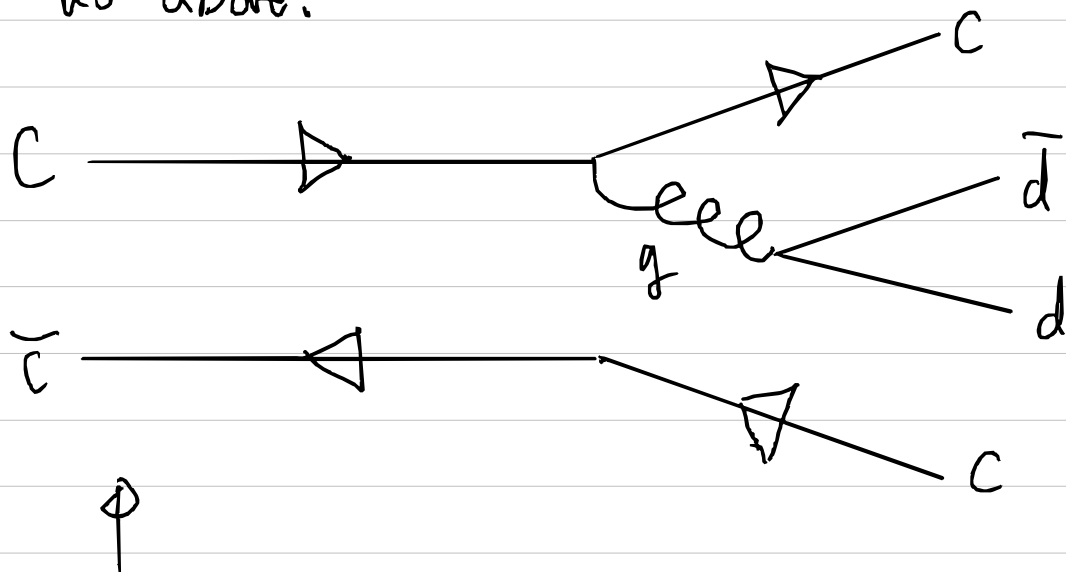
Draw a F.D. for the decay $\rho^0 \rightarrow \pi^+\pi^-$
(dominant decay)

e.g., $\mathcal{B}(\rho^0 \rightarrow e^+e^-) \sim 5 \times 10^{-5}$.

(S) #1. The lowest order F.D.: $\rho^0 \rightarrow \pi^+\pi^-$



#2. The J/ψ is a $J^{PC} = 1^{--}$ state, containing $c\bar{c}$. It can't decay via the same diag. as above.



The decay $J/\psi \rightarrow D^+D^-$.

The mass of two D mesons $2m_{D^\pm} = 3739.4 \text{ MeV}$.

This is heavier than the J/ψ mass, so this decay is kinematically disallowed.

$$2m_{D^\pm} > m_{J/\psi} \approx 3097 \text{ MeV}/c^2$$

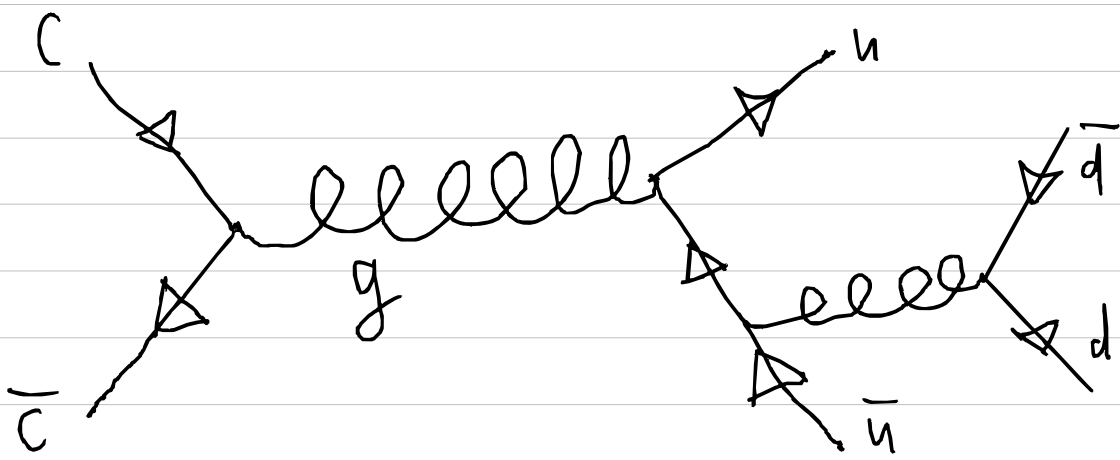
(2)

#3. What color state must the J/ψ be in?

Comment on the prospects of the J/ψ decaying via an annihilation diag. mediated by a single gluon.

(J) #3. J/ψ must be in a colorless state $\frac{r\bar{r} + b\bar{b} + g\bar{g}}{\sqrt{3}}$.

The diag. for the J/ψ annihilating to a single gluon:



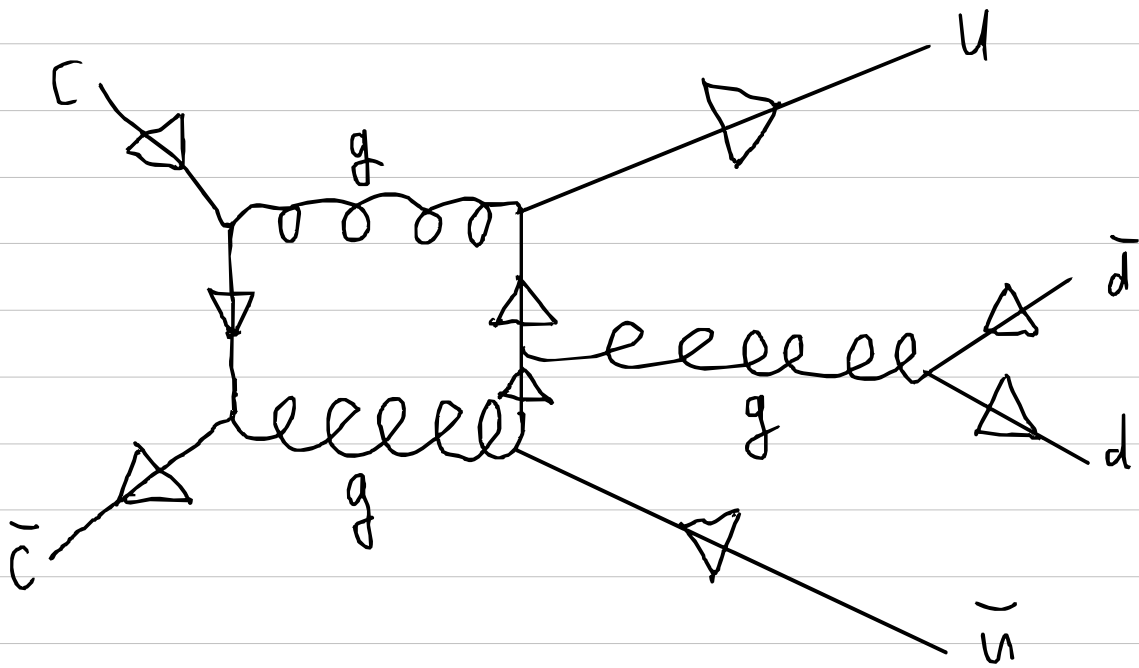
The init. & final sts must be in the colorless combination $\frac{1}{\sqrt{3}} (r\bar{r} + b\bar{b} + g\bar{g})$.

As color is conserved, the gluon also needs to be this combination, but there's no colorless gluon to mediate the interaction.

$\therefore$ this decay cannot proceed.

#4. What about the prospects of the J/ψ decaying via an annihilation diag. mediated by 2 gluons.

(J) (#4) If the J/ψ annihilates to 2 gluons, the F.D. is

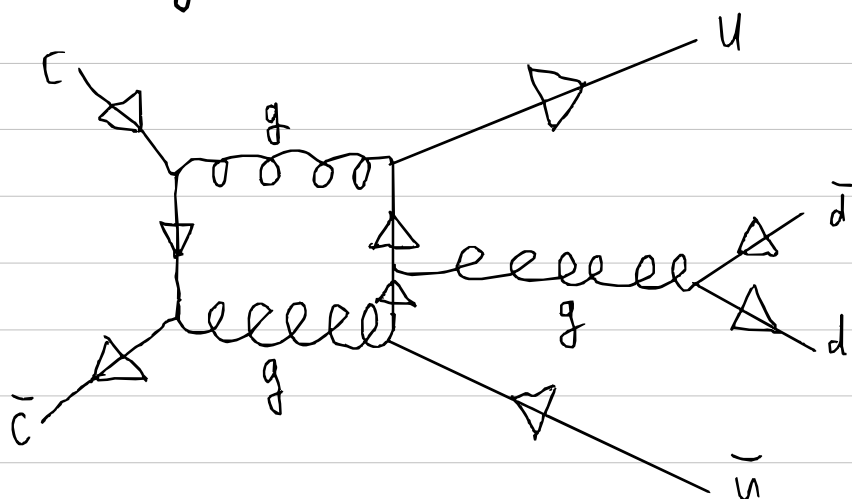


The J^{PC} of the J/ψ is 1^{--} , all the qtm num.s must be conserved in a strong decay.

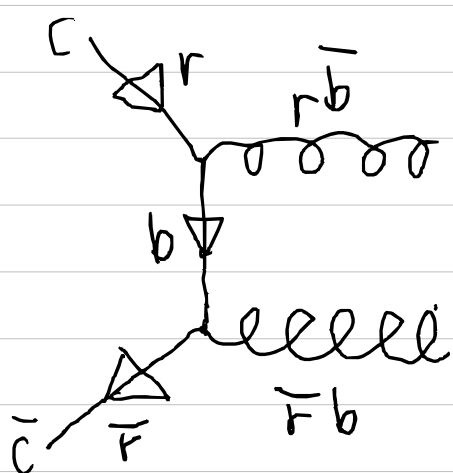
A sys. of 2 gluons is a $C = +1$ state, so this doesn't conserve C . Strictly speaking, gluons aren't necessarily an eigenstate of C .

(3)

Adding color to the F.D.:



(EX) For a $r\bar{r}$ state:



To get a colorless state,
the 2 gluons must be charge
conjugates of each other,
i.e., a st. like

$$|g\rangle |\hat{C}g\rangle + |\hat{C}g\rangle |g\rangle,$$

which, if you apply C to it, gives

$$\hat{C}G = \hat{C}(|g\rangle |\hat{C}g\rangle + |\hat{C}g\rangle |g\rangle)$$

$$= |\hat{C}g\rangle |\hat{C}\hat{C}g\rangle + |\hat{C}\hat{C}g\rangle |\hat{C}g\rangle$$

$$= |\hat{C}g\rangle |g\rangle + |g\rangle |\hat{C}g\rangle = |g\rangle |\hat{C}g\rangle + |\hat{C}g\rangle |g\rangle$$

$$= +G$$

As C can't be conserved, this decay cannot proceed.

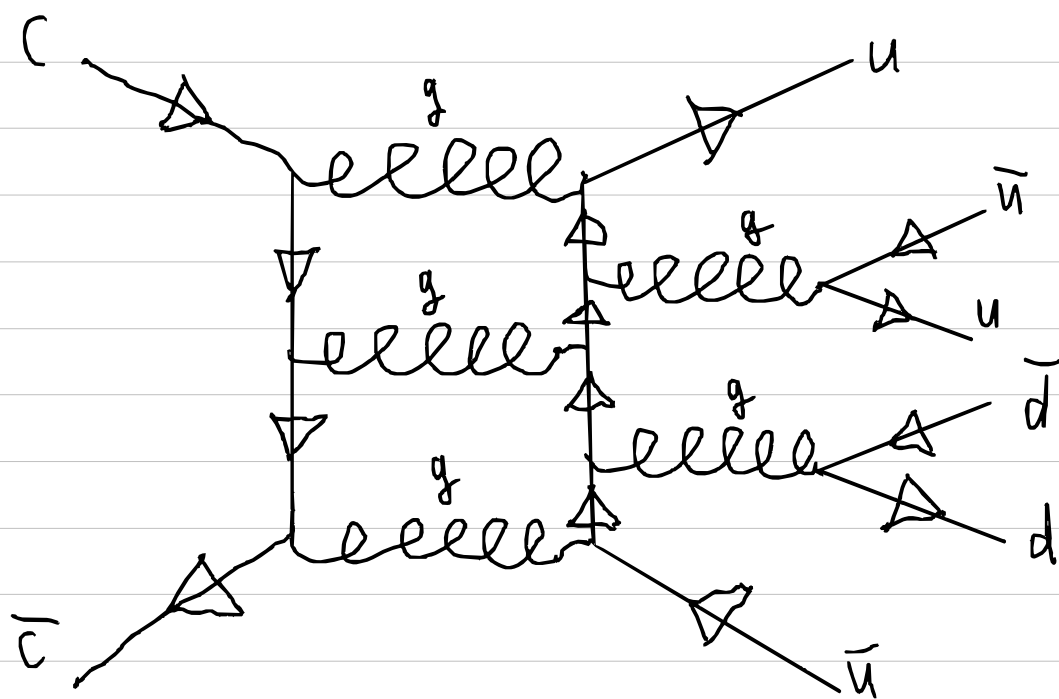
(4)

#5. The strong coupling is $\alpha_s(0.7 \text{ GeV}) \sim 0.5$

& $\alpha_s(3 \text{ GeV}) \sim 0.25$.

Use this to explain the difference in widths of the ρ^0 & J/ψ .

(S)#5. The lowest order QCD diag. I can draw involves 3 gluons, e.g.,



N.B.: We don't really need to consider the gluons emitted as part of the hadronization process of the outgoing qrk.s.

Here, we have 3 gluons, so the width will go as

$$\Gamma_{J/\psi} \propto \left(\alpha_s(m_{J/\psi}) \right)^6$$

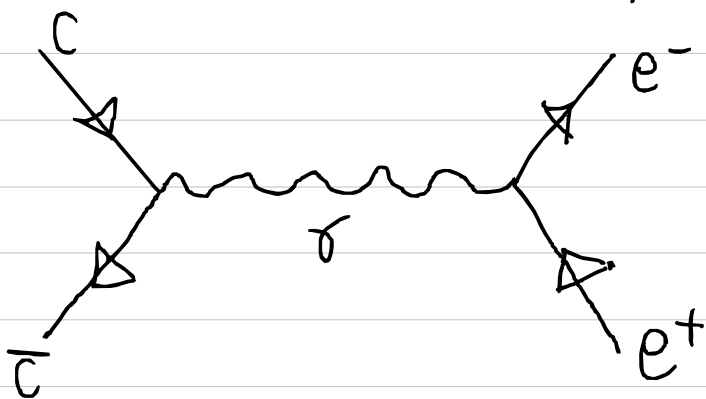
For the ρ^0 , the width will go as $\Gamma_{\rho^0} \propto \left(\alpha_s(m_{\rho^0}) \right)^2$

$$\text{so } \frac{\Gamma_{\rho^0}}{\Gamma_{J/\psi}} \sim \frac{(0.5)^2}{(0.25)^6} \sim 1000, \text{ which is roughly what we see.}$$

#6. Estimate the relative order of magnitude of the $J/\psi \rightarrow e^+e^-$ branching fraction.

(S)(#6.) The decay $J/\psi \rightarrow e^+e^-$ would be an EM process.

Annihilation w/ a single photon.



This comes in w/ an α^2 . One would, again roughly,

we expect
$$\frac{\Gamma_{e^+e^-}}{\Gamma_{ggg}} = \frac{\left(\frac{2}{3}\right)^2 \alpha^2}{\alpha_s^6} \approx \frac{4}{9} \frac{4^6}{(137)^2} \approx 0.1.$$

The experimentally observed val. :
$$\frac{5.97\%}{64.1\%} \approx 0.1,$$

which is in surprisingly good agreement w/ our estimate given we're comparing loop diag.s to tree diag.s. It's often seen that estimates based on the relative couplings will yield results at the correct order of magnitude.

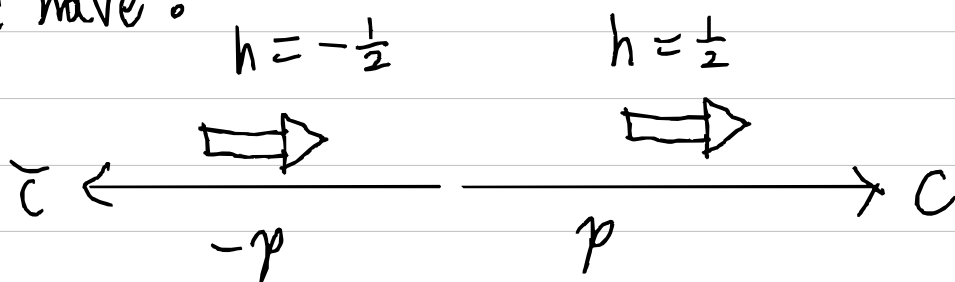
Comparing this to the ρ^0 case, where the EM decay is suppressed by more than 4 orders of magnitude.

(6)

#7. The J/ψ was discovered as a narrow resonance in e^+e^- collisions, but not the $g.s.$ state η_c ($J^{PC} = 0^{-+}$).

(S)(#7) The production mechanism via e^+e^- annihilation is dominantly from a single photon.

We saw previously that the vector nature of QED int. $\bar{\psi} \gamma^\mu \psi$ req's the fermion & antifermion to be opposite helicity, e.g., for 1 helicity configuration, we have:



We can see that the spins end up aligned, $\rightarrow$ must be a $J=1$ state.

The J/ψ is a $J=1$ state, but the η_c is a $J=0$ state,

$\hookrightarrow$ This disallows direct production of η_c in e^+e^- collisions (thru a single σ).

The identical argument $\rightarrow$ we don't see

$$\eta_c \rightarrow e^+e^- \text{ decays}$$